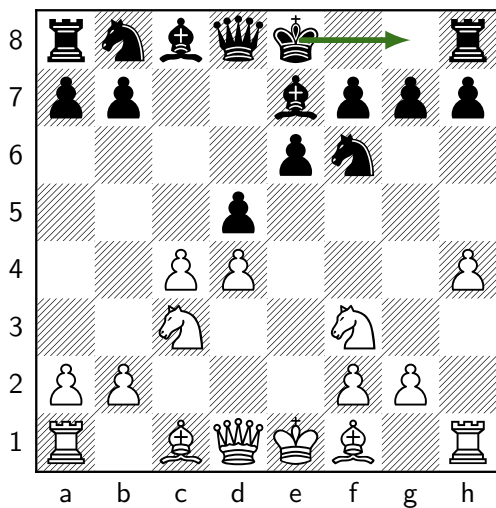


# Melanie-ANALOGY — Caro-Kann, Panov

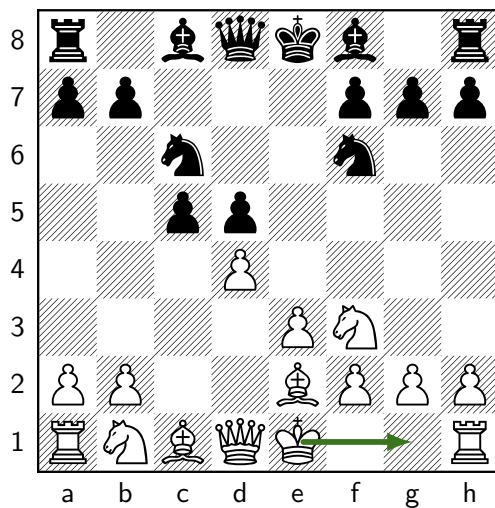
Melanie (Deep\_Vision)

## Abstract

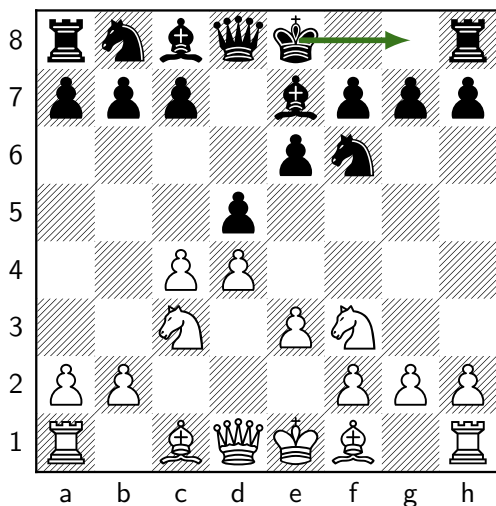
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



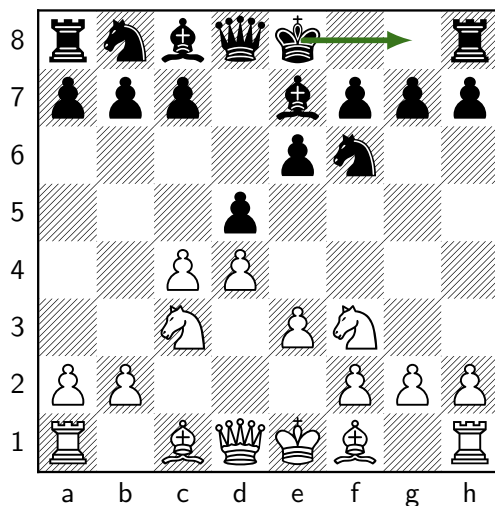
(a)



(b)

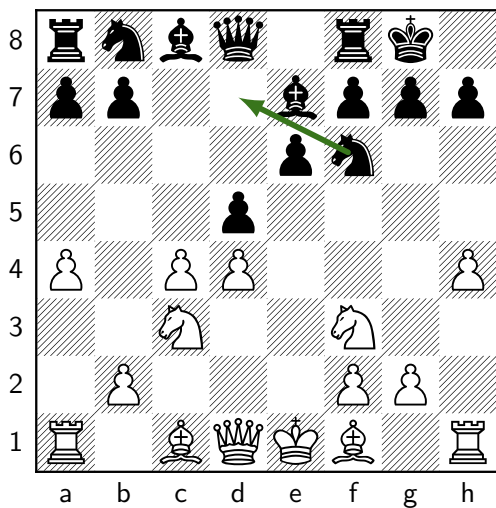


(c)

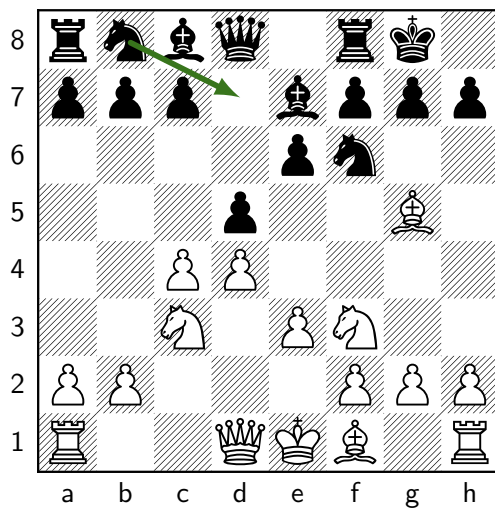


(d)

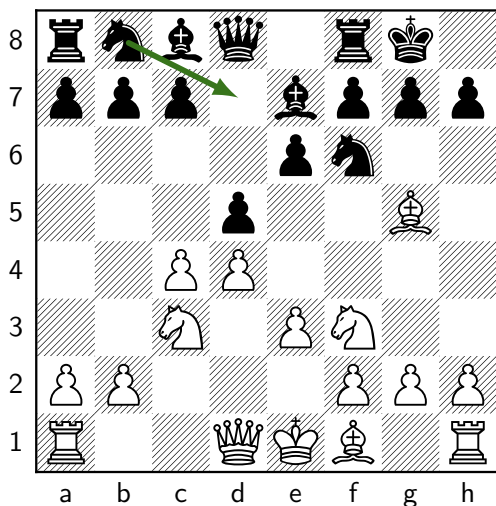
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0001#2417` (ply 12) [structural distance  $d=1$ , QAP obj=108]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Be7* as their *Ra1*, our *ra1* as their *Nb1*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `cctl_top_p0000#1490` (ply 9) [structural distance  $d=30$ , QAP obj=107]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *ph4* as their *Ra8*, our *Be7* as their *pd4*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#3836` (ply 9) [structural distance  $d=30$ , QAP obj=107]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *ph4* as their *Ra8*, our *Be7* as their *pd4*; transferred move e8g8 (**endorses this move**). Synthesis: 22 retrieved analogs transferred to 0-0 (summed QAP objective 2072, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



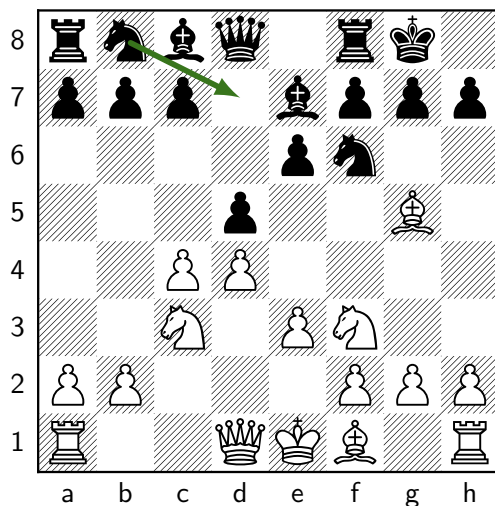
(a)



(b)

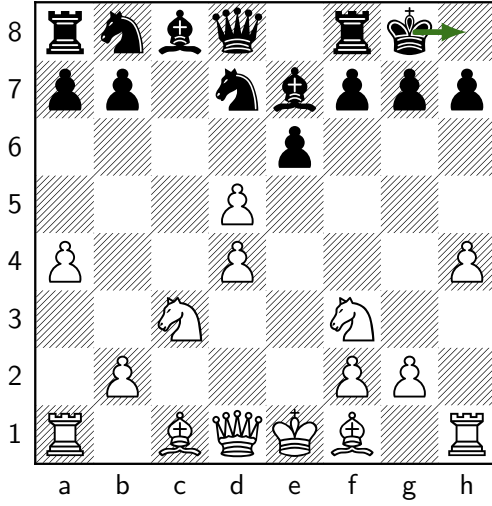


(c)

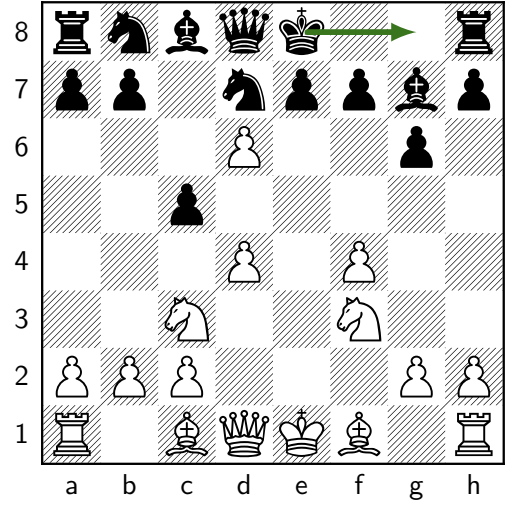


(d)

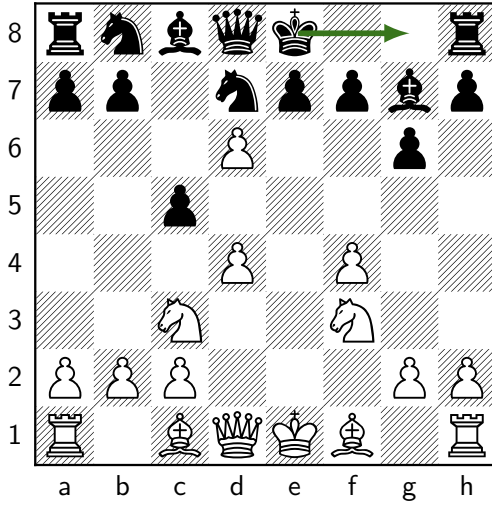
Figure 2: **Move 8, Black: Nfd7.** (a) Our position; Melanie plays Nfd7 (f6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#21 (ply 11) [structural distance  $d=34$ , QAP obj=49]: the master played Nbd7; under  $\mu$  our Nf3 plays the role of that game’s Nb1; cross-type roles: our pd4 as their Bc8, our Be7 as their Pa7; transferred move f6d7 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#22 (ply 11) [structural distance  $d=34$ , QAP obj=49]: the master played Nbd7; under  $\mu$  our Nf3 plays the role of that game’s Nb1; cross-type roles: our pd4 as their Bc8, our Be7 as their Pa7; transferred move f6d7 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1170 (ply 11) [structural distance  $d=34$ , QAP obj=49]: the master played Nbd7; under  $\mu$  our Nf3 plays the role of that game’s Nb1; cross-type roles: our pd4 as their Bc8, our Be7 as their Pa7; transferred move f6d7 (**endorses this move**). Synthesis: 31 retrieved analogs transferred to Nfd7 (summed QAP objective 1508, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



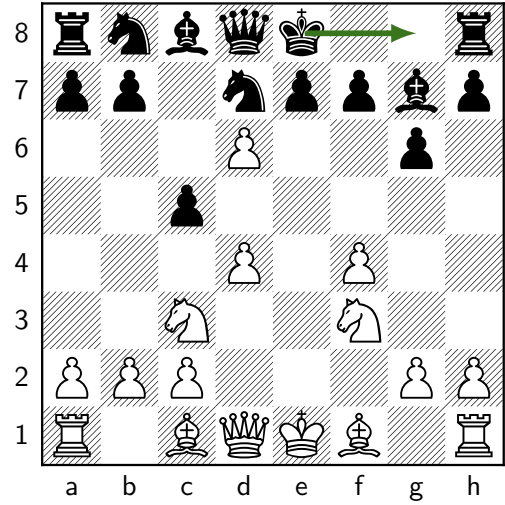
(a)



(b)

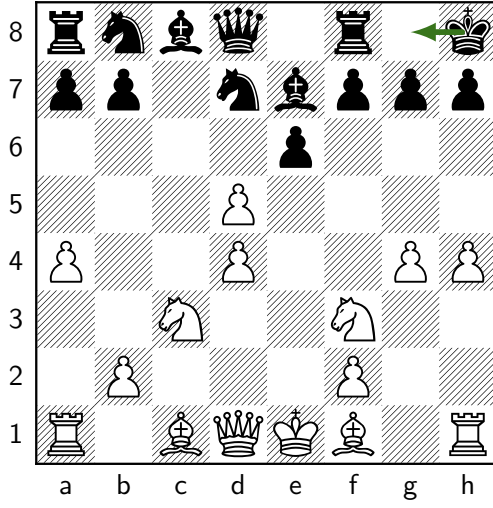


(c)

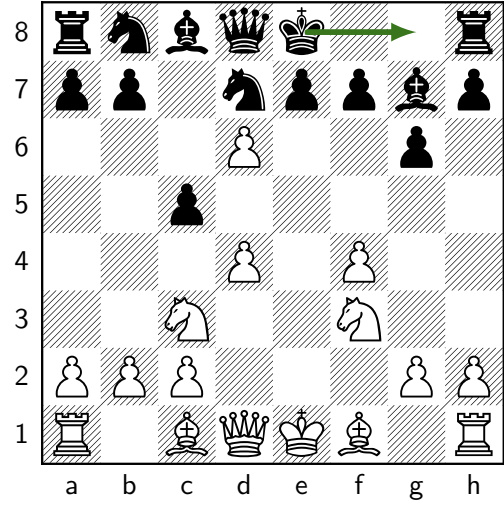


(d)

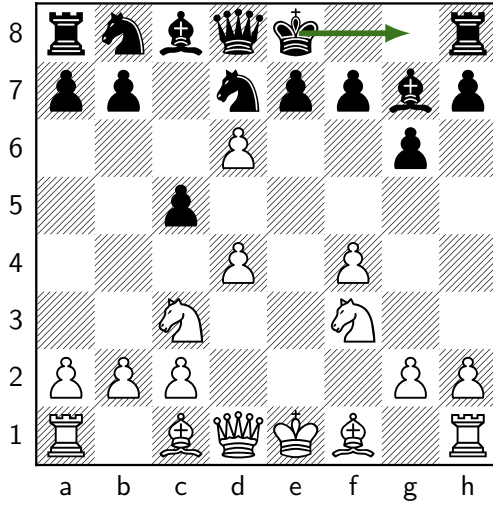
Figure 3: **Move 9, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2495` (ply 13) [structural distance  $d=68$ , QAP obj=71]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2712` (ply 13) [structural distance  $d=68$ , QAP obj=71]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2713` (ply 13) [structural distance  $d=68$ , QAP obj=71]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move g8h8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh8 (summed QAP objective 500, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



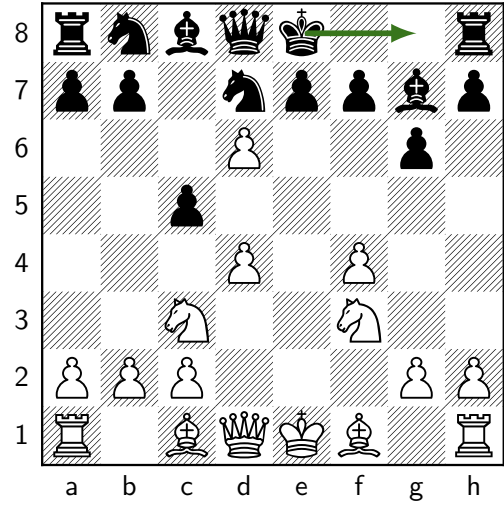
(a)



(b)

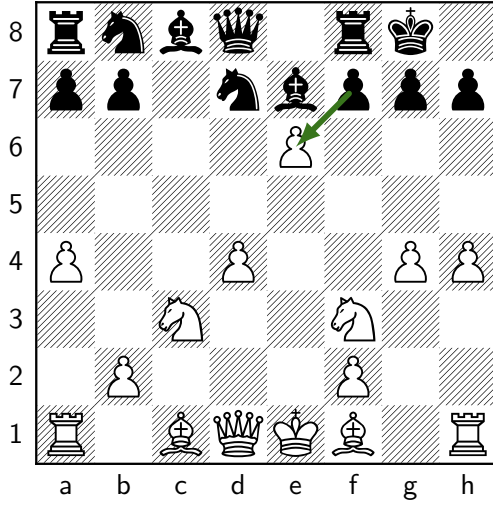


(c)

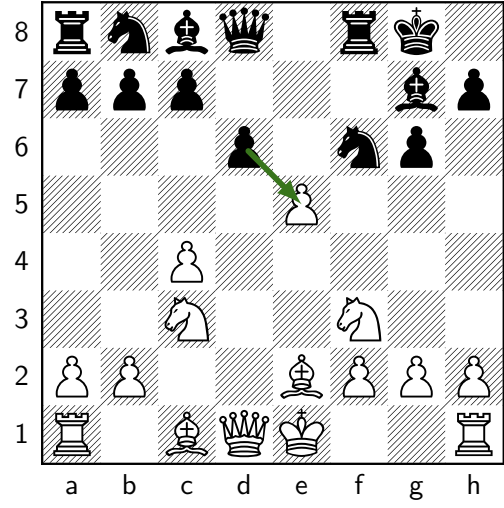


(d)

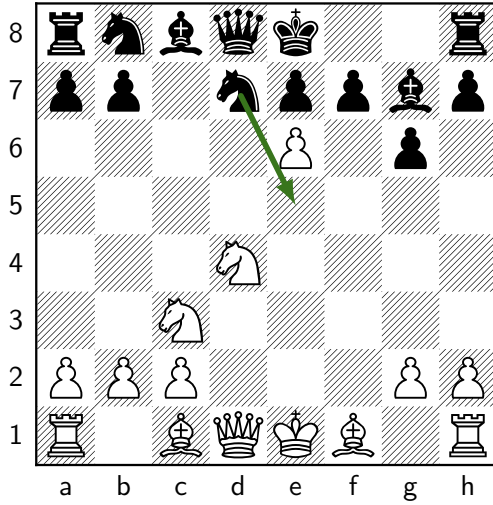
Figure 4: **Move 10, Black: Kg8**. (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2495` (ply 13) [structural distance  $d=74$ , QAP obj=69]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Be7* as their *Nb8*; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2712` (ply 13) [structural distance  $d=74$ , QAP obj=69]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Be7* as their *Nb8*; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2713` (ply 13) [structural distance  $d=74$ , QAP obj=69]: the master played 0-0; under  $\mu$  our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Be7* as their *Nb8*; transferred move h8g8 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kg8 (summed QAP objective 799, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



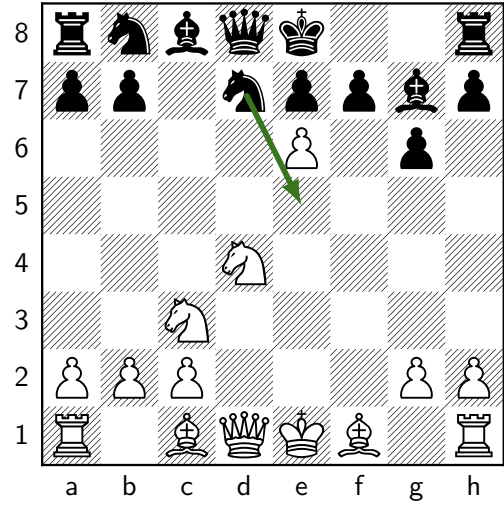
(a)



(b)

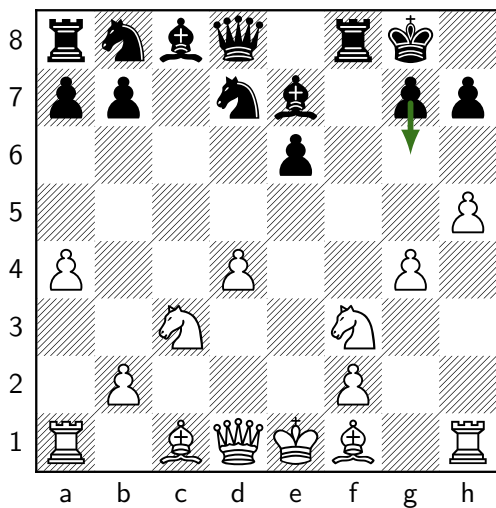


(c)

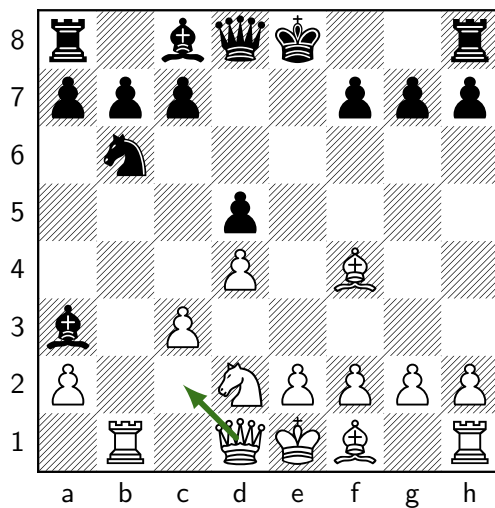


(d)

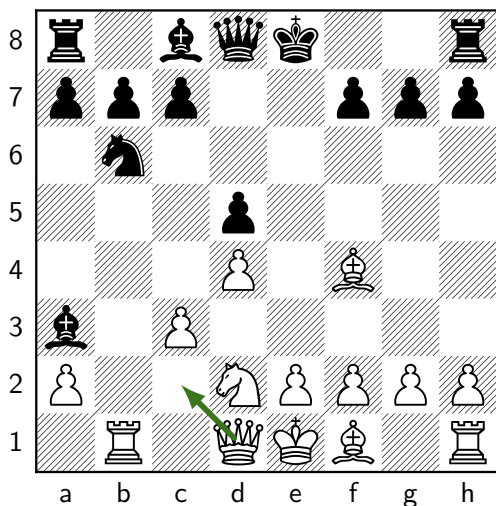
Figure 5: **Move 11, Black: fxe6**. (a) Our position; Melanie plays fxe6 (f7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2378` (ply 17) [structural distance  $d=88$ , QAP obj=58]: the master played dxe5; under  $\mu$  our Pf2 plays the role of that game’s Pd3; cross-type roles: our Bc8 as their Ra8, our Ra8 as their Nb8; transferred move f7e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2894` (ply 17) [structural distance  $d=82$ , QAP obj=61]: the master played Ne5; under  $\mu$  our Nd2 plays the role of that game’s Nd2; cross-type roles: our Rf8 as their Qd8, our Qd8 as their Rh8; transferred move d7f6. (d) Retrieved from `ccr1_top_p0000#2895` (ply 17) [structural distance  $d=82$ , QAP obj=61]: the master played Ne5; under  $\mu$  our Nd2 plays the role of that game’s Nd2; cross-type roles: our Rf8 as their Qd8, our Qd8 as their Rh8; transferred move d7f6. Synthesis: 1 retrieved analog transferred to fxe6 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



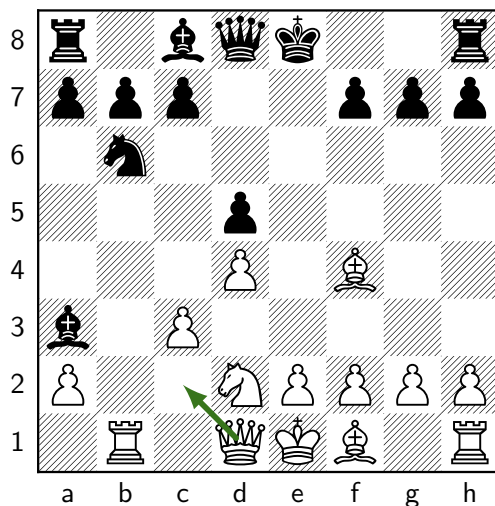
(a)



(b)

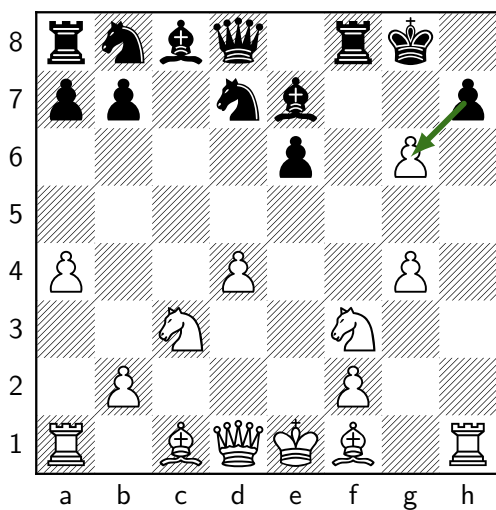


(c)

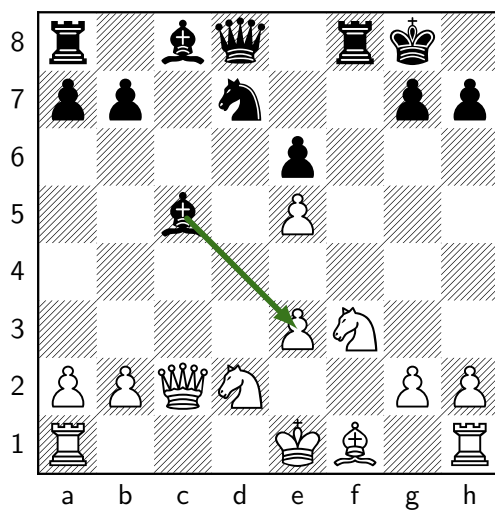


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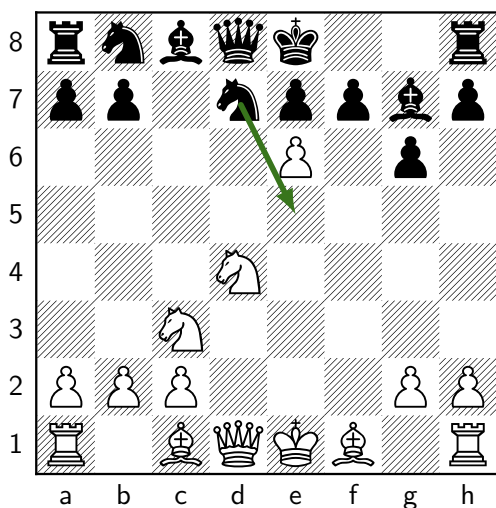
Figure 6: **Move 12, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#1947 (ply 20) [structural distance  $d=93$ , QAP obj=25]: the master played Qc2; under  $\mu$  our Pg2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pg4 as their Bf1, our bf1 as their Rh1; transferred move g7g6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#791 (ply 20) [structural distance  $d=93$ , QAP obj=25]: the master played Qc2; under  $\mu$  our Pg2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pg4 as their Bf1, our bf1 as their Rh1; transferred move g7g6 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#792 (ply 20) [structural distance  $d=93$ , QAP obj=25]: the master played Qc2; under  $\mu$  our Pg2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pg4 as their Bf1, our bf1 as their Rh1; transferred move g7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



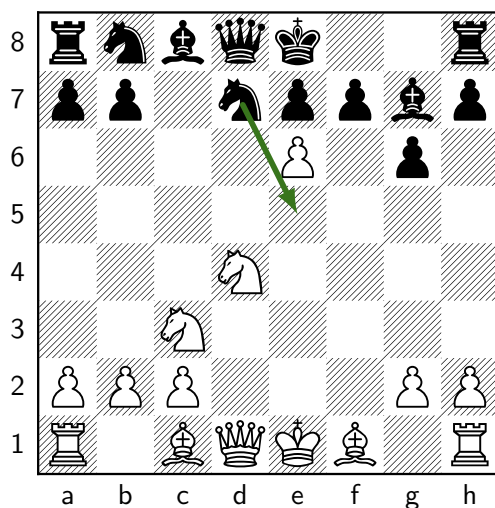
(a)



(b)



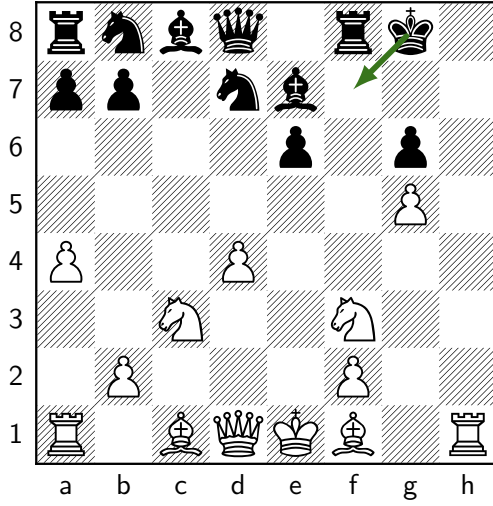
(c)



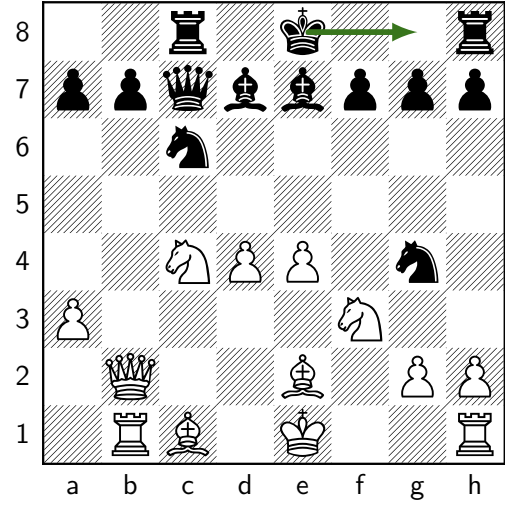
(d)

Figure 7: **Move 13, Black: hxg6**. (a) Our position; Melanie plays **hxg6** (**h7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#728` (ply 23) [structural distance  $d=122$ , QAP obj=46]: the master played **Bxe3**; under  $\mu$  our *Ph2* plays the role of that game’s *Bc4* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Bc8*, our *Be7* as their *Pe6*; transferred move **h7g6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2894` (ply 17) [structural distance  $d=110$ , QAP obj=21]: the master played **Ne5**; under  $\mu$  our *Ph2* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra8*, our *pa4* as their *Nb8*; transferred move **h7g6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2895` (ply 17) [structural distance  $d=110$ , QAP obj=21]: the master played **Ne5**; under  $\mu$  our *Ph2* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra8*, our *pa4* as their *Nb8*; transferred move **h7g6** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **hxg6** (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

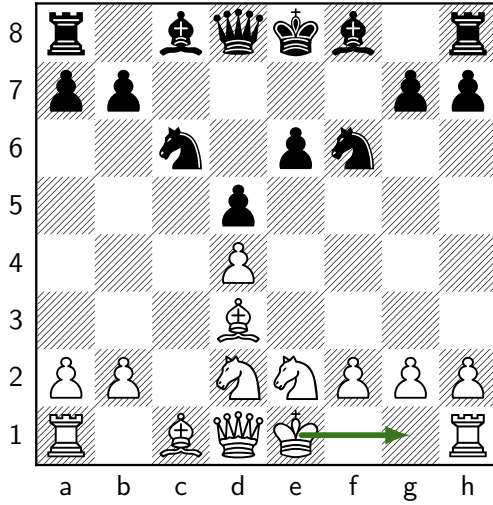




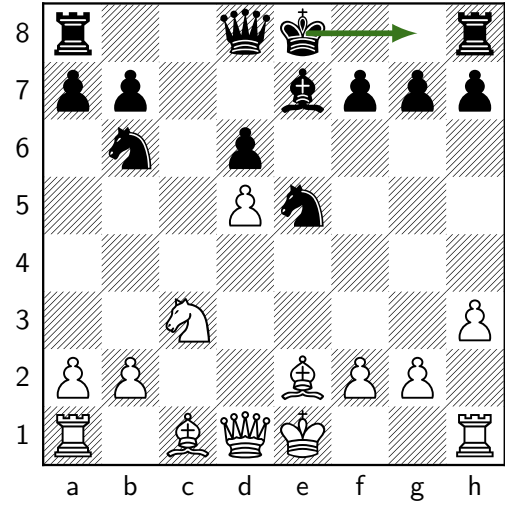
(a)



(b)

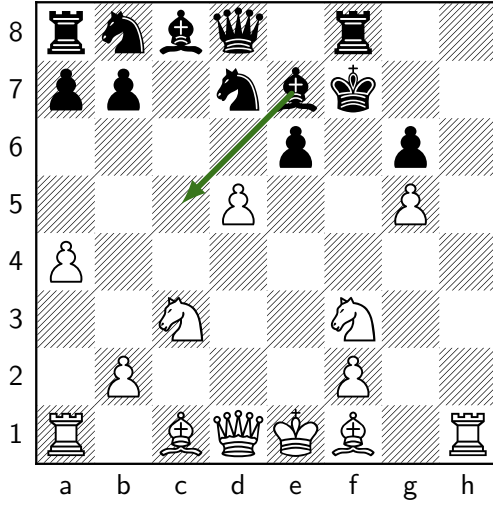


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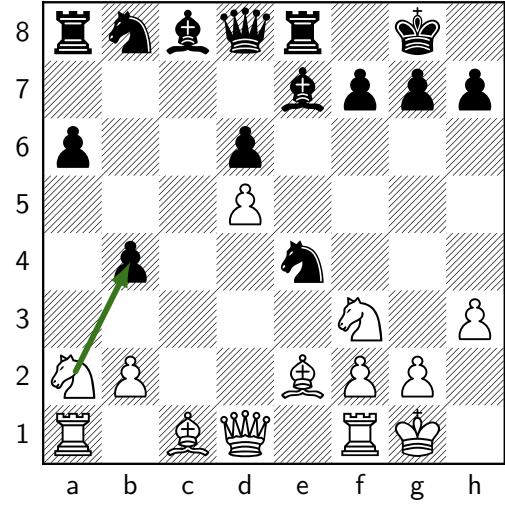


(d)

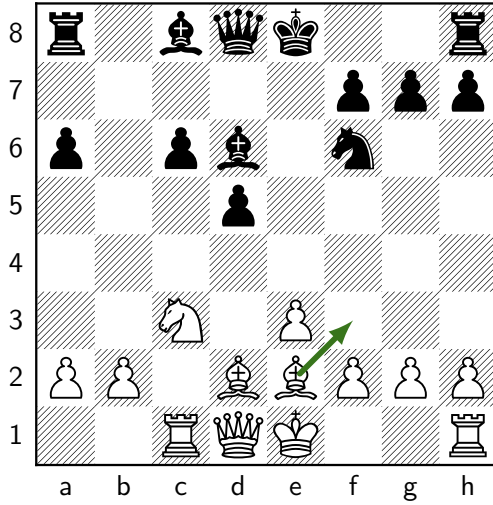
Figure 8: **Move 14, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#638` (ply 35) [structural distance  $d=140$ , QAP obj=39]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa7* as their *Rc8*, our *Bc8* as their *Rh8*; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1150` (ply 18) [structural distance  $d=125$ , QAP obj=26]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nd7* as their *Ra1*, our *Pe6* as their *Bc1*; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2918` (ply 25) [structural distance  $d=132$ , QAP obj=23]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd8* as their *Ra8*, our *Rf8* as their *Qd8*; transferred move g8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



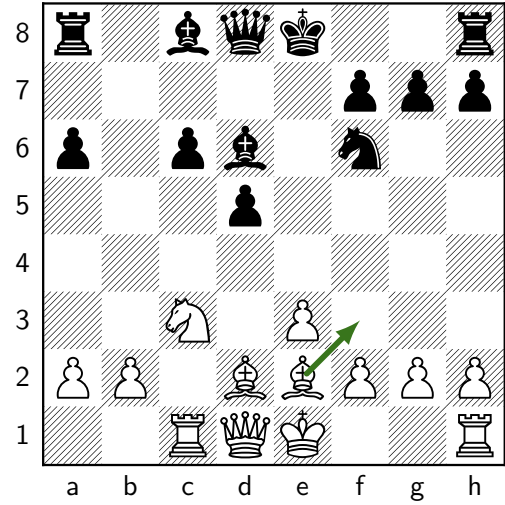
(a)



(b)

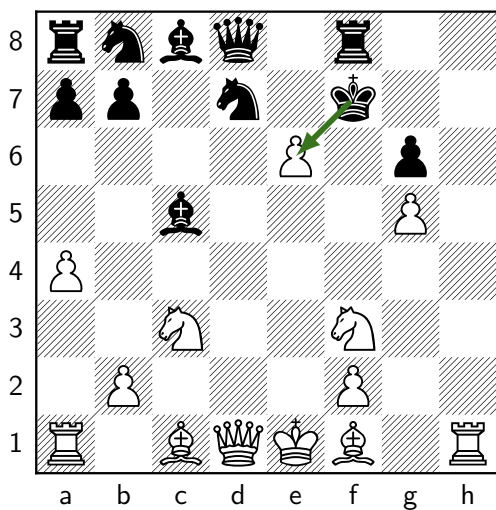


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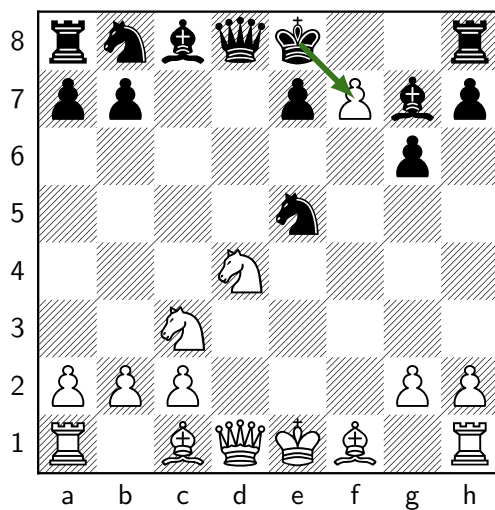


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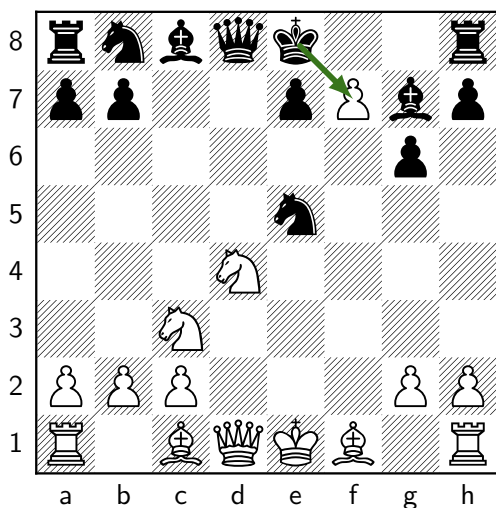
Figure 9: **Move 15, Black: Bc5.** (a) Our position; Melanie plays Bc5 (e7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3547` (ply 28) [structural distance  $d=102$ , QAP obj=24]: the master played Nxb4; under  $\mu$  our *Be2* plays the role of that game’s *Na2* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Ra1*, our *bc1* as their *Qd1*; transferred move e7c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2023` (ply 22) [structural distance  $d=110$ , QAP obj=23]: the master played Bf3; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Rf8* as their *Qd1*, our *Bc8* as their *Pa2*; transferred move e7c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2024` (ply 22) [structural distance  $d=110$ , QAP obj=23]: the master played Bf3; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Rf8* as their *Qd1*, our *Bc8* as their *Pa2*; transferred move e7c5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bc5 (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



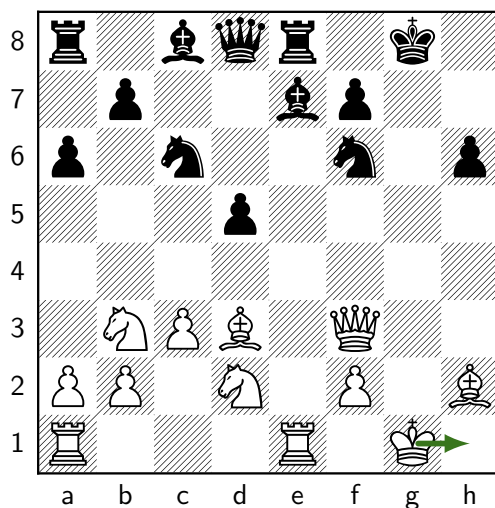
(a)



(b)

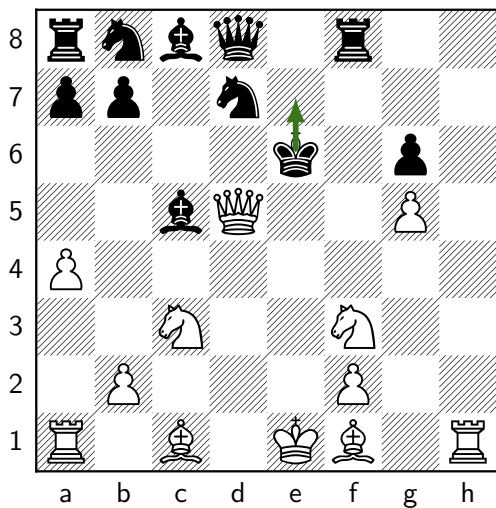


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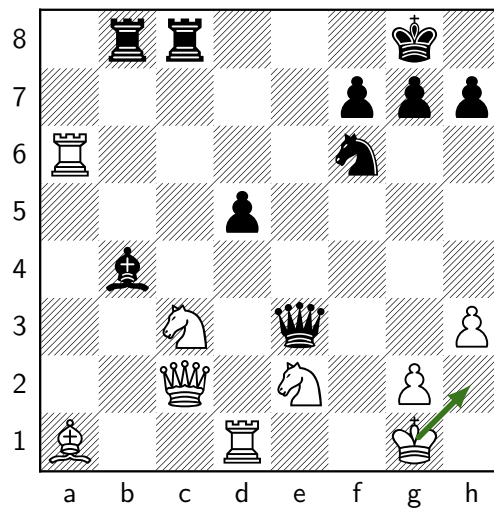


(d)

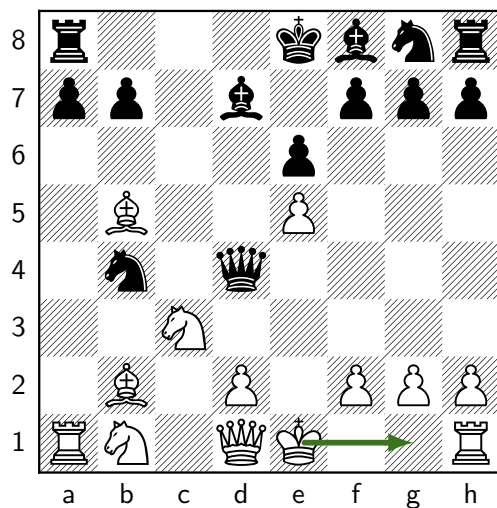
Figure 10: **Move 16, Black: Kxe6.** (a) Our position; Melanie plays Kxe6 (f7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2894` (ply 19) [structural distance  $d=122$ , QAP obj=85]: the master played Kxf7; under  $\mu$  our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Ra8* as their *Nb8*, our *pa4* as their *Bc8*; transferred move f7g8. (c) Retrieved from `ccr1_top_p0000#2895` (ply 19) [structural distance  $d=122$ , QAP obj=85]: the master played Kxf7; under  $\mu$  our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Ra8* as their *Nb8*, our *pa4* as their *Bc8*; transferred move f7g8. (d) Retrieved from `ccr1_top_p0000#1535` (ply 38) [structural distance  $d=119$ , QAP obj=37]: the master played Kh1; under  $\mu$  our *Kf2* plays the role of that game’s *Kg1*; cross-type roles: our *nc3* as their *Ra1*, our *qd1* as their *Re1*; transferred move f7g8. Synthesis: no retrieved analog transferred to Kxe6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



(a)

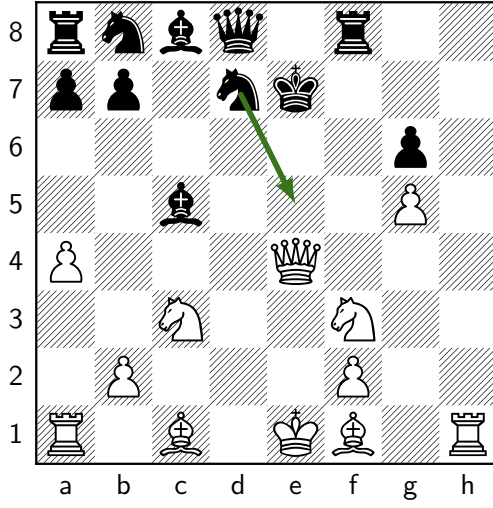


(b)

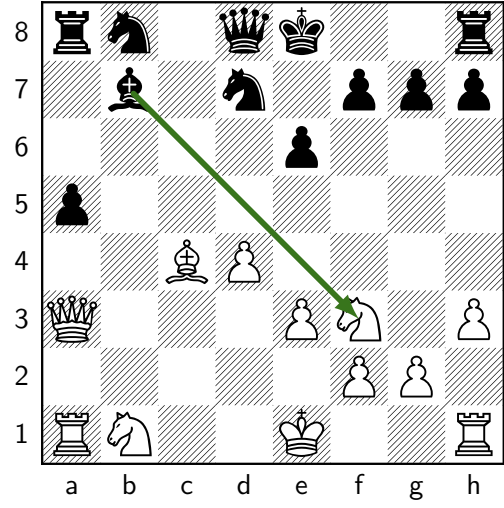


(c)

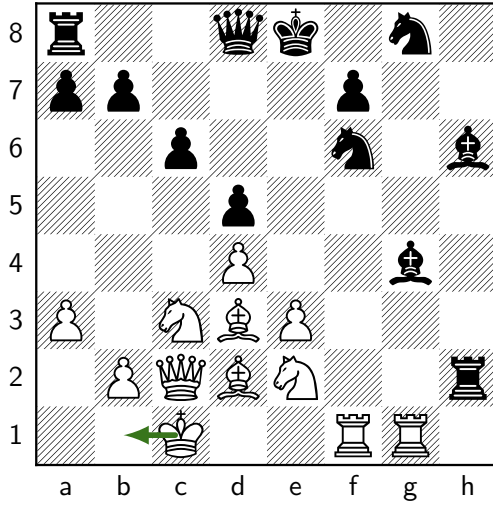
Figure 11: **Move 17, Black: Ke7.** (a) Our position; Melanie plays Ke7 (e6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2186` (ply 46) [structural distance  $d=153$ , QAP obj=45]: the master played Kh2; under  $\mu$  our Ke3 plays the role of that game’s Kg1; cross-type roles: our Nb8 as their Ba1, our Bc5 as their Rd1; transferred move e6e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#747` (ply 22) [structural distance  $d=149$ , QAP obj=25]: the master played O-O; under  $\mu$  our Ke3 plays the role of that game’s Ke1; cross-type roles: our Qd8 as their Ra1, our Pg6 as their Nb1; transferred move e6e7 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Ke7 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



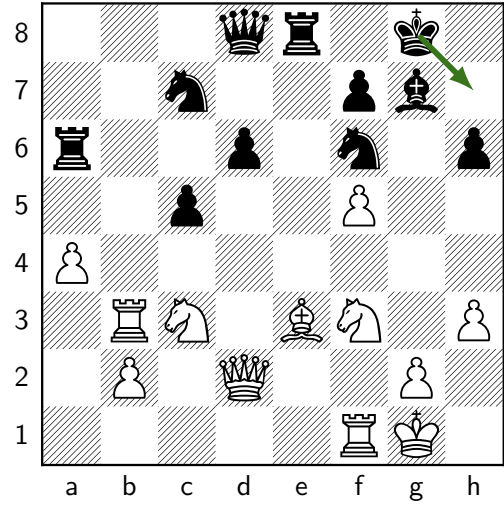
(a)



(b)

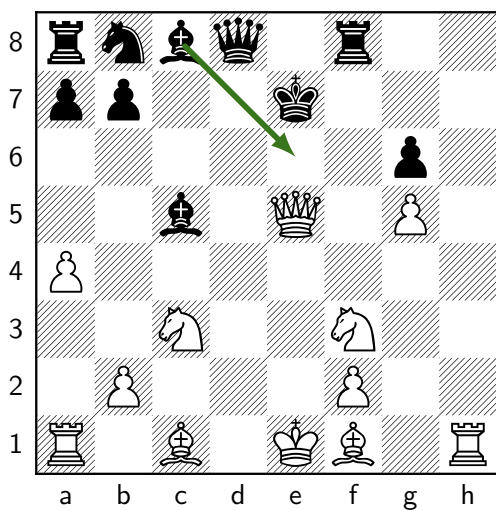


(c)

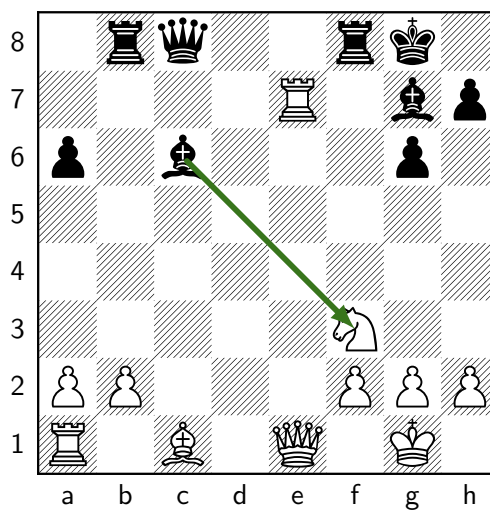


(d)

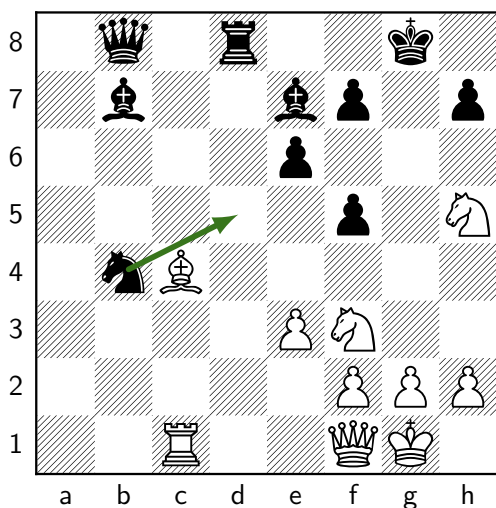
Figure 12: **Move 18, Black: Ne5**. (a) Our position; Melanie plays Ne5 (d7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3086` (ply 25) [structural distance  $d=131$ , QAP obj=19]: the master played Bxf3; under  $\mu$  our Nd2 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra8, our pb2 as their Nb8; transferred move d7e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#437` (ply 46) [structural distance  $d=132$ , QAP obj=26]: the master played Kb1; under  $\mu$  our Ke2 plays the role of that game’s Kc1; cross-type roles: our Nd7 as their Rg1, our bfl as their Ne2; transferred move e7f7. (d) Retrieved from `ccrl_top_p0003#851` (ply 41) [structural distance  $d=116$ , QAP obj=25]: the master played Kh7; under  $\mu$  our Ke2 plays the role of that game’s Kg1; cross-type roles: our Bc5 as their Qd8, our Bc8 as their Nc7; transferred move e7f7. Synthesis: 1 retrieved analog transferred to Ne5 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



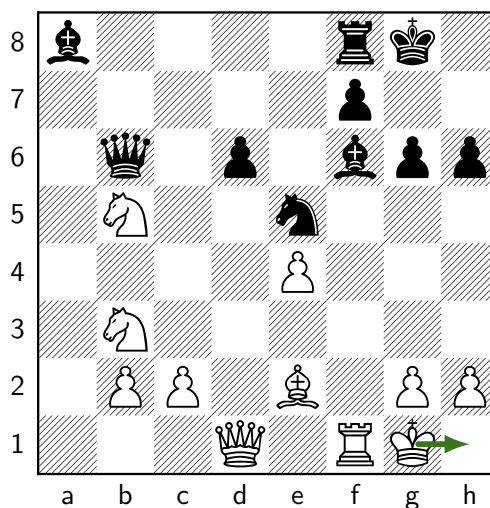
(a)



(b)

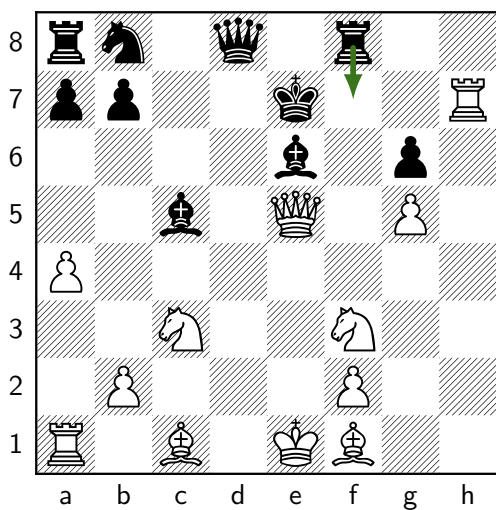


(c)

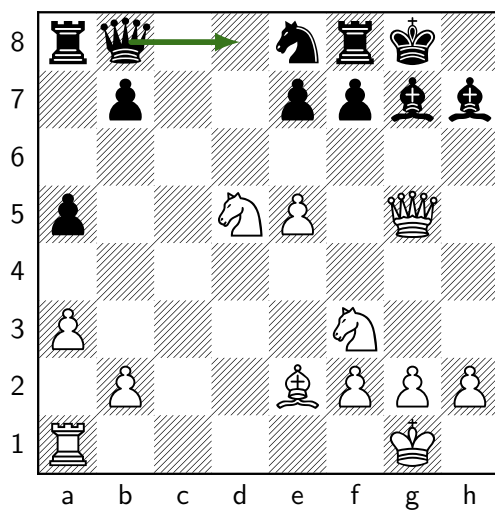


(d)

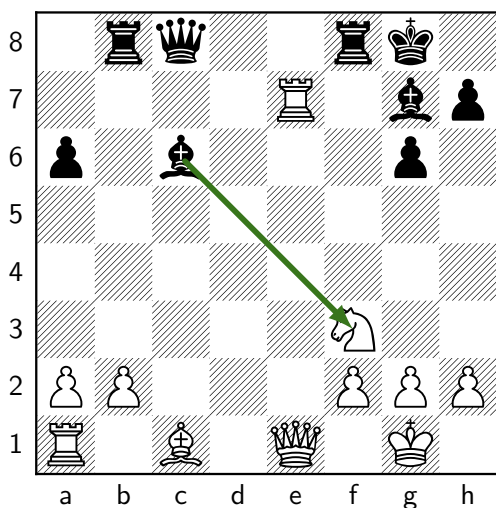
Figure 13: **Move 19, Black: Be6.** (a) Our position; Melanie plays **Be6** (c8e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1206` (ply 45) [structural distance  $d=82$ , QAP obj=16]: the master played **Bxf3**; under  $\mu$  our *Bc1* plays the role of that game’s *Bc3*; cross-type roles: our *Bc5* as their *Rb8*, our *bf1* as their *Qc8*; transferred move **c8e6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1530` (ply 45) [structural distance  $d=103$ , QAP obj=6]: the master played **Nd5**; under  $\mu$  our *Bc1* plays the role of that game’s *Nb5* (a **cross-type** correspondence); cross-type roles: our *Bc5* as their *Qb8*, our *Qd8* as their *Rd8*; transferred move **c8e6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2328` (ply 36) [structural distance  $d=84$ , QAP obj=47]: the master played **Kh1**; under  $\mu$  our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *Rf8* as their *Qd1*, our *Qd8* as their *Rf1*; transferred move **e7f7**. Synthesis: 2 retrieved analogs transferred to **Be6** (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



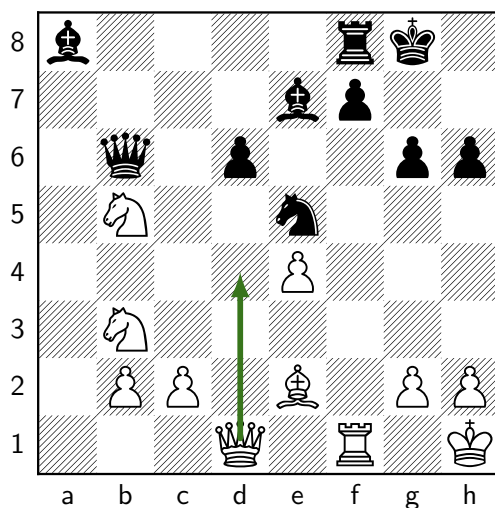
(a)



(b)



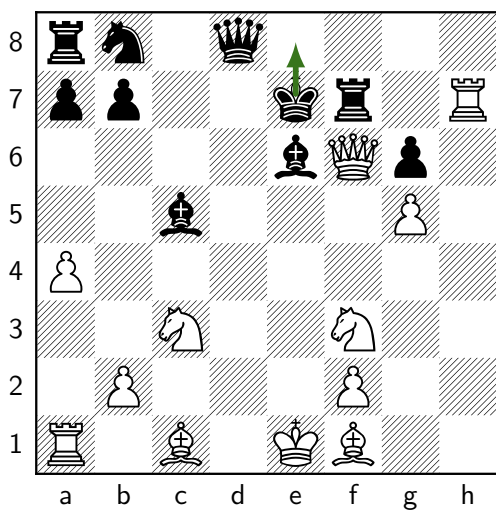
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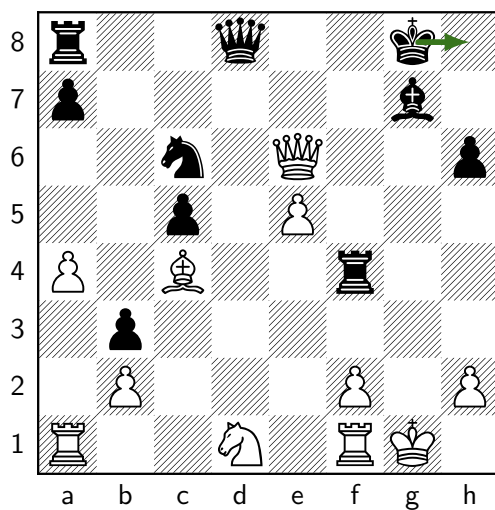
(d)

Figure 14: **Move 20, Black: Rf7.** (a) Our position; Melanie plays Rf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2295` (ply 43) [structural distance  $d=112$ , QAP obj=34]: the master played Qd8; under  $\mu$  our Rf1 plays the role of that game’s Qb1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Ne8, our Pg6 as their Rf8; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1206` (ply 45) [structural distance  $d=84$ , QAP obj=18]: the master played Bxf3; under  $\mu$  our Rf1 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Bc5 as their Qc8, our Qd8 as their Rf8; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2328` (ply 38) [structural distance  $d=107$ , QAP obj=17]: the master played Qd4; under  $\mu$  our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rf1, our bc1 as their Pc2; transferred move f8f7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rf7 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

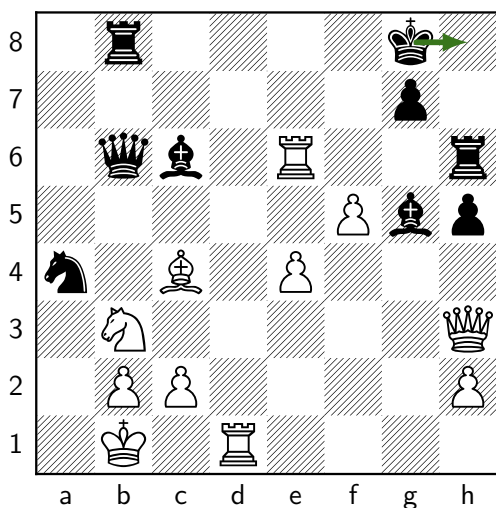




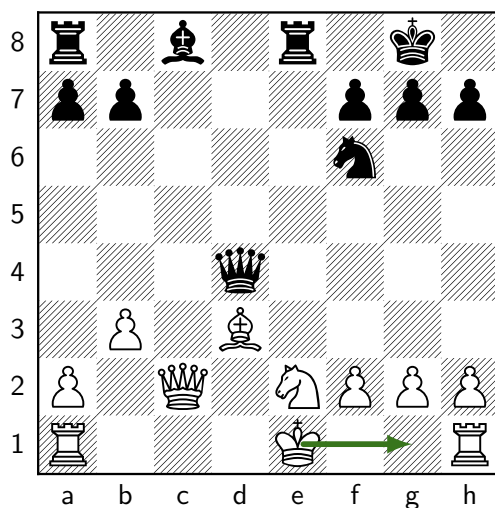
(a)



(b)



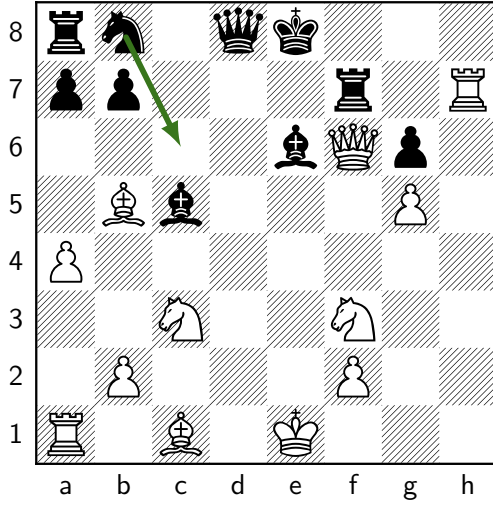
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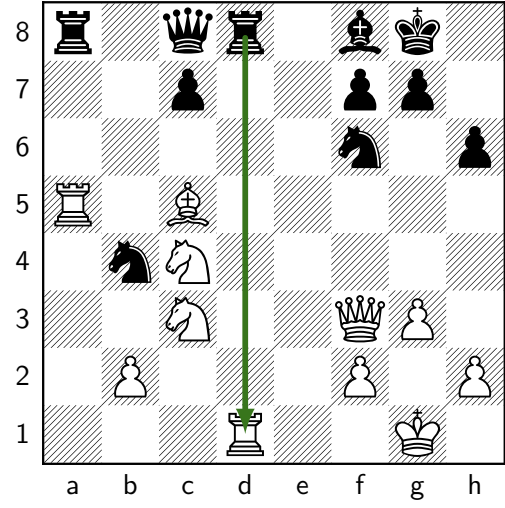
(d)

Figure 15: **Move 21, Black: Ke8.** (a) Our position; Melanie plays Ke8 (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3161` (ply 43) [structural distance  $d=135$ , QAP obj=41]: the master played Kh8; under  $\mu$  our Ke2 plays the role of that game's Kg1; cross-type roles: our Ra8 as their Pa7, our rh7 as their bc4; transferred move e7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1348` (ply 57) [structural distance  $d=124$ , QAP obj=16]: the master played Kh8; under  $\mu$  our Ke2 plays the role of that game's Kg1; cross-type roles: our Be6 as their Qb6, our Rf7 as their Bc6; transferred move e7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3503` (ply 30) [structural distance  $d=104$ , QAP obj=13]: the master played 0-0; under  $\mu$  our Ke2 plays the role of that game's Ke1; cross-type roles: our Qd8 as their Ra1, our Ra8 as their Pa2; transferred move e7e8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke8 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

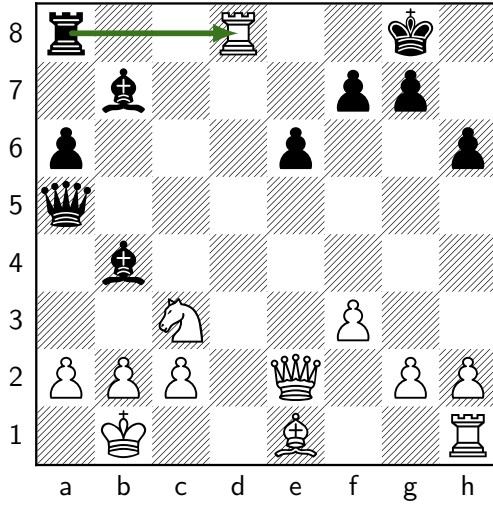




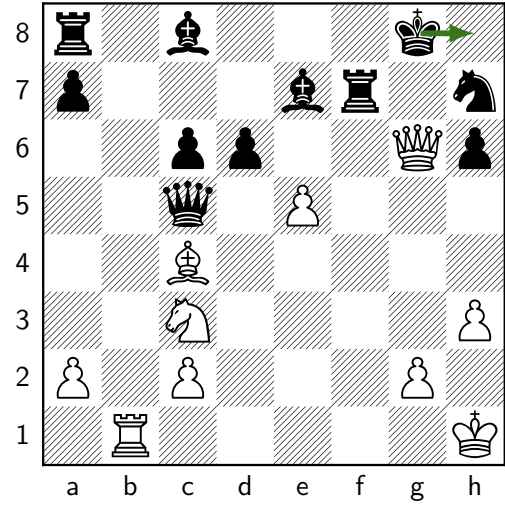
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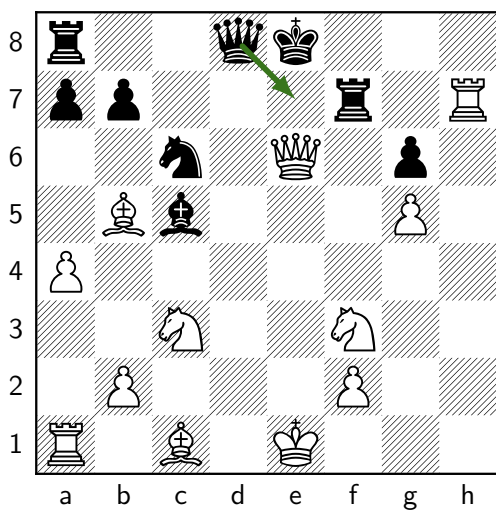


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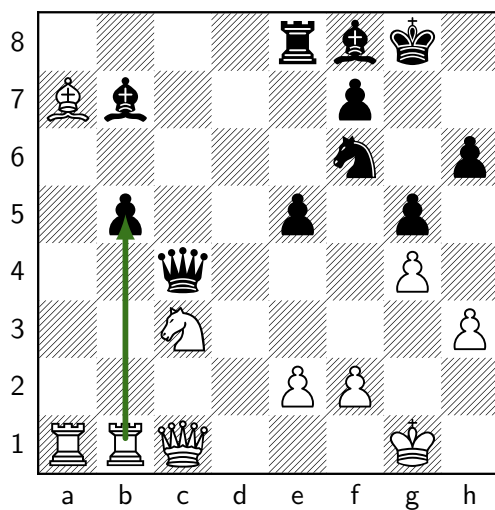


(d)

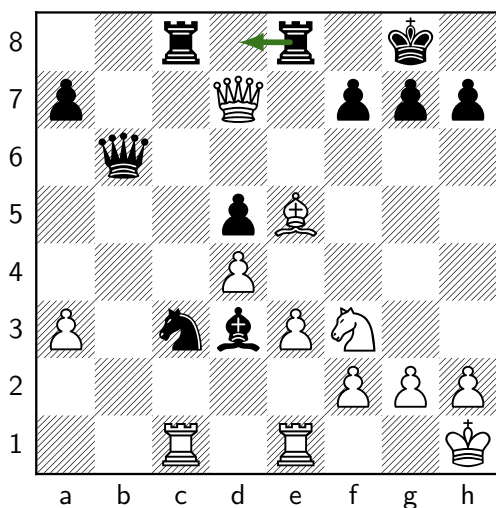
Figure 16: **Move 22, Black: Nc6.** (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1955` (ply 45) [structural distance  $d=138$ , QAP obj=3]: the master played  $Rxd1+$ ; under  $\mu$  our  $Nb1$  plays the role of that game’s  $Rd1$  (a **cross-type** correspondence); cross-type roles: our  $Bc5$  as their  $Qc8$ , our  $Rf7$  as their  $Bf8$ ; transferred move **b8c6 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2571` (ply 37) [structural distance  $d=141$ , QAP obj=71]: the master played  $Rxd8$ ; under  $\mu$  our  $Rf2$  plays the role of that game’s  $Ra1$ ; cross-type roles: our  $bb5$  as their  $rd8$ , our  $nc3$  as their  $Bb7$ ; transferred move **f7d7**. (d) Retrieved from `ccr1_top_p0001#503` (ply 37) [structural distance  $d=145$ , QAP obj=50]: the master played  $Kh8$ ; under  $\mu$  our  $Ke1$  plays the role of that game’s  $Kg1$ ; cross-type roles: our  $Nb8$  as their  $Pa7$ , our  $Pa7$  as their  $Be7$ ; transferred move **e8f8**. Synthesis: 1 retrieved analog transferred to Nc6 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



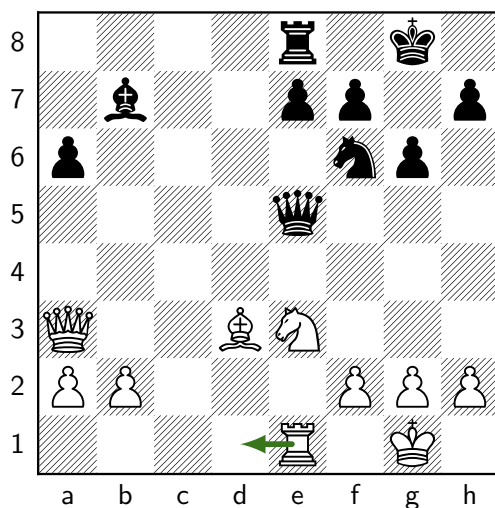
(a)



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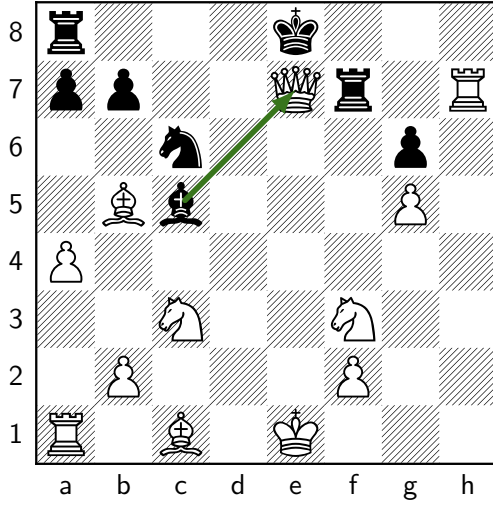


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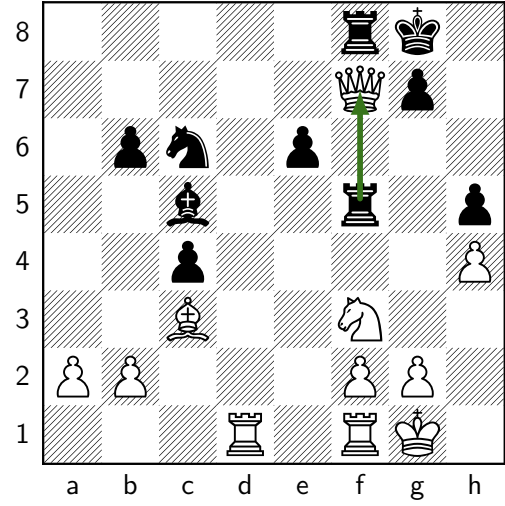


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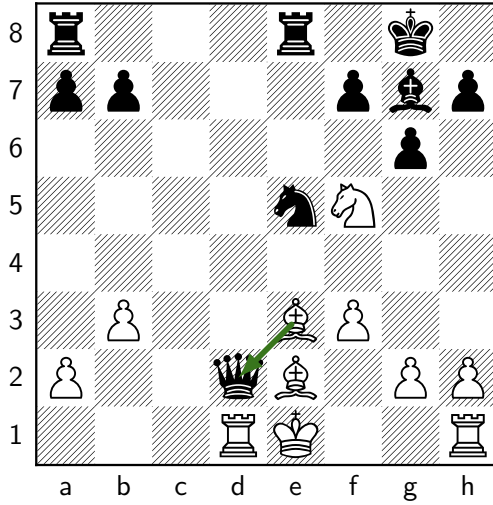
Figure 17: **Move 23, Black: Qe7**. (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2456` (ply 58) [structural distance  $d=114$ , QAP obj=27]: the master played Rxb5; under  $\mu$  our Qd1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Rf7 as their Qc1, our bb5 as their qc4; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2736` (ply 43) [structural distance  $d=118$ , QAP obj=20]: the master played Red8; under  $\mu$  our Qd1 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Bc5 as their Rc8, our nf3 as their Pg7; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2247` (ply 60) [structural distance  $d=126$ , QAP obj=20]: the master played Rd1; under  $\mu$  our Qd1 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pa2, our Ra8 as their Pg2; transferred move d8e7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qe7 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



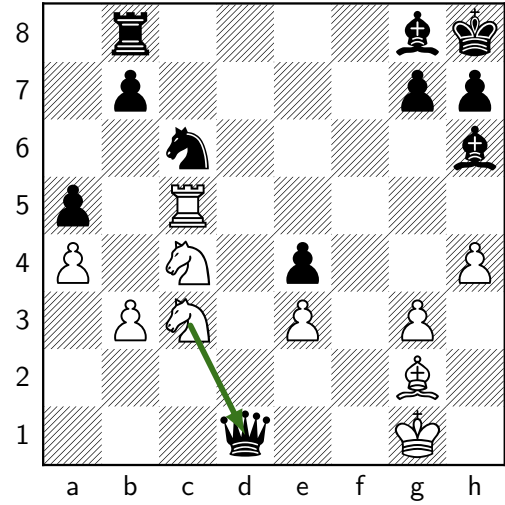
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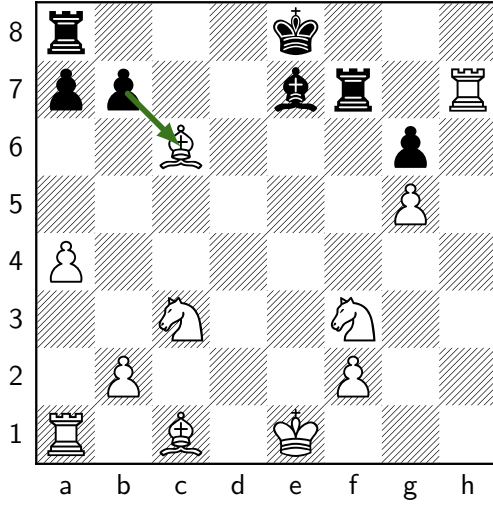


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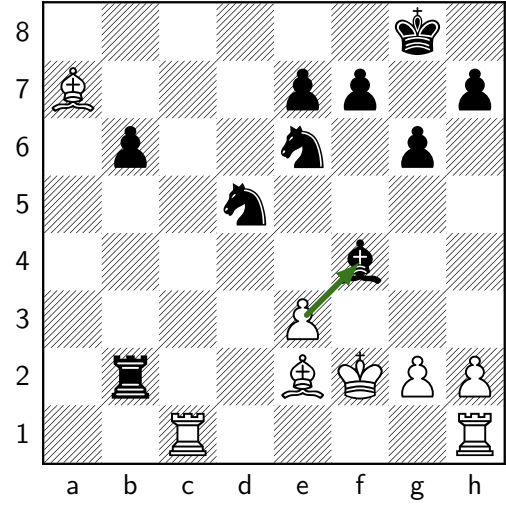


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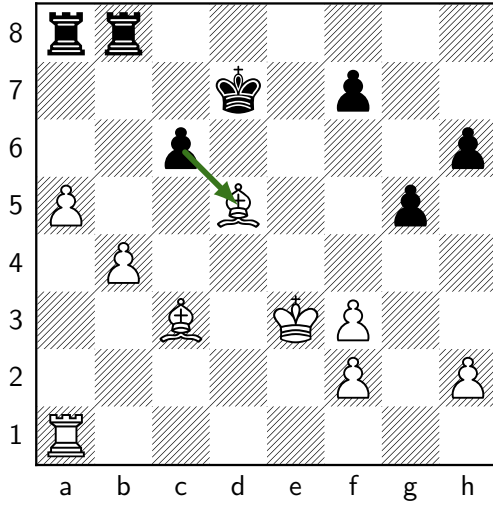
Figure 18: **Move 24, Black: Bxe7 (c5e7).** (a) Our position; Melanie plays Bxe7 (c5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3708` (ply 49) [structural distance  $d=156$ , QAP obj=86]: the master played R5xf7; under  $\mu$  our Bc4 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Bc5, our bc1 as their Ph5; transferred move c5e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1120` (ply 36) [structural distance  $d=139$ , QAP obj=85]: the master played Bxd2; under  $\mu$  our Bc4 plays the role of that game’s Be3; cross-type roles: our rh7 as their Be2, our bb5 as their Pf3; transferred move c5e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1161` (ply 56) [structural distance  $d=156$ , QAP obj=25]: the master played Nxd1; under  $\mu$  our Bc4 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Rf7 as their Bg2, our Ra8 as their Pe3; transferred move c5e7 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Bxe7 (summed QAP objective 270, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



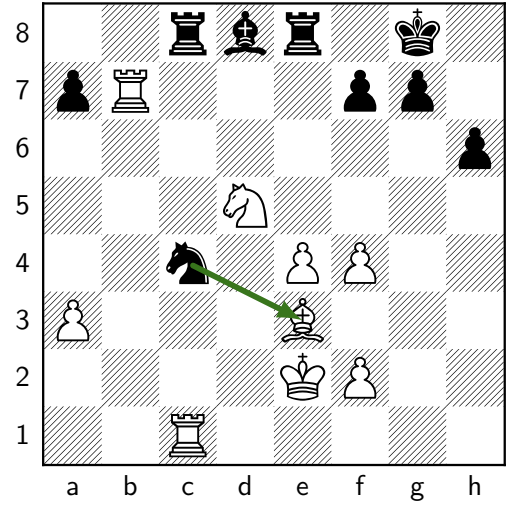
(a)



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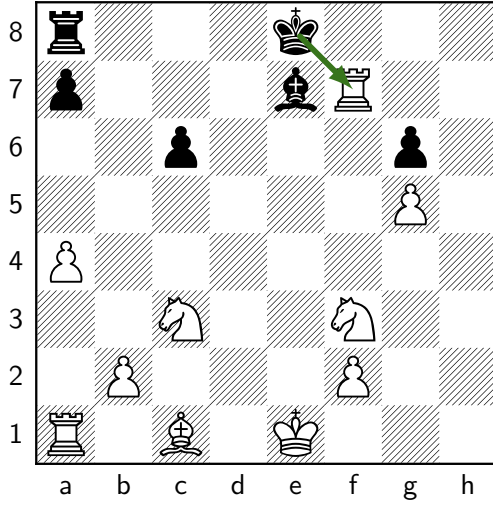


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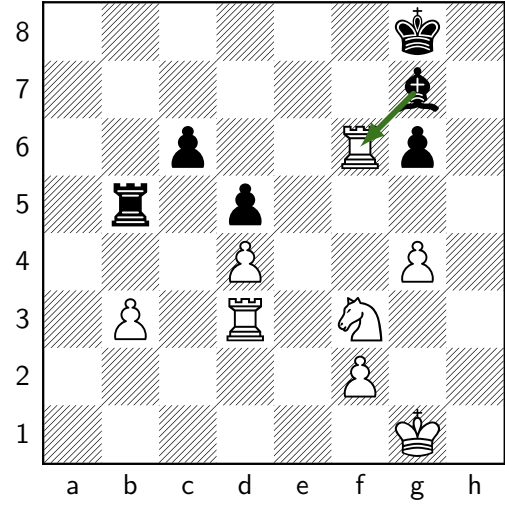


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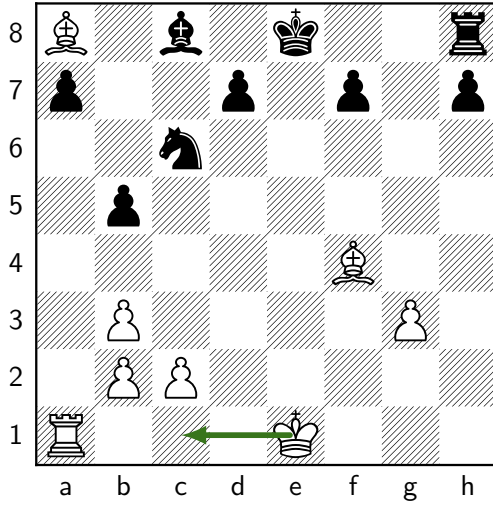
Figure 19: **Move 25, Black: bxc6.** (a) Our position; Melanie plays **bxc6** (**b7c6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3136` (ply 56) [structural distance  $d=97$ , QAP obj=33]: the master played **exf4**; under  $\mu$  our *Pb2* plays the role of that game’s *Pe3*; cross-type roles: our *bc6* as their *rb2*, our *Pb7* as their *Be2*; transferred move **b7c6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#544` (ply 55) [structural distance  $d=99$ , QAP obj=25]: the master played **cxd5**; under  $\mu$  our *Pb2* plays the role of that game’s *Pc3*; cross-type roles: our *Be7* as their *Ra8*, our *Ra8* as their *Pf7*; transferred move **b7c6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1492` (ply 43) [structural distance  $d=95$ , QAP obj=23]: the master played **Nxe3**; under  $\mu$  our *Pb2* plays the role of that game’s *Nc5* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Pa7*, our *nc3* as their *Pf7*; transferred move **b7c6** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **bxc6** (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



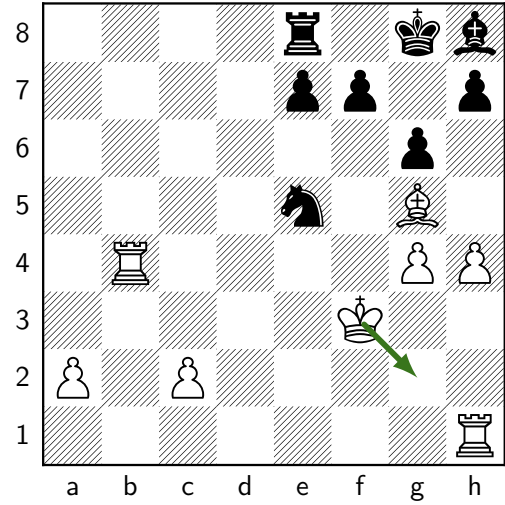
(a)



(b)



(c)



(d)

Figure 20: **Move 26, Black: Kxf7.** (a) Our position; Melanie plays Kxf7 (e8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3544` (ply 61) [structural distance  $d=60$ , QAP obj=26]: the master played Bxf6; under  $\mu$  our Ke1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Be7 as their Pg6, our nc3 as their Rb5; transferred move e8f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2392` (ply 40) [structural distance  $d=57$ , QAP obj=9]: the master played 0-0-0; under  $\mu$  our Ke1 plays the role of that game’s Ke1; cross-type roles: our nc3 as their Pg3, our pb2 as their Bf4; transferred move e8f7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1457` (ply 58) [structural distance  $d=60$ , QAP obj=4]: the master played Kg2; under  $\mu$  our Ke1 plays the role of that game’s Kf3; cross-type roles: our pa4 as their Rh1, our Ra8 as their Pa2; transferred move e8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kxf7 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

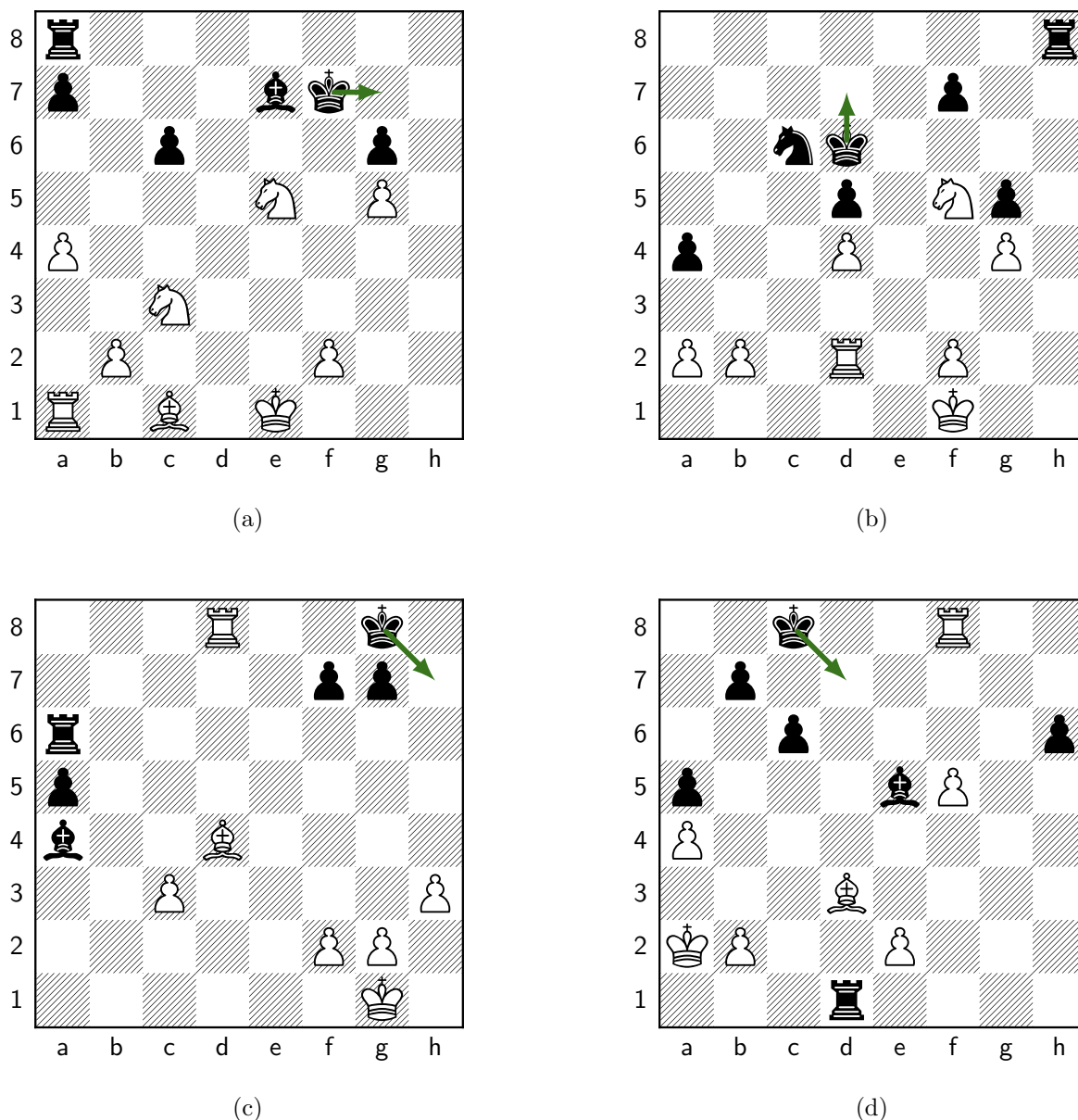
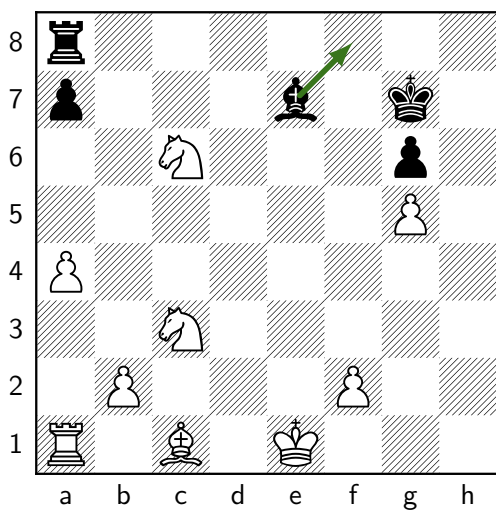
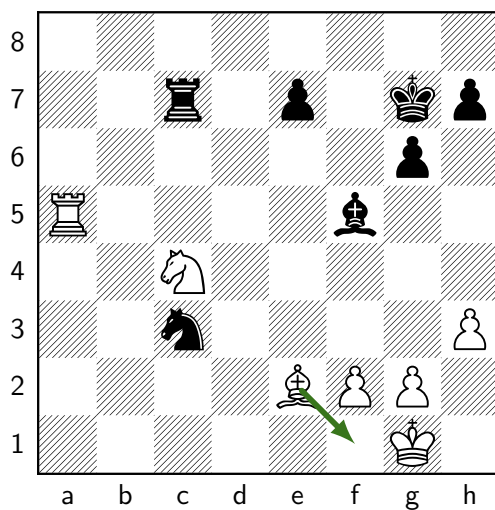


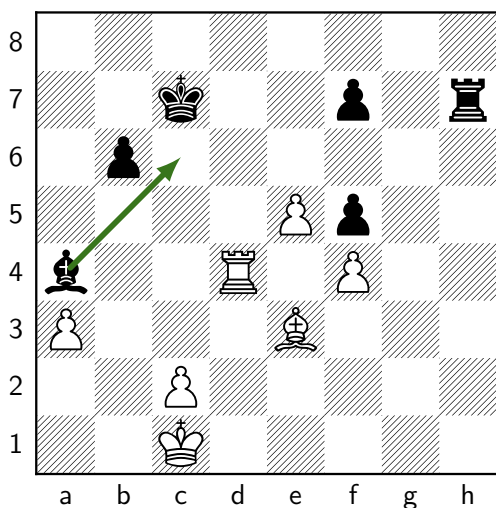
Figure 21: **Move 27, Black: Kg7.** (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3150` (ply 65) [structural distance  $d=47$ , QAP obj=75]: the master played Kd7; under  $\mu$  our Kf2 plays the role of that game’s Kd3; cross-type roles: our Be7 as their Nc6, our Ra8 as their pg4; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1054` (ply 53) [structural distance  $d=47$ , QAP obj=50]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our ne5 as their rd8, our Be7 as their Pf7; transferred move f7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2110` (ply 63) [structural distance  $d=46$ , QAP obj=48]: the master played Kd7; under  $\mu$  our Kf2 plays the role of that game’s Kc1; cross-type roles: our ne5 as their rf8, our Be7 as their Pb7; transferred move f7g7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg7 (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



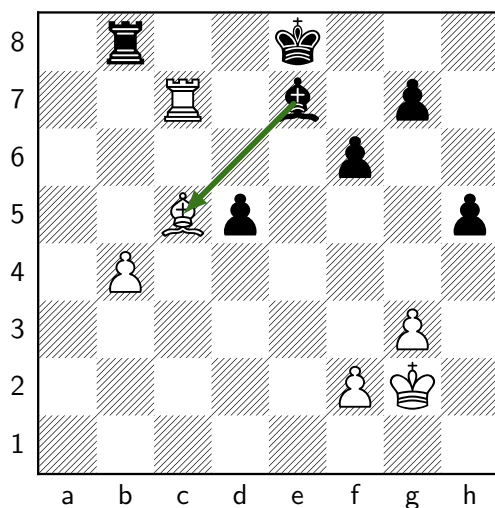
(a)



(b)

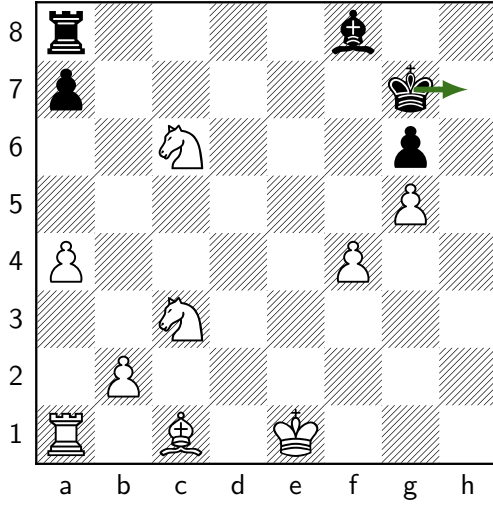


(c)

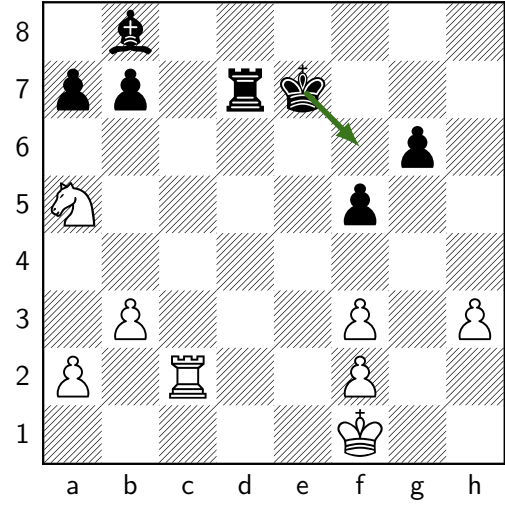


(d)

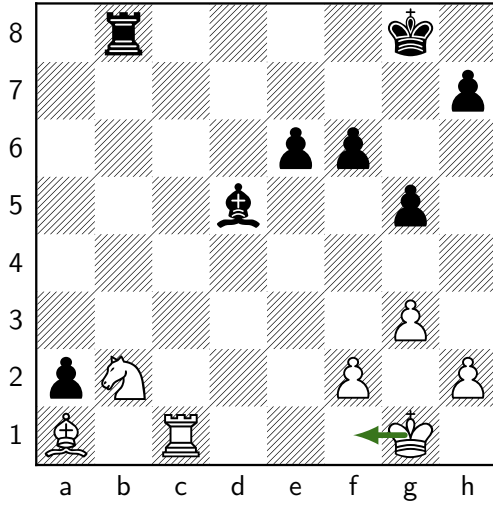
Figure 22: **Move 28, Black: Bf8**. (a) Our position; Melanie plays Bf8 (e7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#2151` (ply 56) [structural distance  $d=43$ , QAP obj=36]: the master played Bf1; under  $\mu$  our Be2 plays the role of that game’s Be2; cross-type roles: our bc1 as their Pg2, our Be7 as their Nc4; transferred move e7f8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#596` (ply 61) [structural distance  $d=45$ , QAP obj=27]: the master played Bc6; under  $\mu$  our Be2 plays the role of that game’s Ba5; cross-type roles: our pa4 as their Rh7, our nc3 as their pe5; transferred move e7f8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#1137` (ply 69) [structural distance  $d=44$ , QAP obj=33]: the master played Bxc5; under  $\mu$  our Be2 plays the role of that game’s Be2; cross-type roles: our nc3 as their Rb8, our nc6 as their rc7; transferred move e7a3. Synthesis: 2 retrieved analogs transferred to Bf8 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



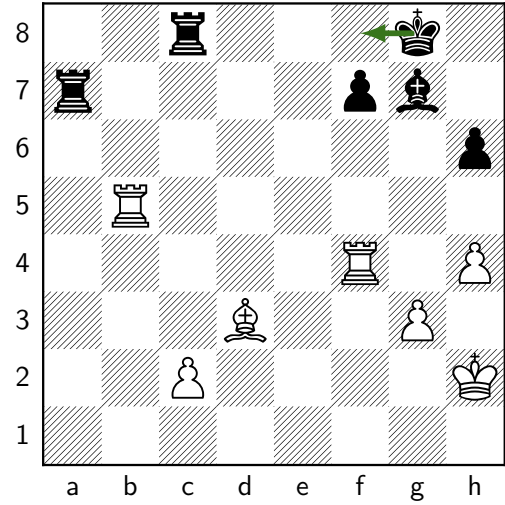
(a)



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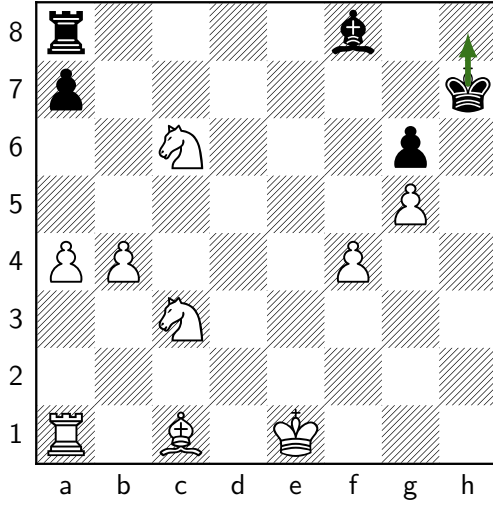
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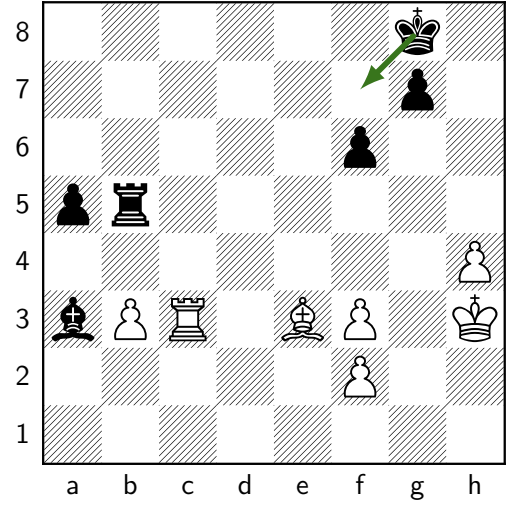
(d)

Figure 23: **Move 29, Black: Kh7 (g7h7).** (a) Our position; Melanie plays Kh7 (g7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3929` (ply 73) [structural distance  $d=45$ , QAP obj=20]: the master played Kf6; under  $\mu$  our Kg2 plays the role of that game’s Ke2; cross-type roles: our pg5 as their Bb8, our Ra8 as their Pa7; transferred move g7h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3339` (ply 72) [structural distance  $d=45$ , QAP obj=20]: the master played Kf1; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Ba1, our Bf8 as their Rc1; transferred move g7h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3482` (ply 71) [structural distance  $d=48$ , QAP obj=14]: the master played Kf8; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Bf8 as their Rc8, our nc3 as their Bg7; transferred move g7h7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh7 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

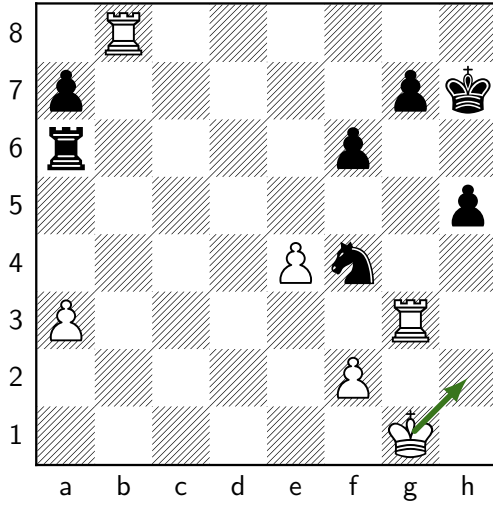




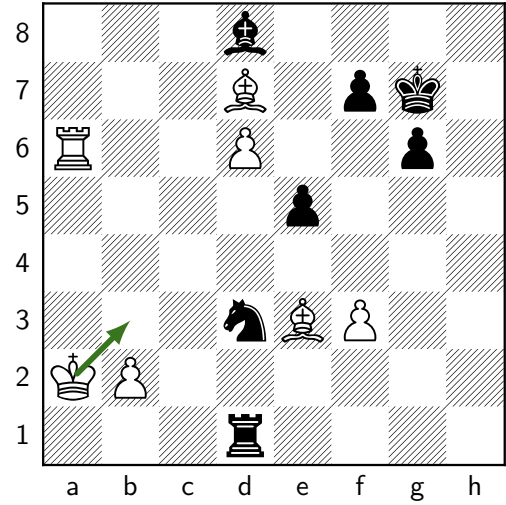
(a)



(b)

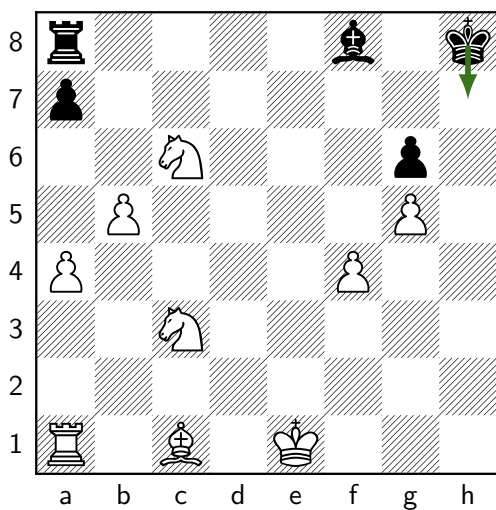


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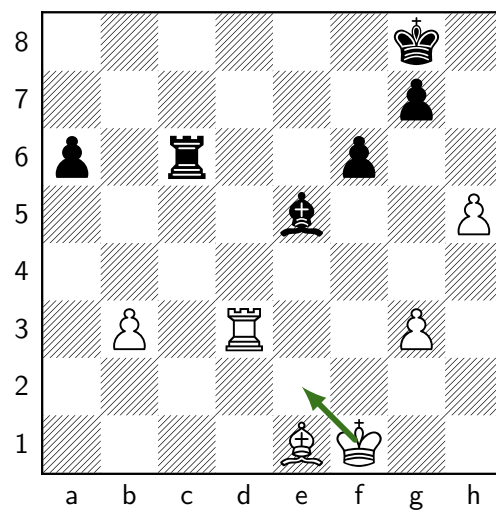


(d)

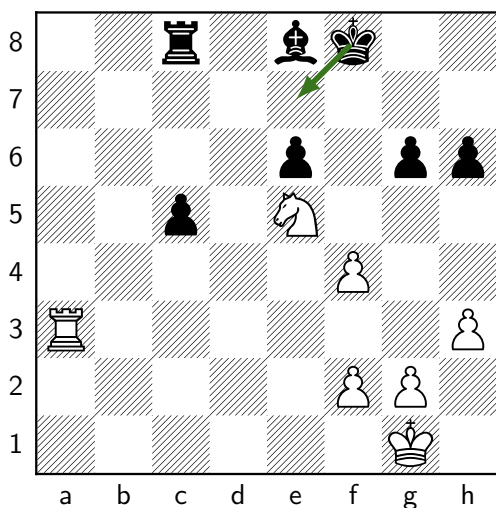
Figure 24: **Move 30, Black: Kh8 (h7h8)**. (a) Our position; Melanie plays Kh8 (h7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1733` (ply 71) [structural distance  $d=42$ , QAP obj=24]: the master played Kf7; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our nc6 as their Rb5, our ral as their Ba3; transferred move h7h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1492` (ply 64) [structural distance  $d=42$ , QAP obj=22]: the master played Kh2; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our Bf8 as their Rg3, our bc1 as their nf4; transferred move h7h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1592` (ply 70) [structural distance  $d=41$ , QAP obj=22]: the master played Kb3; under  $\mu$  our Kh2 plays the role of that game’s Ka2; cross-type roles: our pg5 as their rd1, our Ra8 as their Pf3; transferred move h7h8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh8 (summed QAP objective 98, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



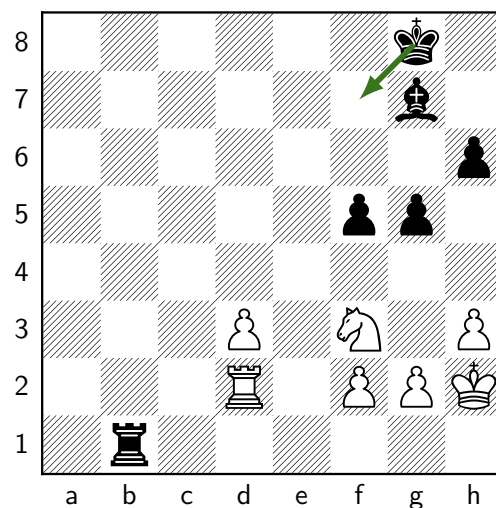
(a)



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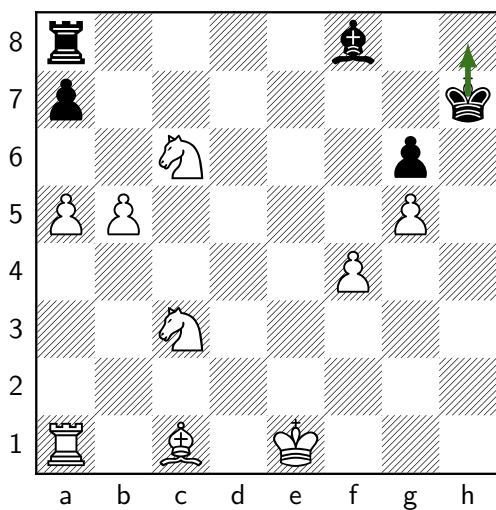


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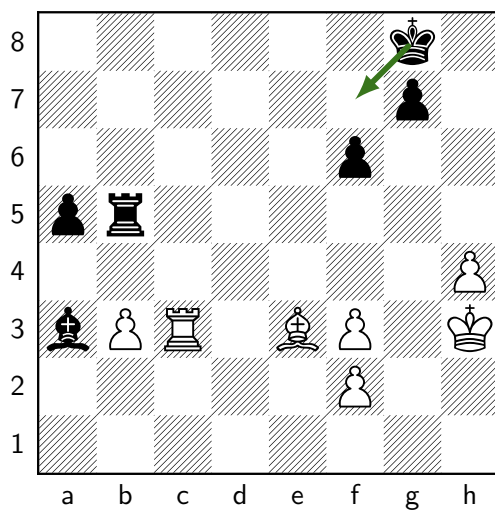


(d)

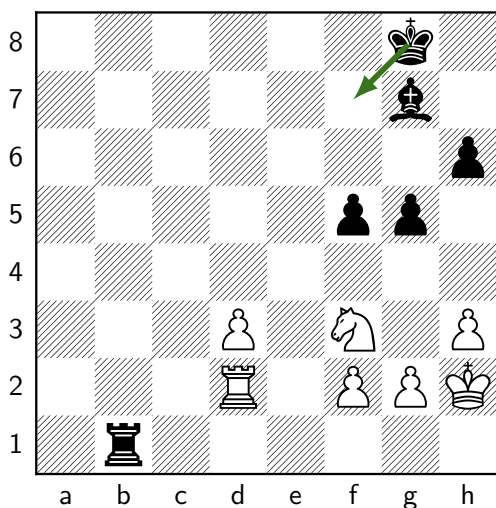
Figure 25: **Move 31, Black: Kh7.** (a) Our position; Melanie plays Kh7 (**h8h7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1667` (ply 74) [structural distance  $d=45$ , QAP obj=24]: the master played Ke2; under  $\mu$  our Kh1 plays the role of that game’s Kf1; cross-type roles: our nc6 as their be5, our nc3 as their Ph5; transferred move h8h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1367` (ply 57) [structural distance  $d=46$ , QAP obj=24]: the master played Ke7; under  $\mu$  our Kh1 plays the role of that game’s Kf1; cross-type roles: our pf4 as their Rc8, our Ra8 as their Be8; transferred move h8h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1546` (ply 65) [structural distance  $d=44$ , QAP obj=21]: the master played Kf7; under  $\mu$  our Kh1 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Ph6, our pg5 as their rd2; transferred move h8h7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh7 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



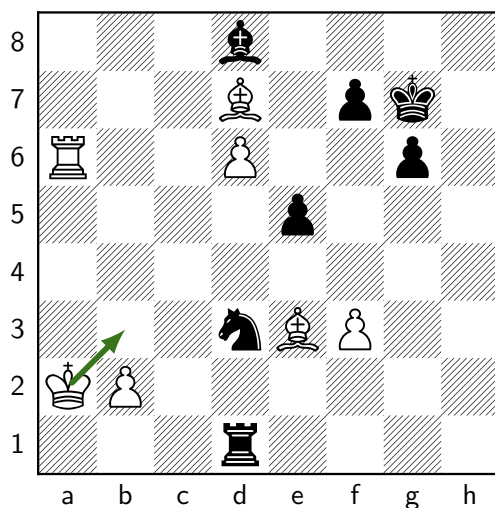
(a)



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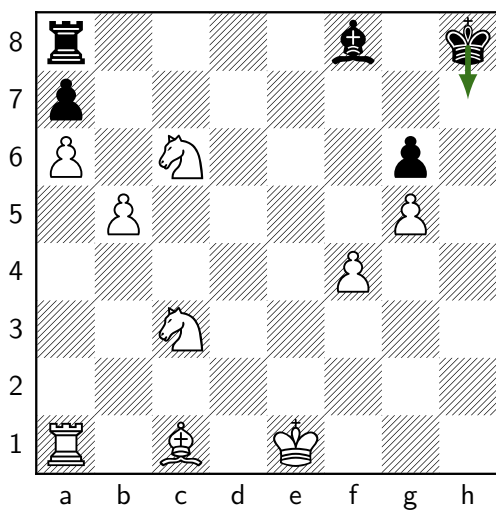


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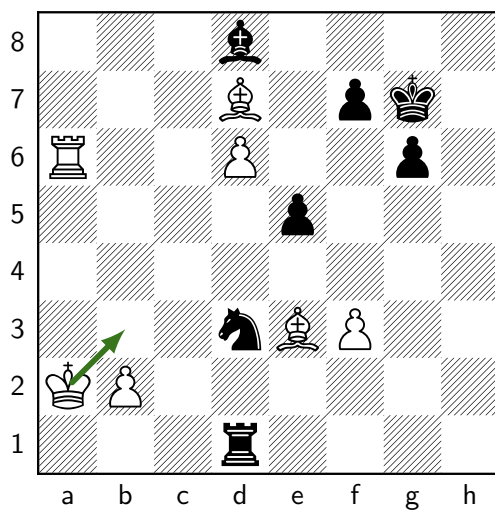


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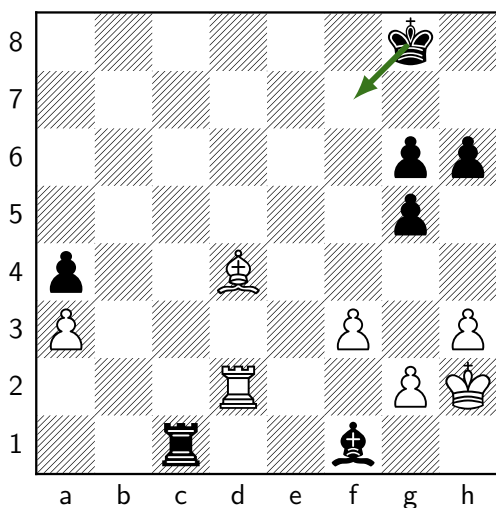
Figure 26: **Move 32, Black: Kh8**. (a) Our position; Melanie plays Kh8 (h7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1733` (ply 71) [structural distance  $d=42$ , QAP obj=26]: the master played Kf7; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our nc6 as their Rb5, our ral as their Ba3; transferred move h7h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1546` (ply 65) [structural distance  $d=39$ , QAP obj=21]: the master played Kf7; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our pg5 as their Bg7, our Bf8 as their Ph6; transferred move h7h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1592` (ply 70) [structural distance  $d=41$ , QAP obj=21]: the master played Kb3; under  $\mu$  our Kh2 plays the role of that game’s Ka2; cross-type roles: our pb5 as their rd1, our Ra8 as their Be3; transferred move h7h8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh8 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



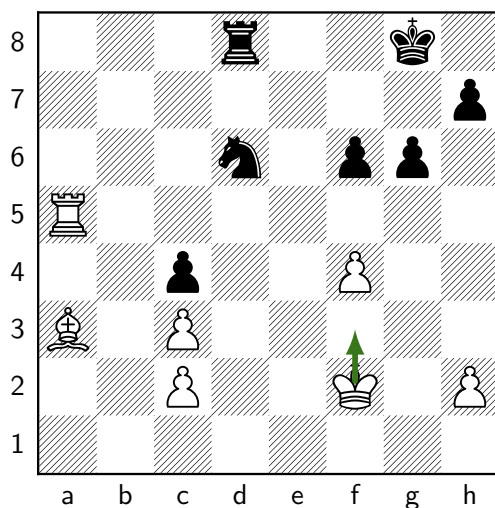
(a)



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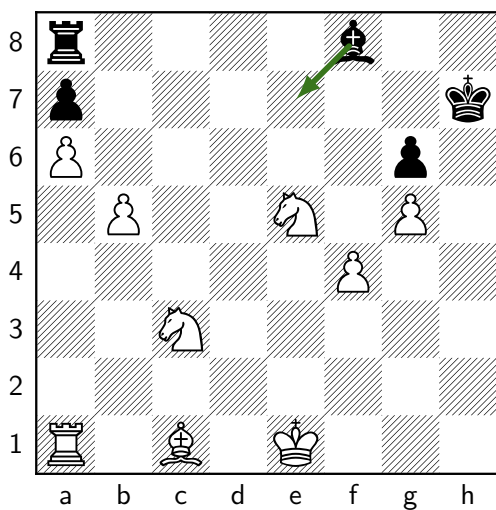


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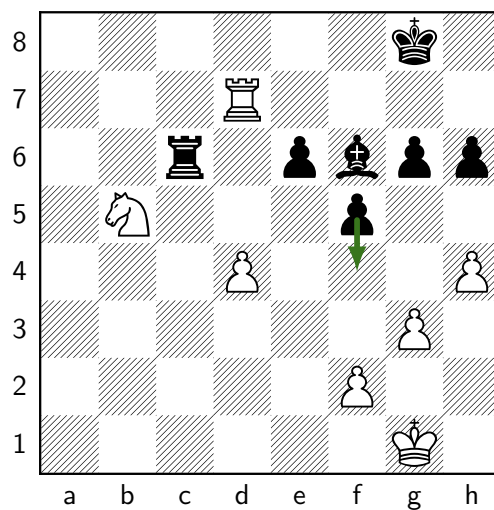


(d)

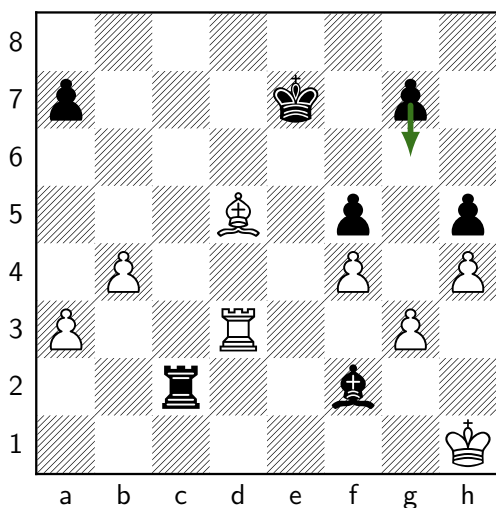
Figure 27: **Move 33, Black: Kh7**. (a) Our position; Melanie plays Kh7 (**h8h7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1592` (ply 70) [structural distance  $d=42$ , QAP obj=21]: the master played Kb3; under  $\mu$  our Kh1 plays the role of that game’s Ka2; cross-type roles: our nc3 as their rd1, our Ra8 as their Be3; transferred move h8h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#721` (ply 75) [structural distance  $d=34$ , QAP obj=13]: the master played Kf7; under  $\mu$  our Kh1 plays the role of that game’s Kg1; cross-type roles: our nc3 as their Ph6, our pf4 as their bd4; transferred move h8h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#554` (ply 70) [structural distance  $d=39$ , QAP obj=12]: the master played Kf3; under  $\mu$  our Kh1 plays the role of that game’s Kf2; cross-type roles: our bc1 as their Pc2, our Bf8 as their Ph2; transferred move h8h7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh7 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



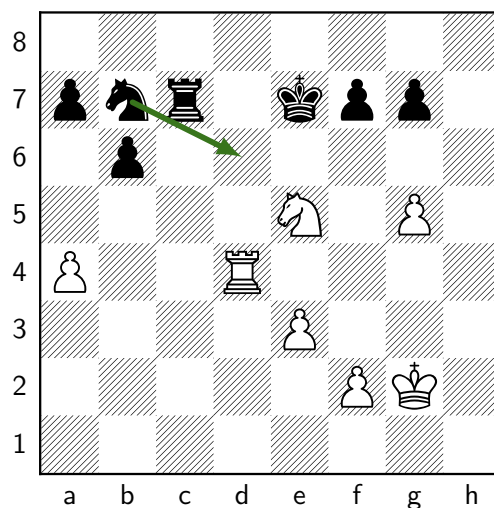
(a)



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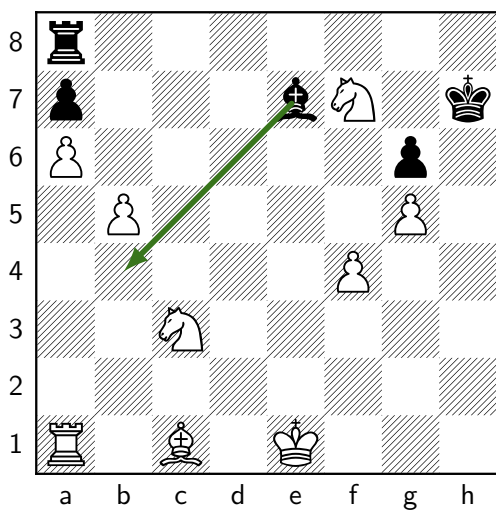


(c)

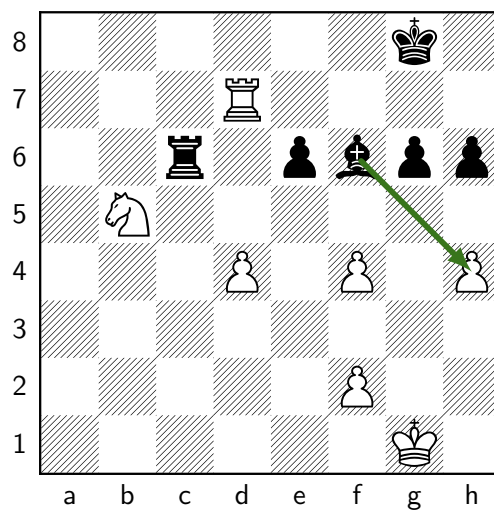


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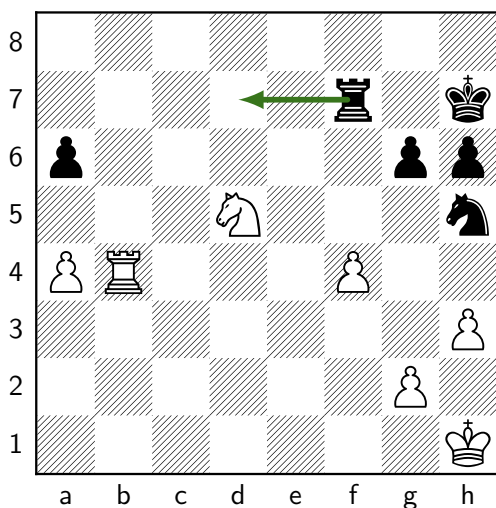
Figure 28: **Move 34, Black: Be7.** (a) Our position; Melanie plays **Be7** (**f8e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2995` (ply 75) [structural distance  $d=41$ , QAP obj=34]: the master played **f4**; under  $\mu$  our *Bf1* plays the role of that game’s *Pf4* (a **cross-type** correspondence); cross-type roles: our *pb5* as their *rd7*, our *Pa7* as their *Rc6*; transferred move **f8e7** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2184` (ply 65) [structural distance  $d=31$ , QAP obj=11]: the master played **g6**; under  $\mu$  our *Bf1* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Pa7*, our *Pa7* as their *bd5*; transferred move **f8e7** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#885` (ply 69) [structural distance  $d=40$ , QAP obj=22]: the master played **Nd6**; under  $\mu$  our *Ra1* plays the role of that game’s *Nb2* (a **cross-type** correspondence); cross-type roles: our *nc3* as their *Rc7*, our *Bf8* as their *pe3*; transferred move **a8c8**. Synthesis: 3 retrieved analogs transferred to **Be7** (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



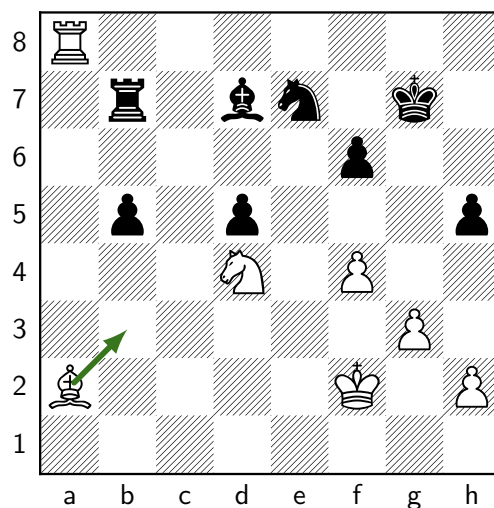
(a)



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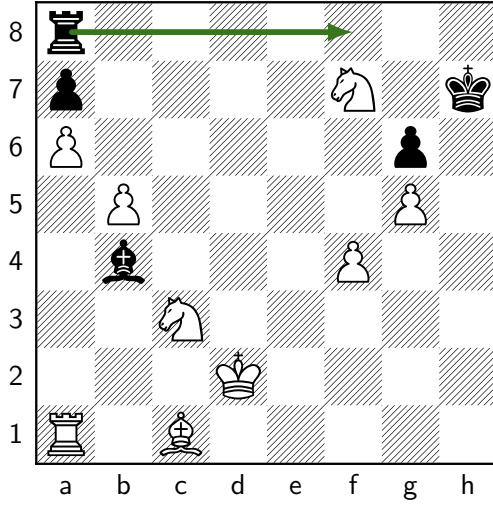


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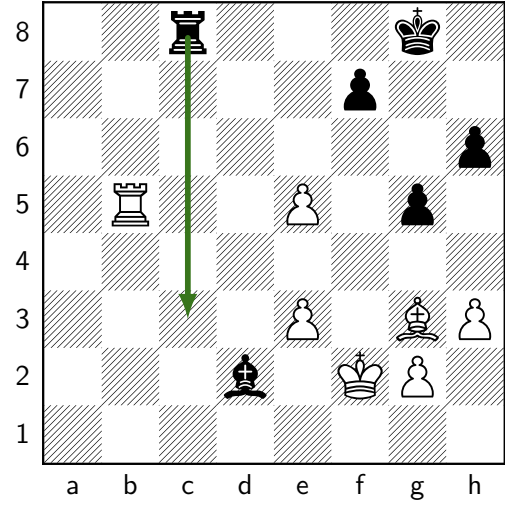


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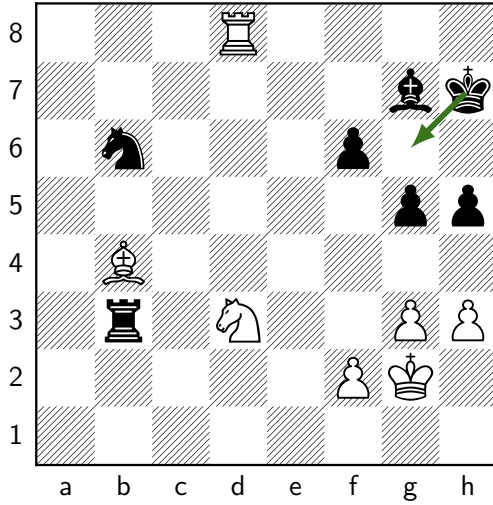
Figure 29: **Move 35, Black: Bb4.** (a) Our position; Melanie plays Bb4 (e7b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2995` (ply 77) [structural distance  $d=39$ , QAP obj=38]: the master played Bxh4; under  $\mu$  our *Be2* plays the role of that game’s *Bf3*; cross-type roles: our *nf7* as their *rd7*, our *ra1* as their *nb5*; transferred move e7b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#996` (ply 73) [structural distance  $d=42$ , QAP obj=31]: the master played Rd7; under  $\mu$  our *Be2* plays the role of that game’s *Rf2* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Nh5*, our *pa6* as their *rb4*; transferred move e7b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1750` (ply 76) [structural distance  $d=42$ , QAP obj=27]: the master played Bb3; under  $\mu$  our *Be2* plays the role of that game’s *Ba2*; cross-type roles: our *Ra8* as their *Ph2*, our *Be7* as their *Nd4*; transferred move e7b4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bb4 (summed QAP objective 178, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



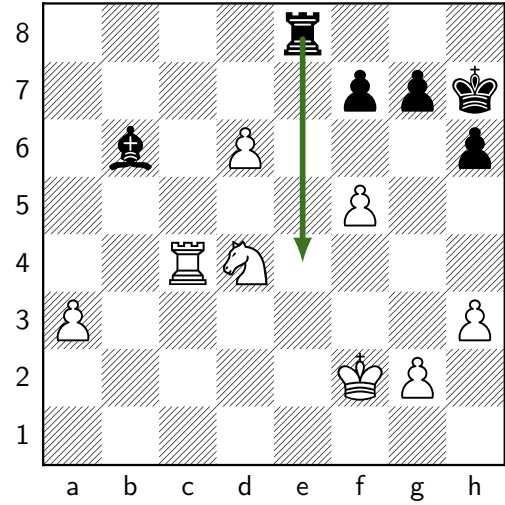
(a)



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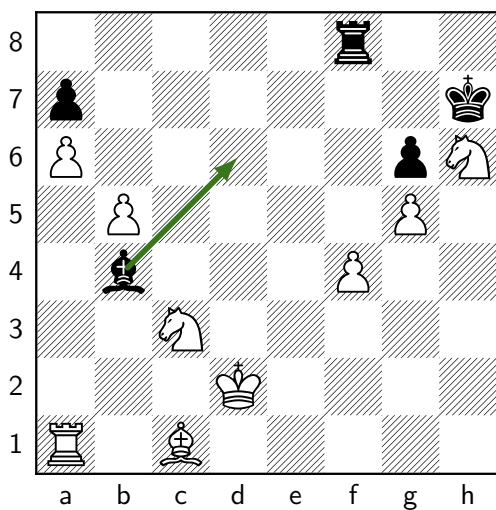


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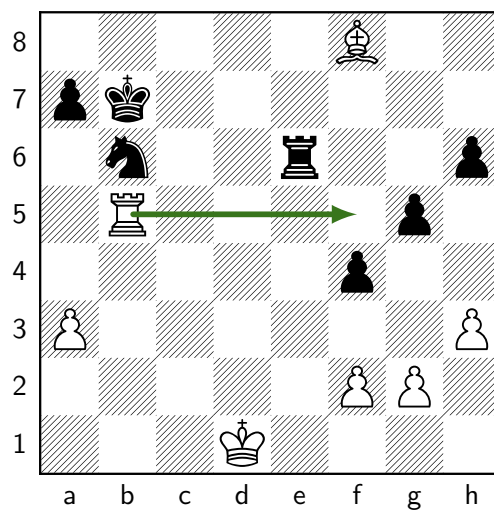


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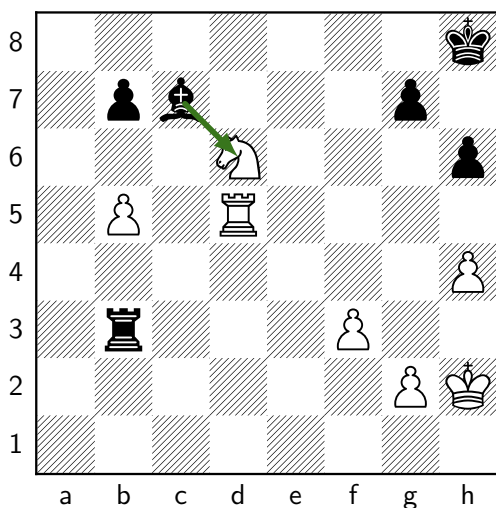
Figure 30: **Move 36, Black: Rf8.** (a) Our position; Melanie plays Rf8 (a8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3139` (ply 67) [structural distance  $d=43$ , QAP obj=12]: the master played Rc3; under  $\mu$  our Ra1 plays the role of that game’s Rc1; cross-type roles: our nf7 as their Ph6, our nc3 as their pe3; transferred move a8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1770` (ply 79) [structural distance  $d=41$ , QAP obj=26]: the master played Kg6; under  $\mu$  our Kh2 plays the role of that game’s Kh2; cross-type roles: our Pg6 as their rd8, our pg5 as their Bg7; transferred move h7g7. (d) Retrieved from `ccrl_top_p0000#3034` (ply 83) [structural distance  $d=40$ , QAP obj=23]: the master played Re4; under  $\mu$  our Ra1 plays the role of that game’s Re1; cross-type roles: our ra1 as their pd6, our bc1 as their rc4; transferred move a8d8. Synthesis: 1 retrieved analog transferred to Rf8 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



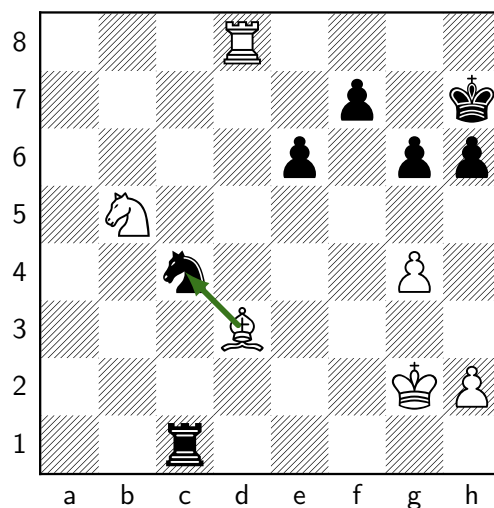
(a)



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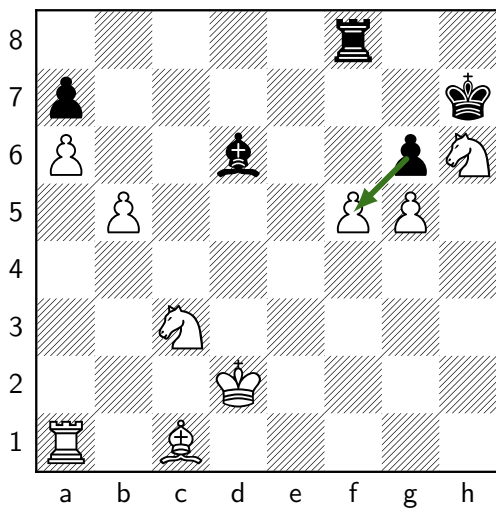
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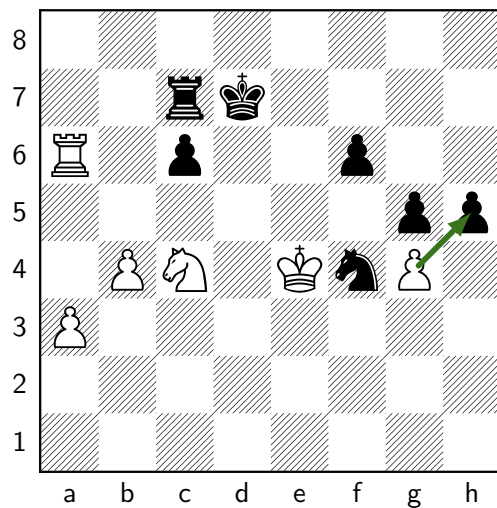
(d)

Figure 31: **Move 37, Black: Bd6**. (a) Our position; Melanie plays Bd6 (b4d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#589` (ply 78) [structural distance  $d=39$ , QAP obj=42]: the master played Rf5; under  $\mu$  our Bb5 plays the role of that game’s Rb5 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pa3, our nh6 as their Ph3; transferred move b4d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2439` (ply 81) [structural distance  $d=48$ , QAP obj=41]: the master played Bxd6+; under  $\mu$  our Bb5 plays the role of that game’s Bc2; cross-type roles: our Rf8 as their Pb7, our nh6 as their Pg7; transferred move b4d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2144` (ply 76) [structural distance  $d=48$ , QAP obj=21]: the master played Bxc4; under  $\mu$  our Bb5 plays the role of that game’s Bd3; cross-type roles: our nh6 as their rc1, our Rf8 as their Ph2; transferred move b4d6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bd6 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

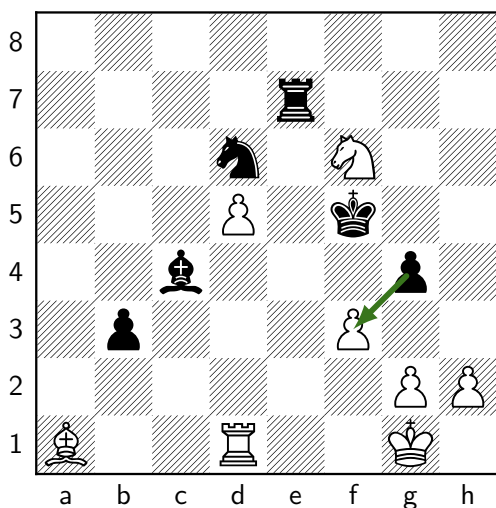




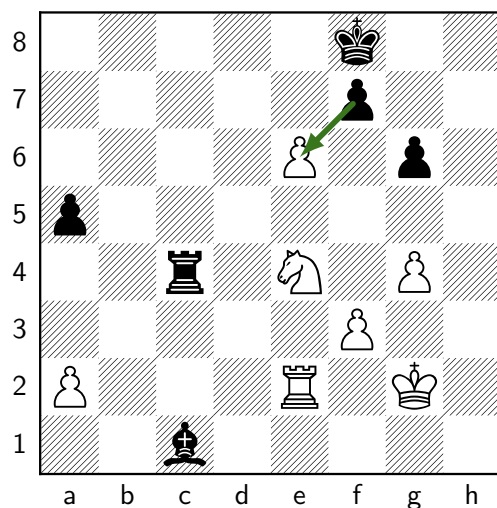
(a)



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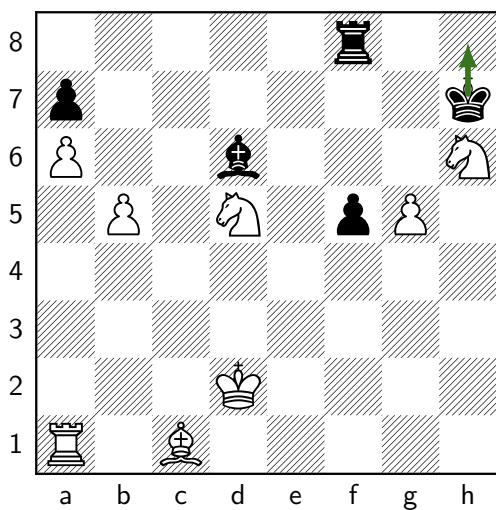


(c)

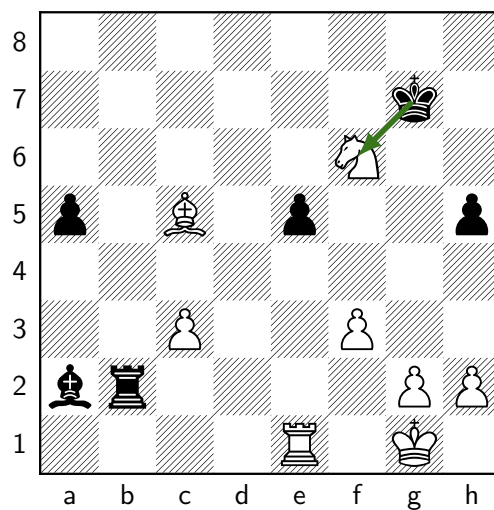


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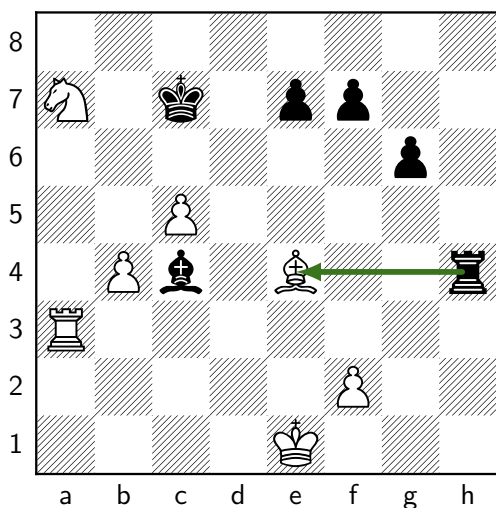
Figure 32: **Move 38, Black: gxf5.** (a) Our position; Melanie plays **gxf5** (**g6f5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#675` (ply 86) [structural distance  $d=48$ , QAP obj=39]: the master played **gxh5**; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Pg4$ ; cross-type roles: our  $Rf8$  as their  $Nc4$ , our  $Bd6$  as their  $Ra6$ ; transferred move **g6f5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3400` (ply 79) [structural distance  $d=37$ , QAP obj=34]: the master played **gxf3**; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Pg5$ ; cross-type roles: our  $ra1$  as their  $ba1$ , our  $bc1$  as their  $rd1$ ; transferred move **g6f5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2954` (ply 81) [structural distance  $d=49$ , QAP obj=22]: the master played **fxe6**; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Pf2$ ; cross-type roles: our  $Bd6$  as their  $Pa5$ , our  $Kh7$  as their  $Rc4$ ; transferred move **g6f5** (**endorses this move**). Synthesis: 16 retrieved analogs transferred to **gxf5** (summed QAP objective 234, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



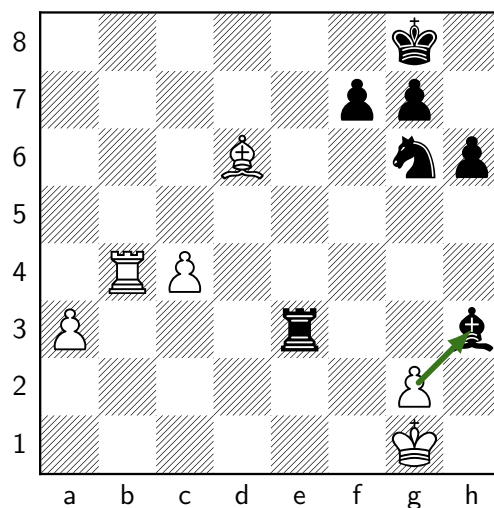
(a)



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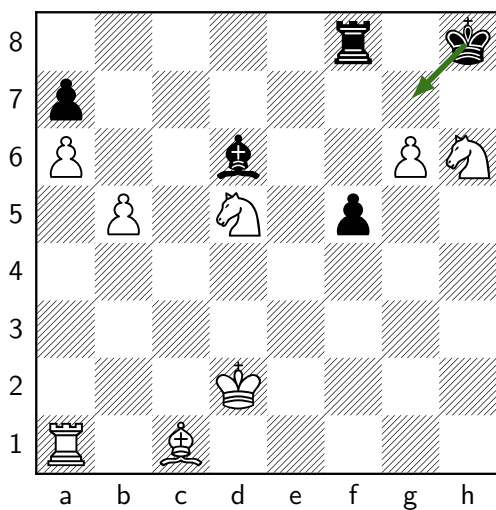


(c)

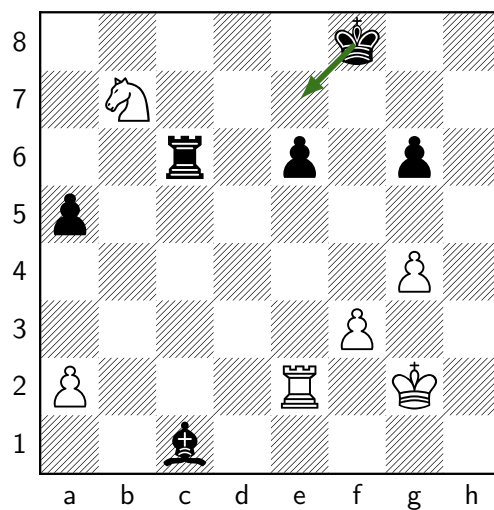


(d)

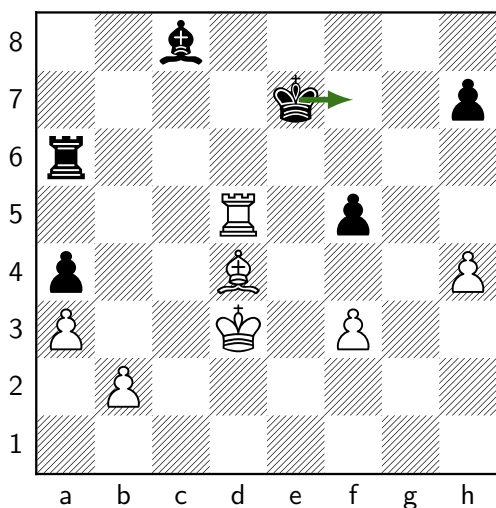
Figure 33: **Move 39, Black: Kh8 (h7h8)**. (a) Our position; Melanie plays Kh8 (h7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#33` (ply 61) [structural distance  $d=44$ , QAP obj=37]: the master played Kxf6; under  $\mu$  our Kh2 plays the role of that game’s Kg2; cross-type roles: our nd5 as their Pa5, our Rf8 as their pf3; transferred move h7h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#123` (ply 61) [structural distance  $d=44$ , QAP obj=19]: the master played Rxe4+; under  $\mu$  our Kh2 plays the role of that game’s Rh5 (a **cross-type** correspondence); cross-type roles: our pg5 as their na7, our Rf8 as their Pf7; transferred move h7h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3591` (ply 76) [structural distance  $d=42$ , QAP obj=15]: the master played gxh3; under  $\mu$  our Kh2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pa3, our pg5 as their re3; transferred move h7h8 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kh8 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



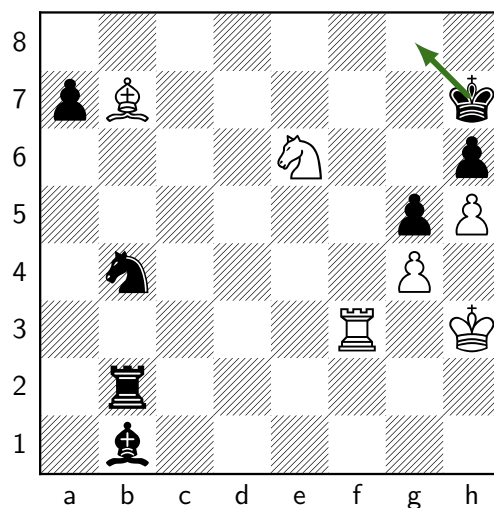
(a)



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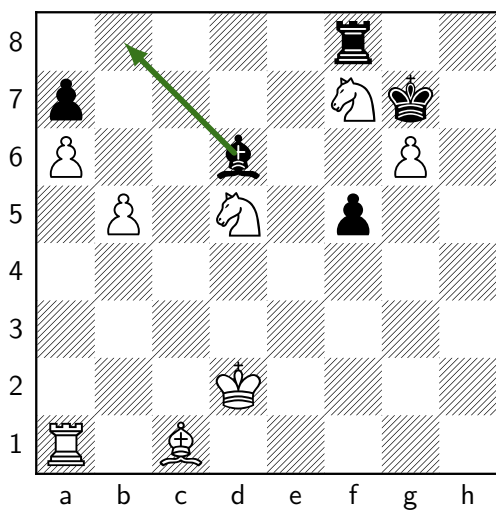


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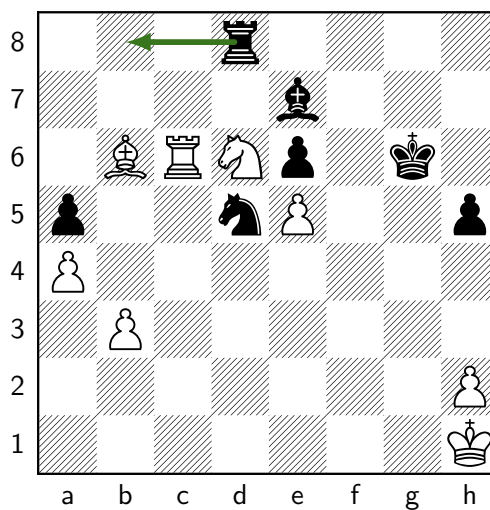


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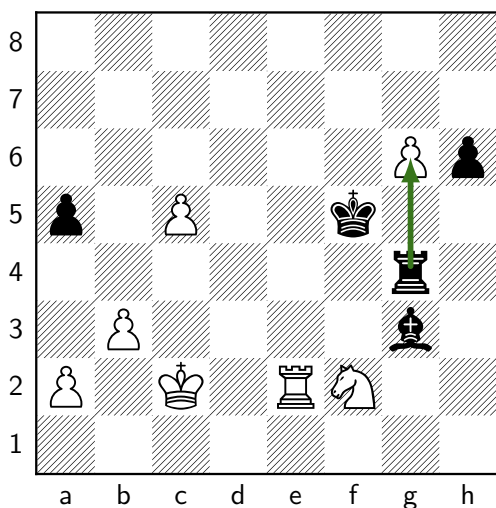
Figure 34: **Move 40, Black: Kg7**. (a) Our position; Melanie plays Kg7 (**h8g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2954` (ply 85) [structural distance  $d=39$ , QAP obj=21]: the master played Ke7; under  $\mu$  our Kh1 plays the role of that game’s Kf1; cross-type roles: our Bd6 as their Pg6, our nd5 as their pg4; transferred move h8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2356` (ply 79) [structural distance  $d=33$ , QAP obj=21]: the master played Kf7; under  $\mu$  our Kh1 plays the role of that game’s Ke2; cross-type roles: our Rf8 as their Bc8, our nd5 as their Ra6; transferred move h8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3717` (ply 85) [structural distance  $d=39$ , QAP obj=21]: the master played Kg8; under  $\mu$  our Kh1 plays the role of that game’s Kh2; cross-type roles: our pb5 as their bb7, our Rf8 as their Ph6; transferred move h8g7 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Kg7 (summed QAP objective 198, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



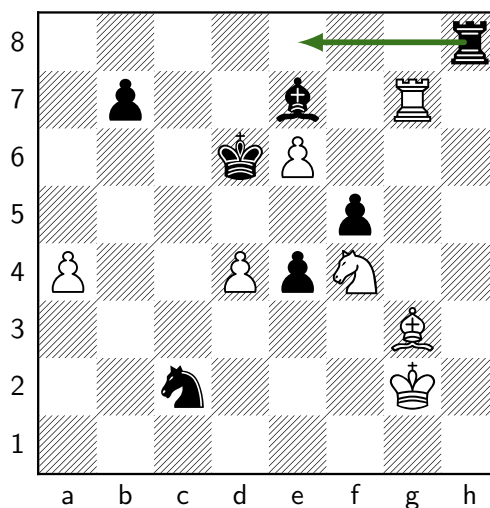
(a)



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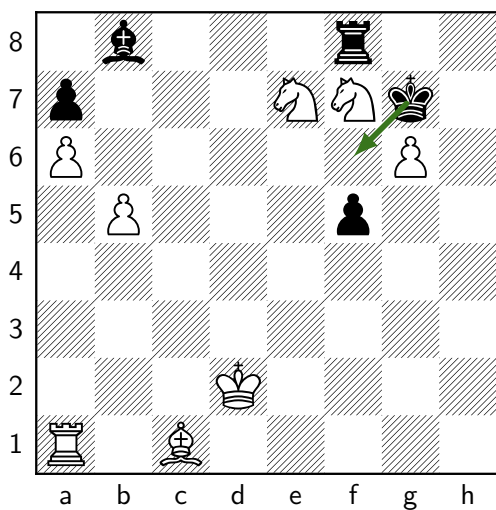


(c)

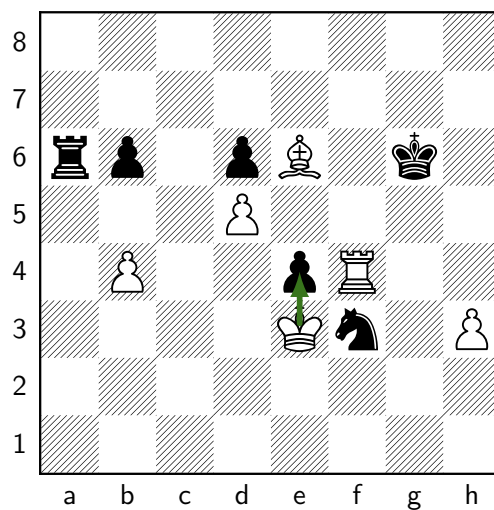


(d)

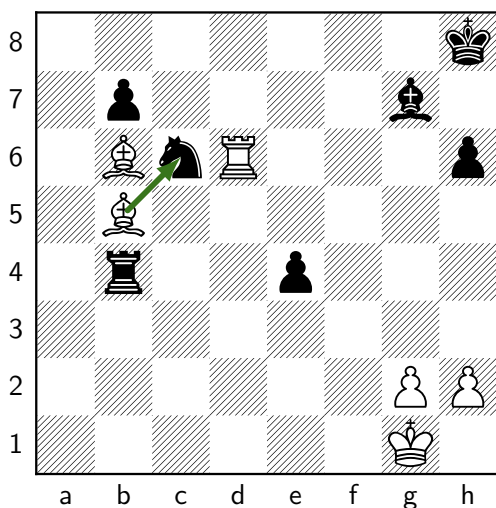
Figure 35: **Move 41, Black: Bb8.** (a) Our position; Melanie plays Bb8 (d6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1692` (ply 91) [structural distance  $d=53$ , QAP obj=47]: the master played Rb8; under  $\mu$  our Bd3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Kg7 as their Be7, our nf7 as their bb6; transferred move d6b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2372` (ply 91) [structural distance  $d=46$ , QAP obj=42]: the master played Rxg6; under  $\mu$  our Bd3 plays the role of that game’s Rg5 (a **cross-type** correspondence); cross-type roles: our nd5 as their pb3, our Rf8 as their Bg3; transferred move d6b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3450` (ply 83) [structural distance  $d=52$ , QAP obj=17]: the master played Re8; under  $\mu$  our Bd3 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our nf7 as their Be7, our ra1 as their pa4; transferred move d6b8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bb8 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



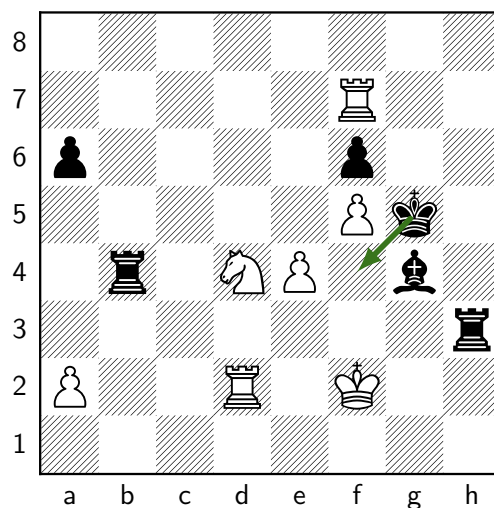
(a)



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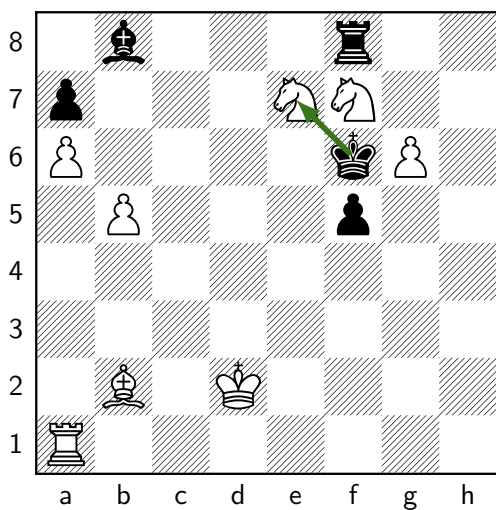


(c)

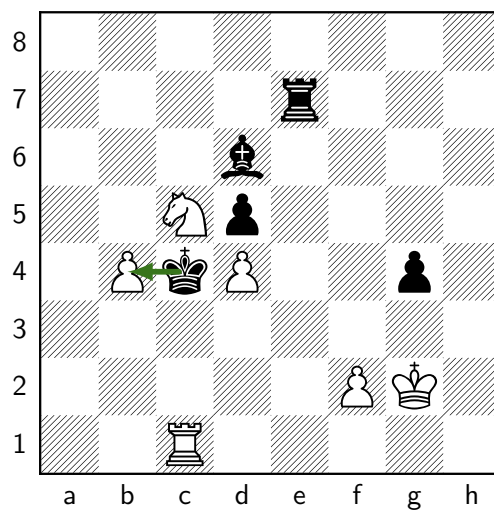


(d)

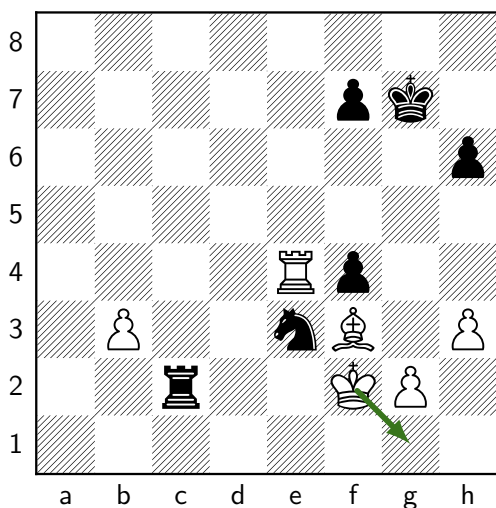
Figure 36: **Move 42, Black: Kf6.** (a) Our position; Melanie plays Kf6 (g7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#453` (ply 94) [structural distance  $d=45$ , QAP obj=70]: the master played Kxe4; under  $\mu$  our Kg2 plays the role of that game’s Ke3; cross-type roles: our Bb8 as their Pd5, our ne7 as their Be6; transferred move g7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2581` (ply 84) [structural distance  $d=45$ , QAP obj=35]: the master played Bxc6; under  $\mu$  our Kg2 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Pf5 as their rb4, our ne7 as their ph6; transferred move g7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1965` (ply 81) [structural distance  $d=40$ , QAP obj=26]: the master played Kf4; under  $\mu$  our Kg2 plays the role of that game’s Kg4; cross-type roles: our Kg7 as their rf7, our nf7 as their pf5; transferred move g7f6 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kf6 (summed QAP objective 275, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



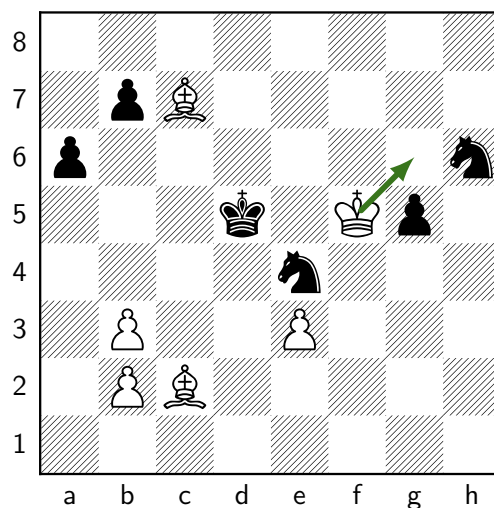
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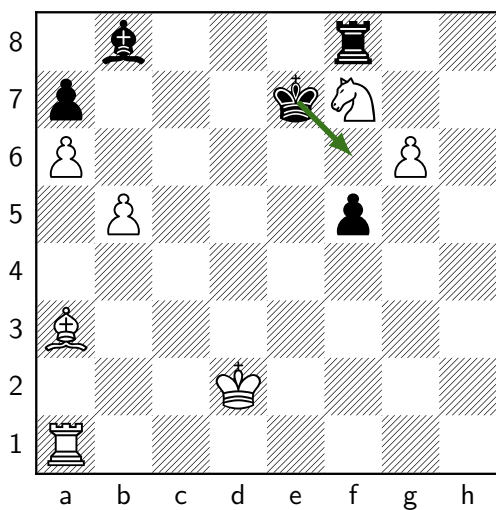


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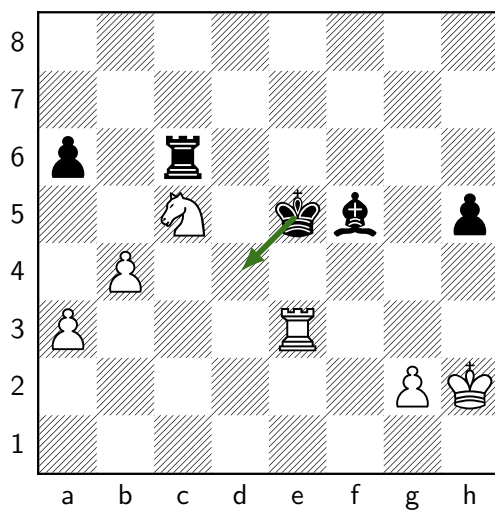


(d)

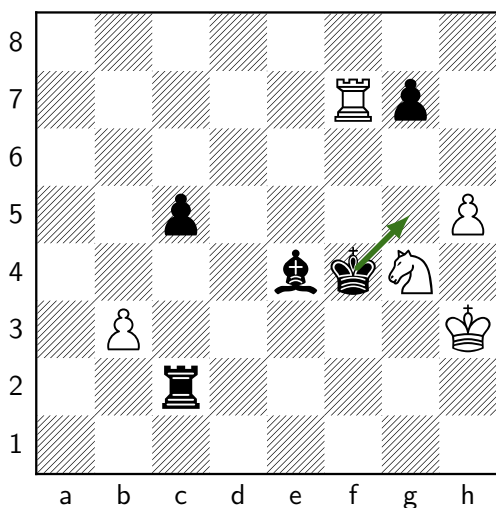
Figure 37: **Move 43, Black: Kxe7.** (a) Our position; Melanie plays Kxe7 (f6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#843 (ply 93) [structural distance  $d=50$ , QAP obj=76]: the master played Kxb4; under  $\mu$  our Kf3 plays the role of that game’s Kc5; cross-type roles: our Bb8 as their Re7, our Rf8 as their Bd6; transferred move f6e7 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1100 (ply 88) [structural distance  $d=44$ , QAP obj=66]: the master played Kg1; under  $\mu$  our Kf3 plays the role of that game’s Kf2; cross-type roles: our bb2 as their rc2, our ne7 as their Pg2; transferred move f6e7 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#2120 (ply 94) [structural distance  $d=44$ , QAP obj=62]: the master played Kg6; under  $\mu$  our Kf3 plays the role of that game’s Kf5; cross-type roles: our Rf8 as their Pb2, our pg6 as their Bc2; transferred move f6e7 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kxe7 (summed QAP objective 443, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



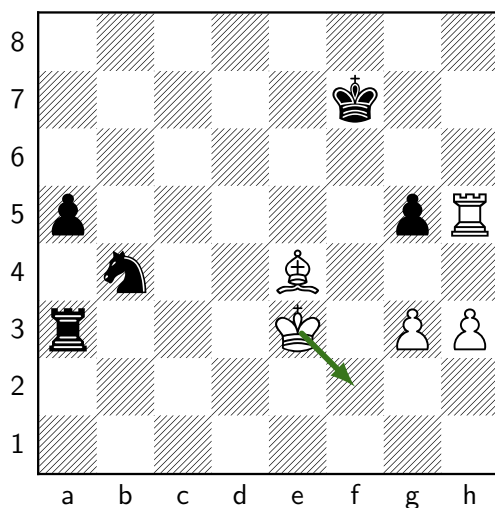
(a)



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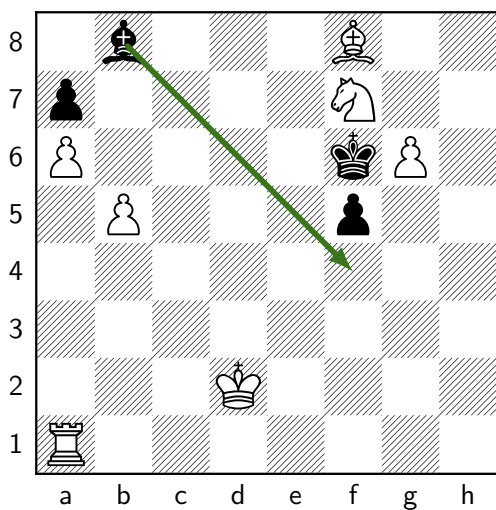


(c)

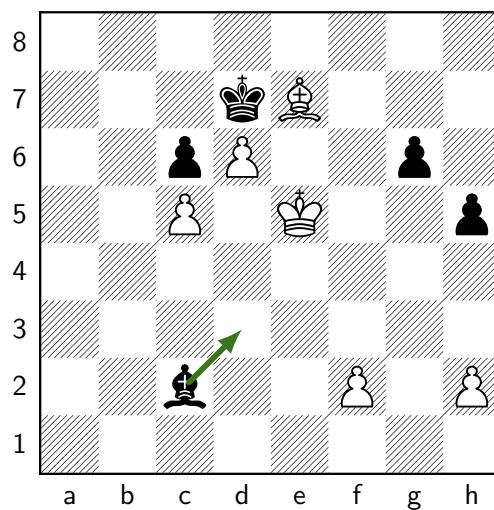


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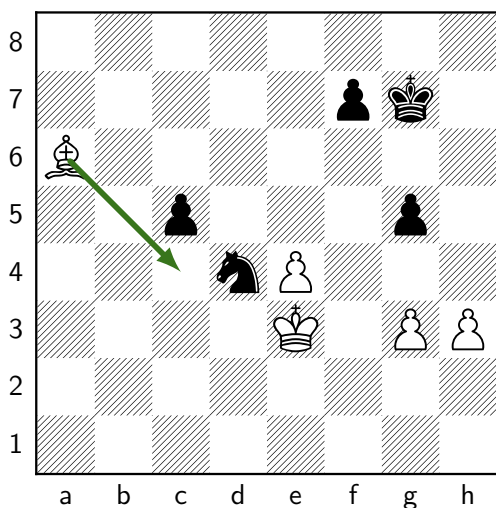
Figure 38: **Move 44, Black: Kf6 (e7f6)**. (a) Our position; Melanie plays Kf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2412` (ply 91) [structural distance  $d=27$ , QAP obj=49]: the master played Kd4; under  $\mu$  our Ke2 plays the role of that game’s Ke4; cross-type roles: our nf7 as their Pa6, our ba3 as their Rc6; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1636` (ply 95) [structural distance  $d=29$ , QAP obj=73]: the master played Kg5; under  $\mu$  our Ke2 plays the role of that game’s Kf5; cross-type roles: our ba3 as their rf7, our Bb8 as their Pg7; transferred move e7d7. (d) Retrieved from `ccrl_top_p0003#1569` (ply 74) [structural distance  $d=25$ , QAP obj=72]: the master played Kf2; under  $\mu$  our Ke2 plays the role of that game’s Ke3; cross-type roles: our ba3 as their ra3, our Rf8 as their Be4; transferred move e7d7. Synthesis: 1 retrieved analog transferred to Kf6 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



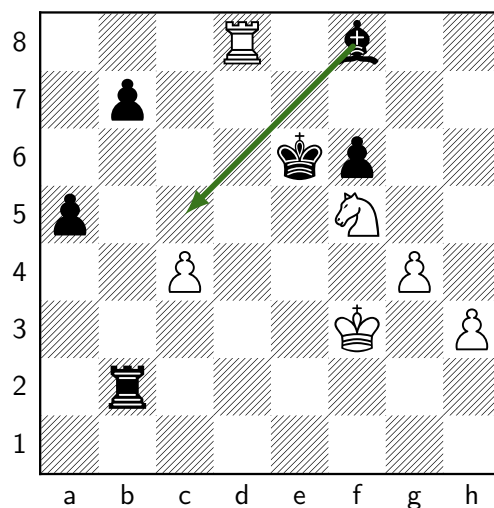
(a)



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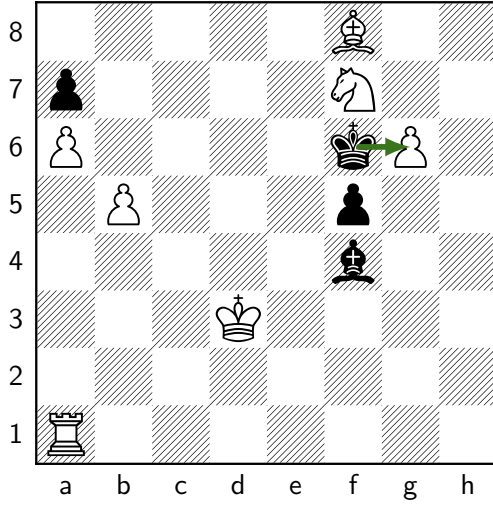
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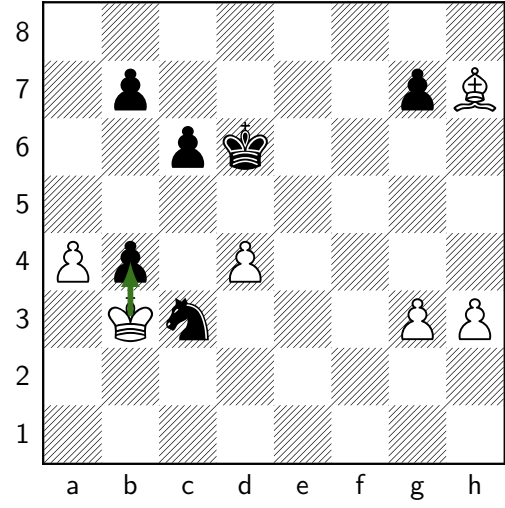
(d)

Figure 39: **Move 45, Black: Bf4+.** (a) Our position; Melanie plays Bf4+ (b8f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3995` (ply 103) [structural distance  $d=23$ , QAP obj=43]: the master played Bd3; under  $\mu$  our Bb1 plays the role of that game’s Bc7; cross-type roles: our nf7 as their be7, our bf8 as their pc5; transferred move b8f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#317` (ply 90) [structural distance  $d=18$ , QAP obj=24]: the master played Bc4; under  $\mu$  our Bb1 plays the role of that game’s Ba6; cross-type roles: our bf8 as their pf7; transferred move b8f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2760` (ply 73) [structural distance  $d=19$ , QAP obj=24]: the master played Bc5; under  $\mu$  our Bb1 plays the role of that game’s Bf1; cross-type roles: our bf8 as their rd8; transferred move b8f4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bf4+ (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

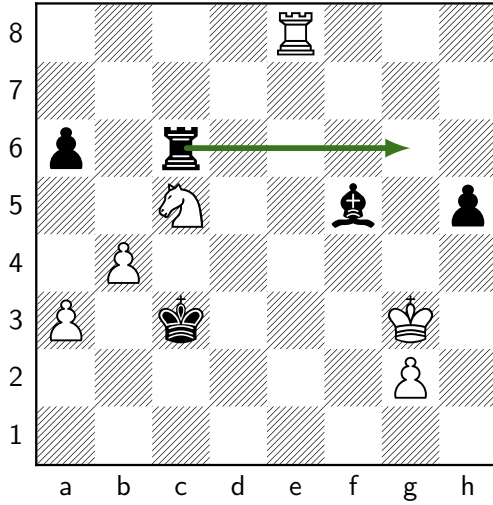




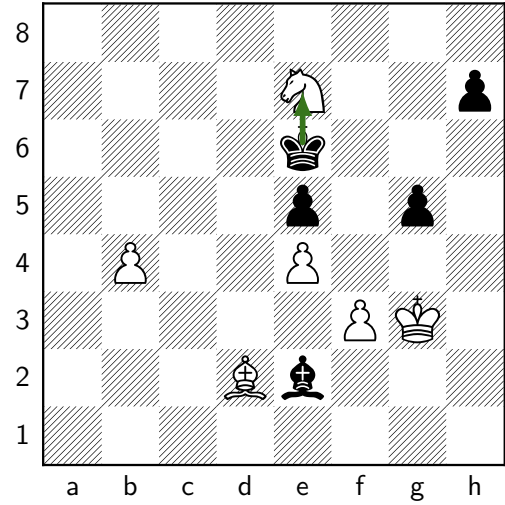
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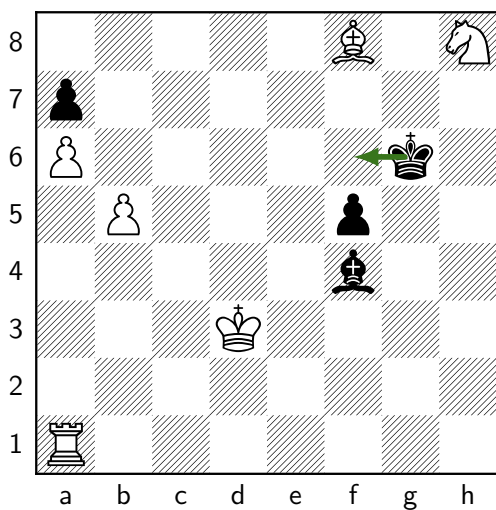


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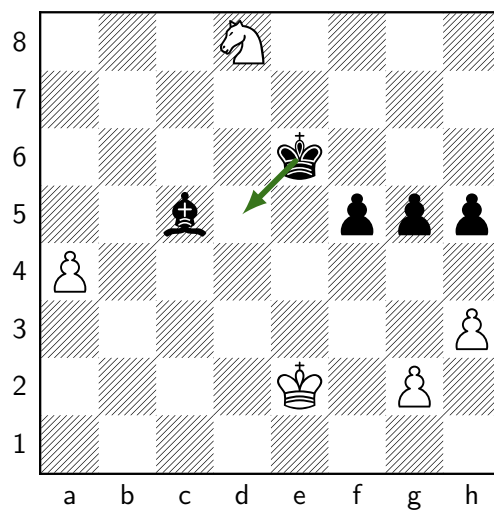


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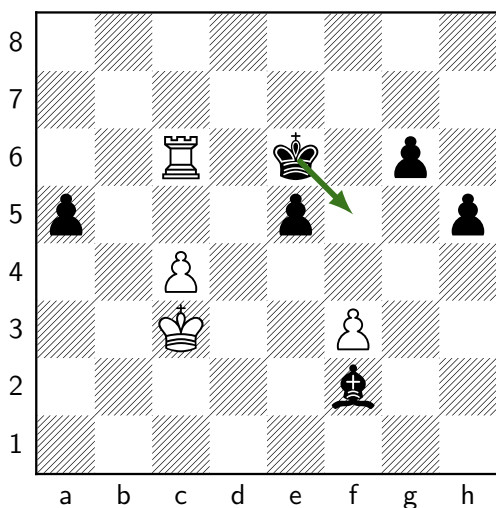
Figure 40: **Move 46, Black: Kxg6.** (a) Our position; Melanie plays Kxg6 (f6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1625` (ply 86) [structural distance  $d=23$ , QAP obj=41]: the master played Kxb4; under  $\mu$  our Kf3 plays the role of that game’s Kb3; cross-type roles: our bf8 as their Pg3, our ra1 as their pg7; transferred move f6g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2412` (ply 95) [structural distance  $d=23$ , QAP obj=24]: the master played Rg6+; under  $\mu$  our Kf3 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our pb5 as their re8, our bf8 as their pa3; transferred move f6g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2563` (ply 85) [structural distance  $d=23$ , QAP obj=23]: the master played Kxe7; under  $\mu$  our Kf3 plays the role of that game’s Ke3; cross-type roles: our Bf4 as their Ph7, our bf8 as their pe4; transferred move f6g6 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kxg6 (summed QAP objective 210, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



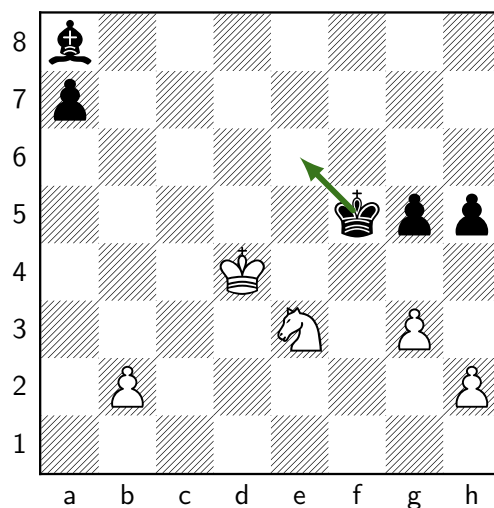
(a)



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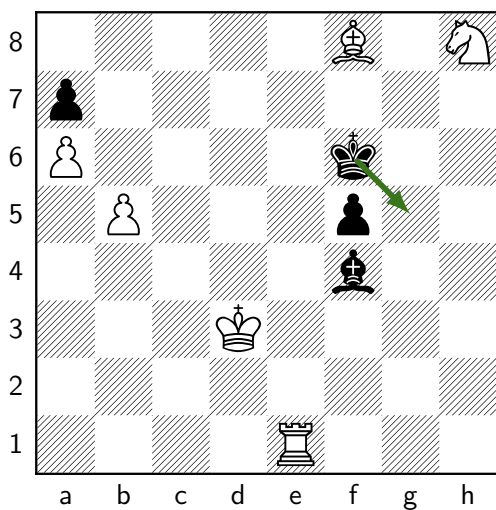


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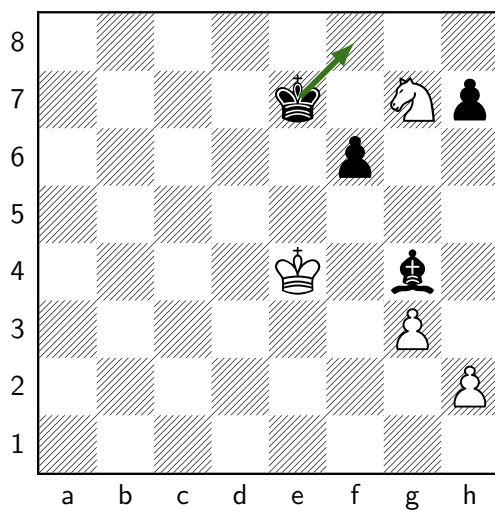


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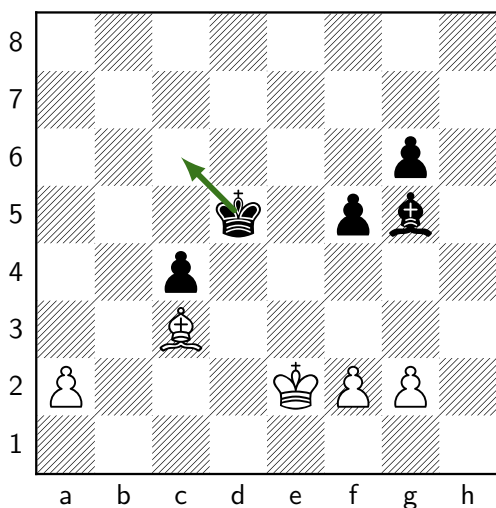
Figure 41: **Move 47, Black: Kf6.** (a) Our position; Melanie plays Kf6 (g6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#3465` (ply 95) [structural distance  $d=14$ , QAP obj=70]: the master played Kd5; under  $\mu$  our Kg3 plays the role of that game’s Ke3; cross-type roles: our bf8 as their Pg5, our ra1 as their Ph5; transferred move g6f6 (**endorses this move**). (c) Retrieved from `ccl1_top_p0002#933` (ply 91) [structural distance  $d=19$ , QAP obj=69]: the master played Kf5; under  $\mu$  our Kg3 plays the role of that game’s Ke3; cross-type roles: our nh8 as their rc6, our bf8 as their pc4; transferred move g6f6 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#2100` (ply 91) [structural distance  $d=18$ , QAP obj=69]: the master played Ke6; under  $\mu$  our Kg3 plays the role of that game’s Kf4; cross-type roles: our ra1 as their Pa7, our bf8 as their pb2; transferred move g6f6 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kf6 (summed QAP objective 764, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



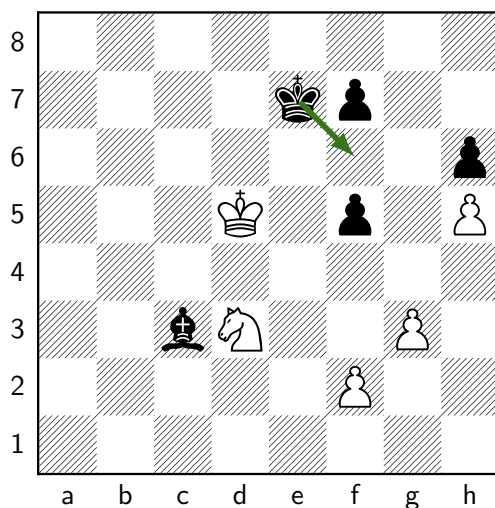
(a)



(b)

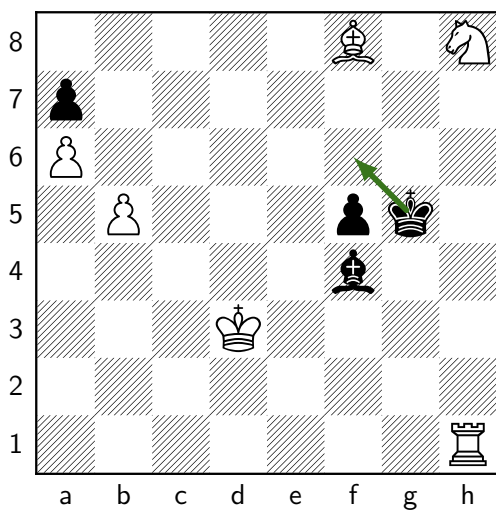


(c)

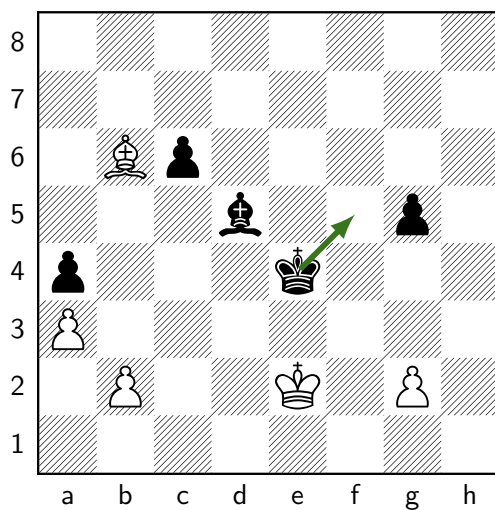


(d)

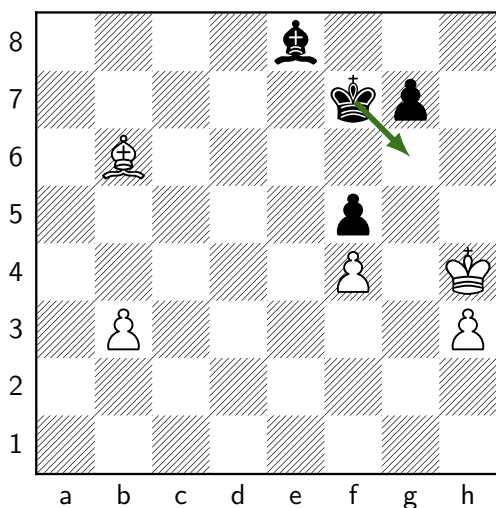
Figure 42: **Move 48, Black: Kg5 (f6g5).** (a) Our position; Melanie plays Kg5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1655` (ply 101) [structural distance  $d=15$ , QAP obj=3]: the master played Kf8; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our re1 as their ng7, our bf8 as their Ph7; transferred move f6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#972` (ply 89) [structural distance  $d=15$ , QAP obj=3]: the master played Kc6; under  $\mu$  our Kf3 plays the role of that game’s Kd4; cross-type roles: our re1 as their bc3, our nh8 as their pa2; transferred move f6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1757` (ply 101) [structural distance  $d=15$ , QAP obj=3]: the master played Kf6; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our bf8 as their Ph6, our nh8 as their ph5; transferred move f6g5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kg5 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



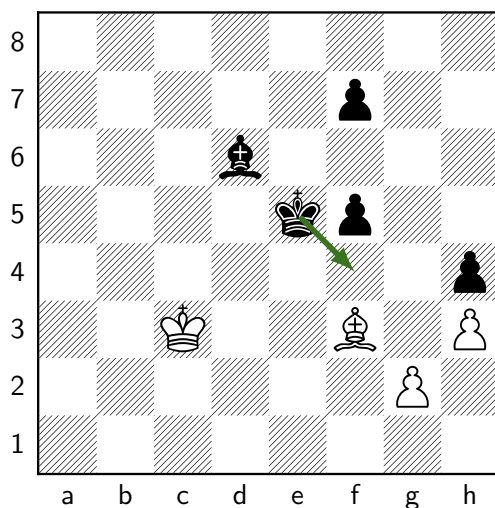
(a)



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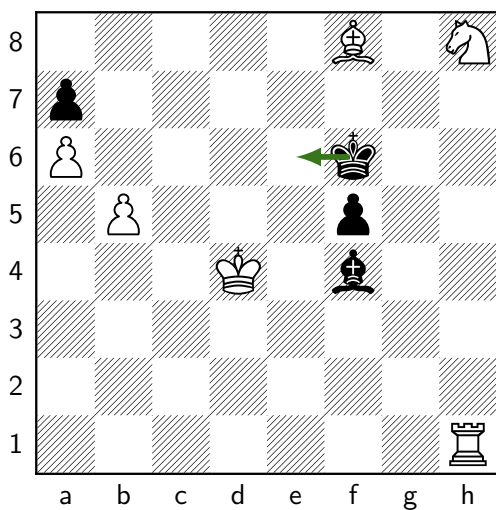


(c)

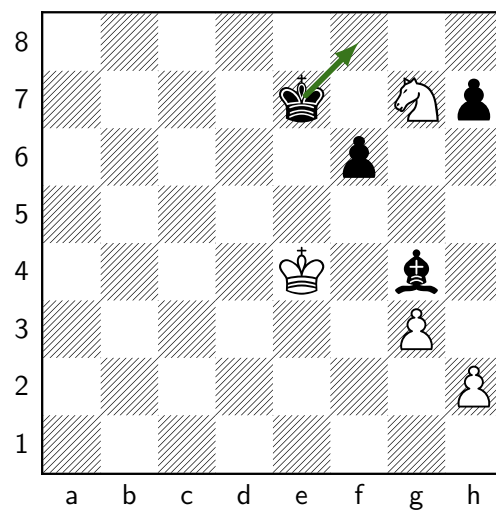


(d)

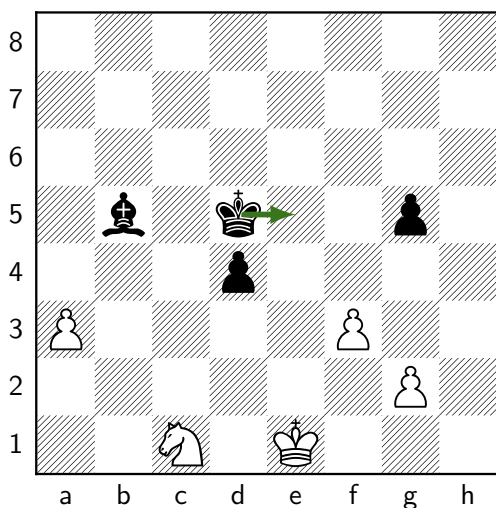
Figure 43: **Move 49, Black: Kf6**. (a) Our position; Melanie plays Kf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2281` (ply 81) [structural distance  $d=18$ , QAP obj=5]: the master played Kf5; under  $\mu$  our  $Kg4$  plays the role of that game’s  $Ke5$ ; cross-type roles: our  $rh1$  as their  $bb6$ , our  $bf8$  as their  $Pc6$ ; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1006` (ply 93) [structural distance  $d=20$ , QAP obj=4]: the master played Kg6; under  $\mu$  our  $Kg4$  plays the role of that game’s  $Kf2$ ; cross-type roles: our  $bf8$  as their  $Pg7$ , our  $nh8$  as their  $bb6$ ; transferred move g5f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1086` (ply 95) [structural distance  $d=20$ , QAP obj=4]: the master played Kf4; under  $\mu$  our  $Kg4$  plays the role of that game’s  $Ke4$ ; cross-type roles: our  $bf8$  as their  $Pf5$ , our  $rh1$  as their  $bf3$ ; transferred move g5f6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kf6 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



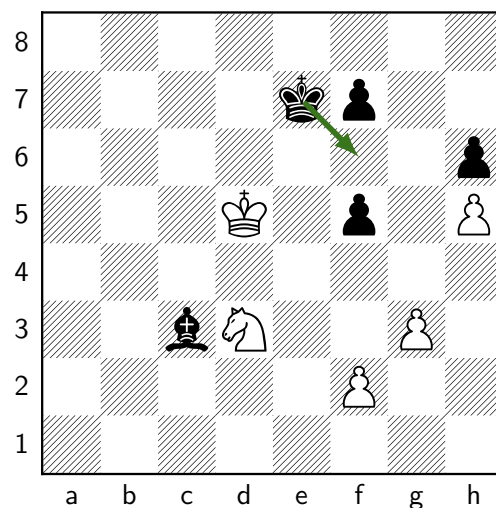
(a)



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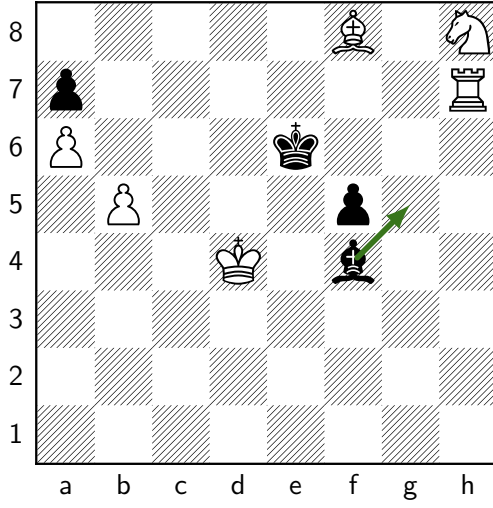


(c)

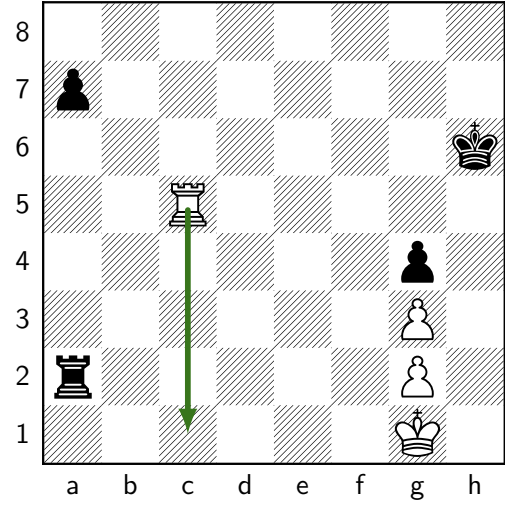


(d)

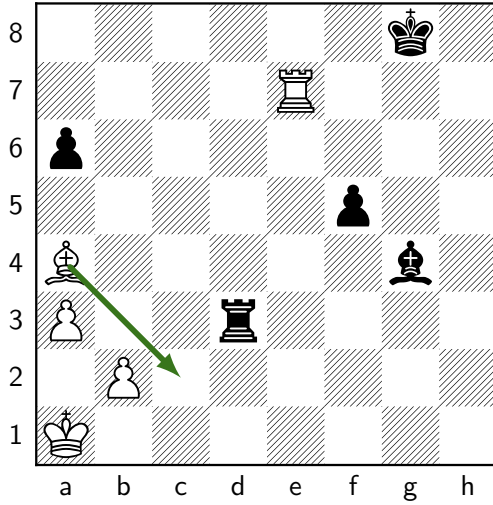
Figure 44: **Move 50, Black: Ke6**. (a) Our position; Melanie plays Ke6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1655` (ply 101) [structural distance  $d=17$ , QAP obj=3]: the master played Kf8; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our bf8 as their Ph7, our Pa7 as their Bg4; transferred move f6e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1102` (ply 115) [structural distance  $d=15$ , QAP obj=3]: the master played Ke5; under  $\mu$  our Kf3 plays the role of that game’s Kd4; cross-type roles: our bf8 as their pa3, our rh1 as their nc1; transferred move f6e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1757` (ply 101) [structural distance  $d=18$ , QAP obj=3]: the master played Kf6; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our nh8 as their Ph6, our bf8 as their Pf5; transferred move f6e6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ke6 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



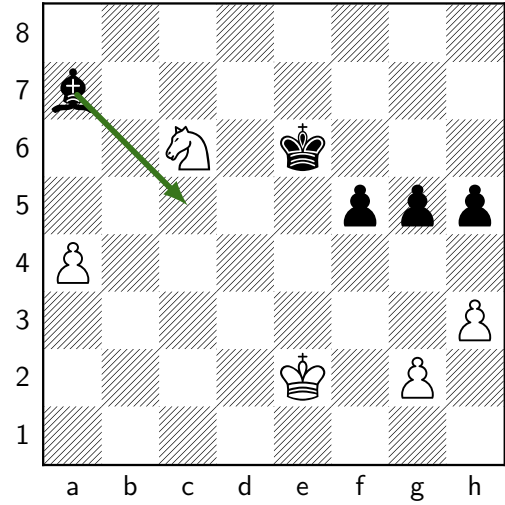
(a)



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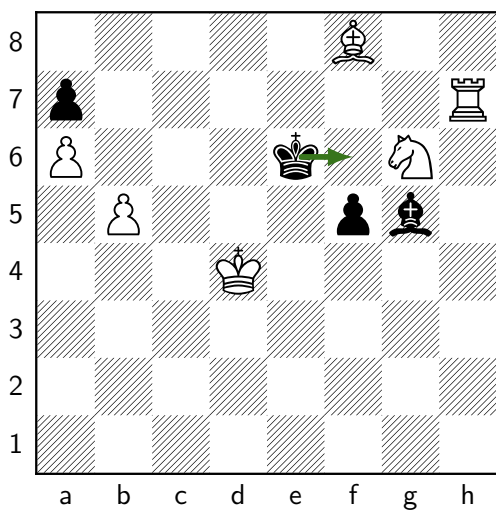


(c)

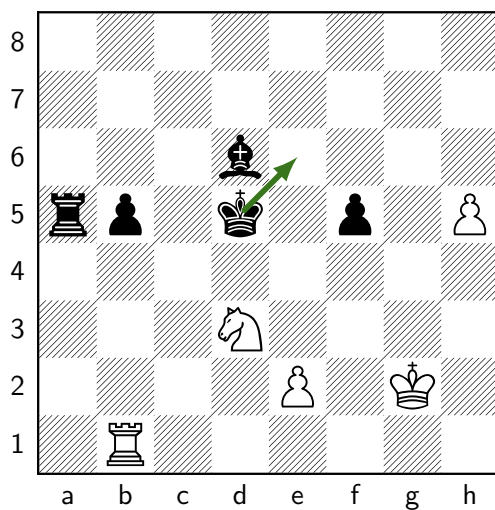


(d)

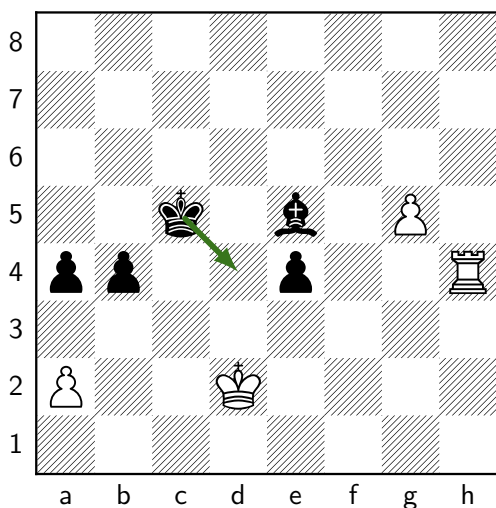
Figure 45: **Move 51, Black: Bg5 (f4g5).** (a) Our position; Melanie plays Bg5 (f4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3199` (ply 92) [structural distance  $d=19$ , QAP obj=19]: the master played Rc1; under  $\mu$  our Bf5 plays the role of that game’s Rc5 (a **cross-type** correspondence); cross-type roles: our nh8 as their pa7; transferred move f4g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1625` (ply 94) [structural distance  $d=19$ , QAP obj=9]: the master played Bc2; under  $\mu$  our Bf5 plays the role of that game’s Ba4; cross-type roles: our pa6 as their bg4, our bf8 as their pa6; transferred move f4g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3465` (ply 93) [structural distance  $d=18$ , QAP obj=4]: the master played Bc5; under  $\mu$  our Bf5 plays the role of that game’s Ba2; cross-type roles: our bf8 as their nc6, our nh8 as their Pg5; transferred move f4g5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bg5 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



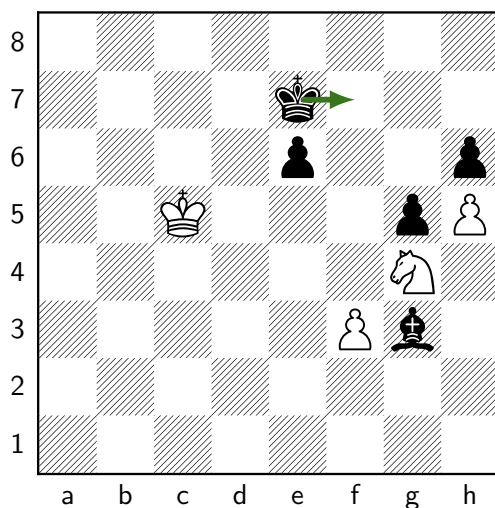
(a)



(b)

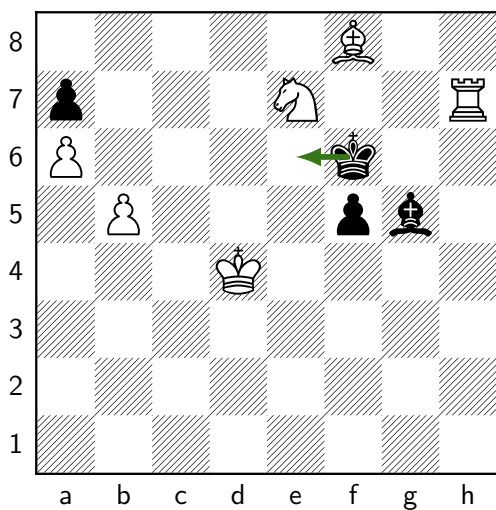


(c)

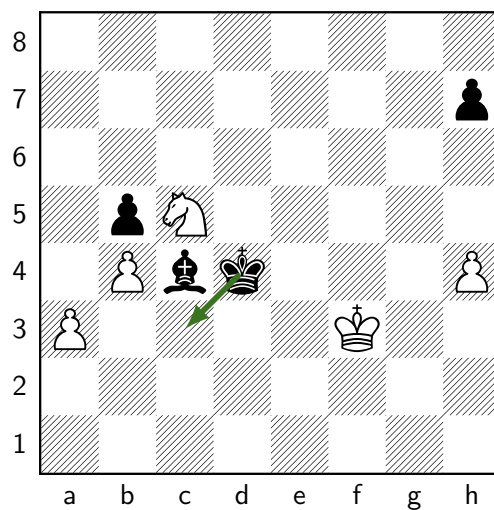


(d)

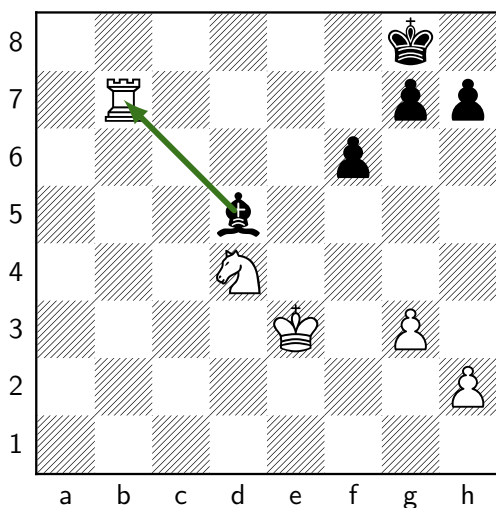
Figure 46: **Move 52, Black: Kf6**. (a) Our position; Melanie plays Kf6 (e6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2796` (ply 115) [structural distance  $d=19$ , QAP obj=13]: the master played Ke6; under  $\mu$  our Ke3 plays the role of that game’s Kd4; cross-type roles: our Pf5 as their Bd6, our Bg5 as their Ra5; transferred move e6f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3998` (ply 95) [structural distance  $d=16$ , QAP obj=11]: the master played Kd4; under  $\mu$  our Ke3 plays the role of that game’s Kc4; cross-type roles: our pa6 as their Be5, our Bg5 as their Pa4; transferred move e6f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1556` (ply 103) [structural distance  $d=19$ , QAP obj=10]: the master played Kf7; under  $\mu$  our Ke3 plays the role of that game’s Ke2; cross-type roles: our rh7 as their ng4, our ng6 as their pf3; transferred move e6f6 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kf6 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



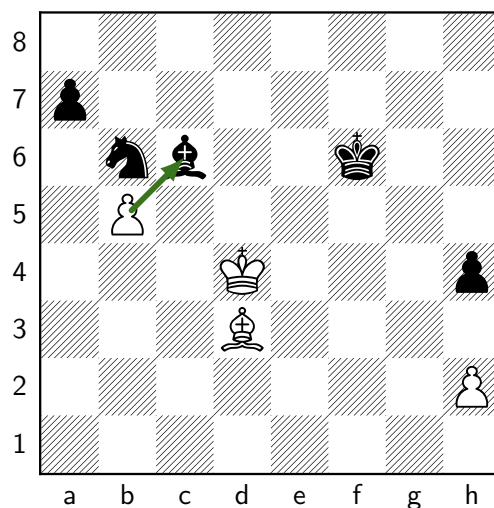
(a)



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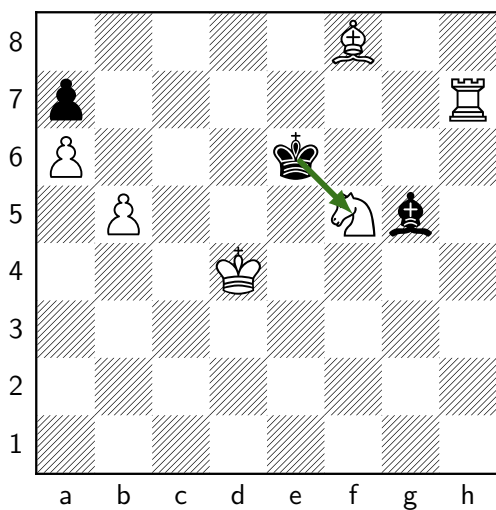
(c)



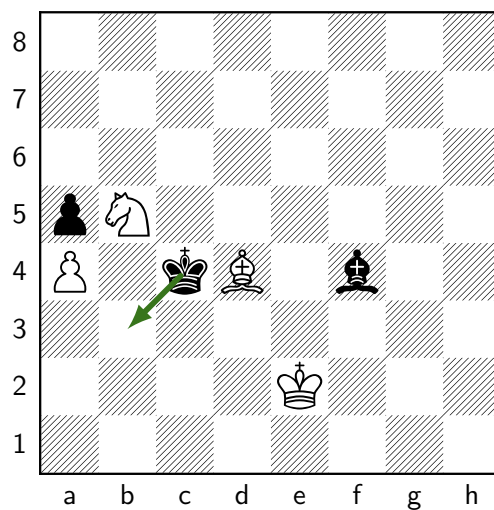
(d)

Figure 47: **Move 53, Black: Ke6.** (a) Our position; Melanie plays Ke6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3902` (ply 103) [structural distance  $d=27$ , QAP obj=22]: the master played Kc3; under  $\mu$  our Kf3 plays the role of that game’s Kd5; cross-type roles: our bf8 as their pb4, our rh7 as their ph4; transferred move f6e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#440` (ply 95) [structural distance  $d=28$ , QAP obj=19]: the master played Bxb7; under  $\mu$  our Kf3 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our ne7 as their rb7, our rh7 as their Ph7; transferred move f6e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1838` (ply 100) [structural distance  $d=24$ , QAP obj=19]: the master played bxc6; under  $\mu$  our Kf3 plays the role of that game’s Pb5 (a **cross-type** correspondence); cross-type roles: our rh7 as their nb6, our ne7 as their bc6; transferred move f6e6 (**endorses this move**). Synthesis: 24 retrieved analogs transferred to Ke6 (summed QAP objective 230, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

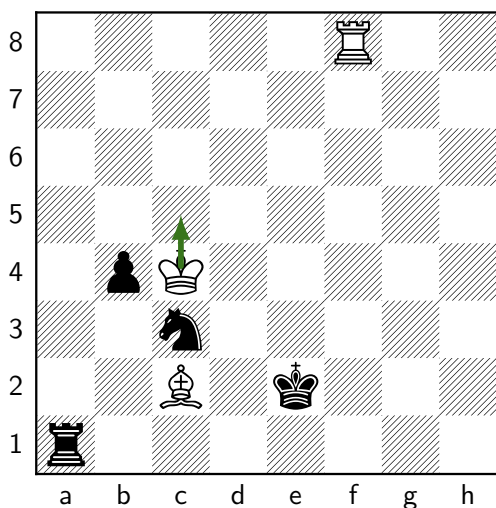




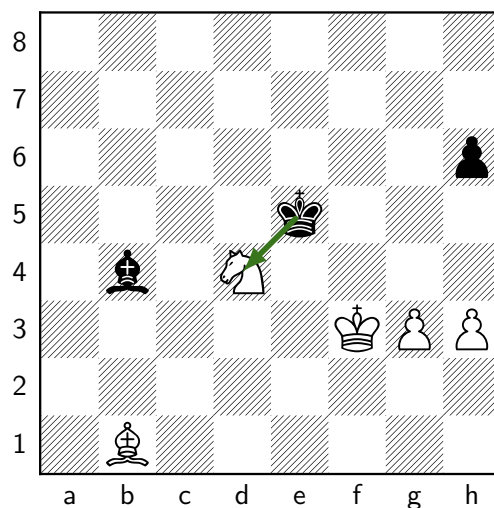
(a)



(b)

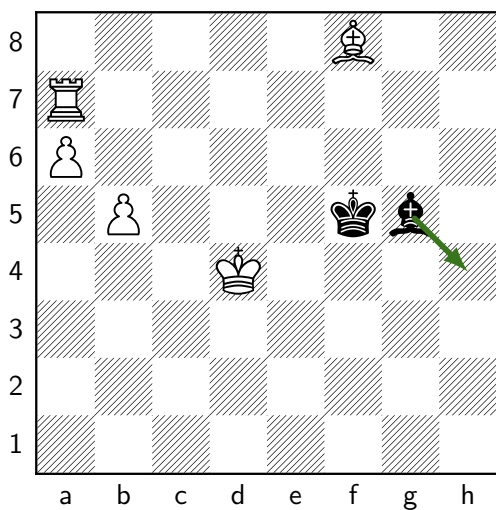


(c)

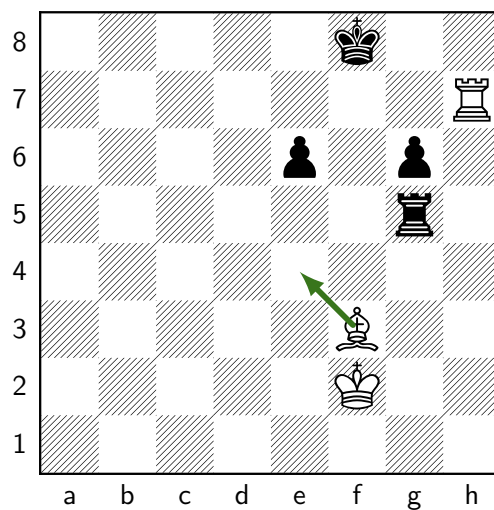


(d)

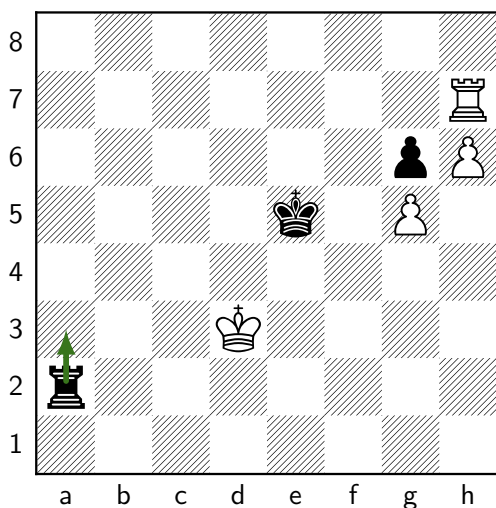
Figure 48: **Move 54, Black: Kxf5**. (a) Our position; Melanie plays Kxf5 (e6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2911` (ply 113) [structural distance  $d=19$ , QAP obj=35]: the master played Kb3; under  $\mu$  our Ke3 plays the role of that game’s Kc5; cross-type roles: our nf5 as their bd4; transferred move e6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3238` (ply 110) [structural distance  $d=19$ , QAP obj=25]: the master played Kc5; under  $\mu$  our Ke3 plays the role of that game’s Kc4; cross-type roles: our nf5 as their pb4, our bf8 as their Rf8; transferred move e6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1894` (ply 111) [structural distance  $d=17$ , QAP obj=25]: the master played Kxd4; under  $\mu$  our Ke3 plays the role of that game’s Ke4; cross-type roles: our Pa7 as their Bb4, our rh7 as their pg3; transferred move e6f5 (**endorses this move**). Synthesis: 26 retrieved analogs transferred to Kxf5 (summed QAP objective 329, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



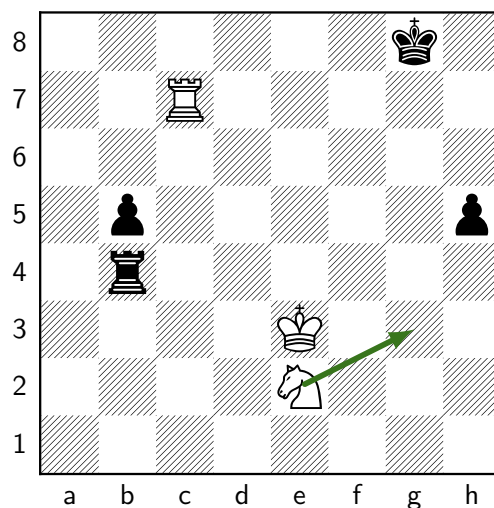
(a)



(b)



(c)



(d)

Figure 49: **Move 55, Black: Bh4**. (a) Our position; Melanie plays Bh4 (g5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3574` (ply 106) [structural distance  $d=7$ , QAP obj=3]: the master played Be4; under  $\mu$  our Bg4 plays the role of that game’s Bf3; cross-type roles: our bf8 as their pe6, our pb5 as their Rh7; transferred move g5h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#770` (ply 103) [structural distance  $d=10$ , QAP obj=3]: the master played Ra3+; under  $\mu$  our Bg4 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our bf8 as their Pg6; transferred move g5h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1171` (ply 104) [structural distance  $d=13$ , QAP obj=3]: the master played Ng3; under  $\mu$  our Bg4 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our bf8 as their ph5, our pb5 as their Rc7; transferred move g5h4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bh4 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

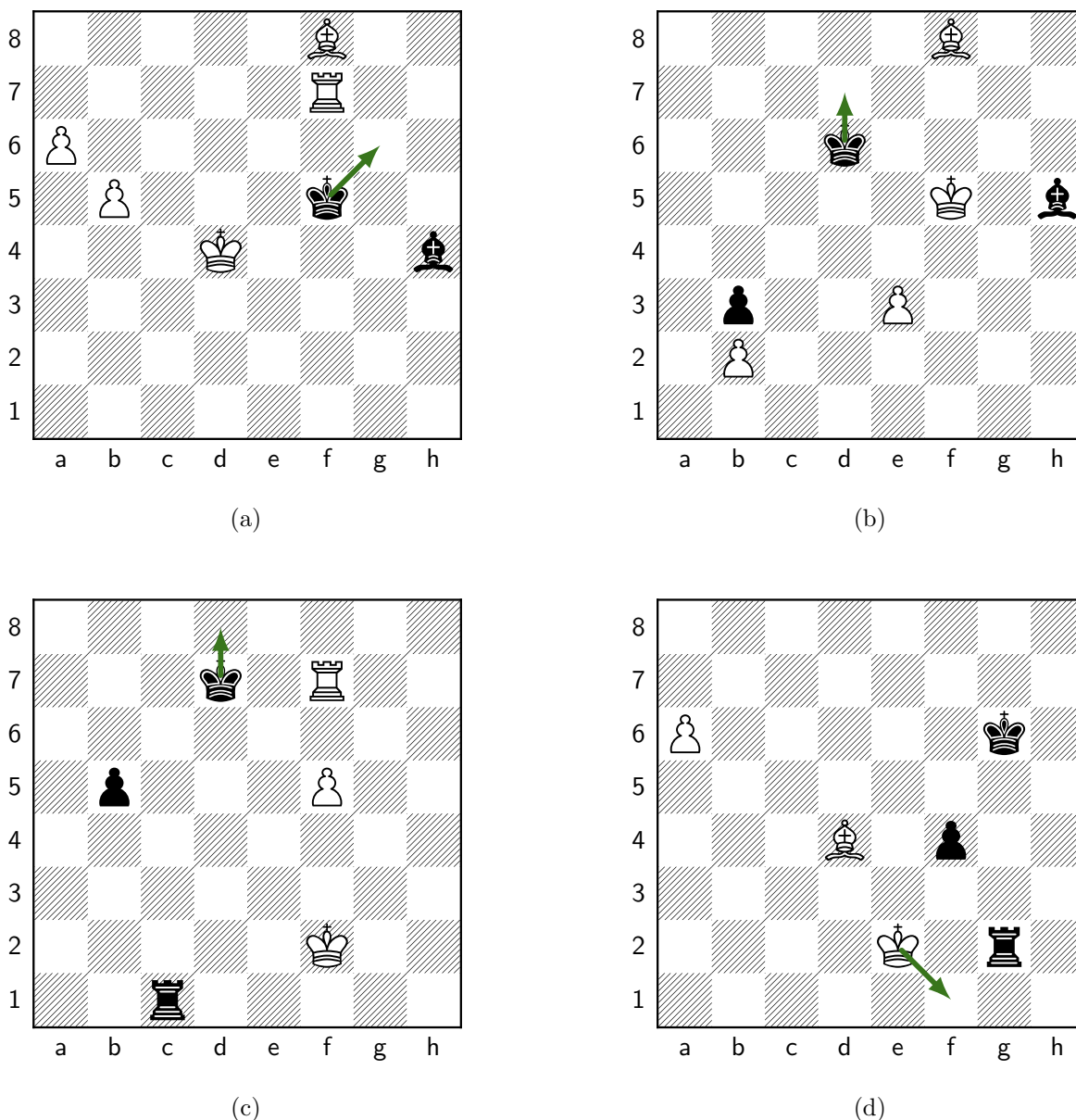
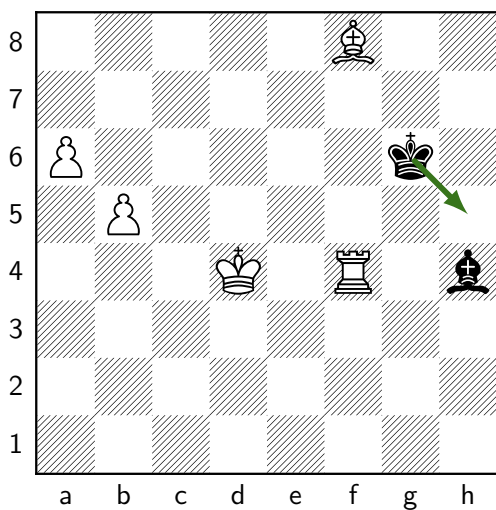
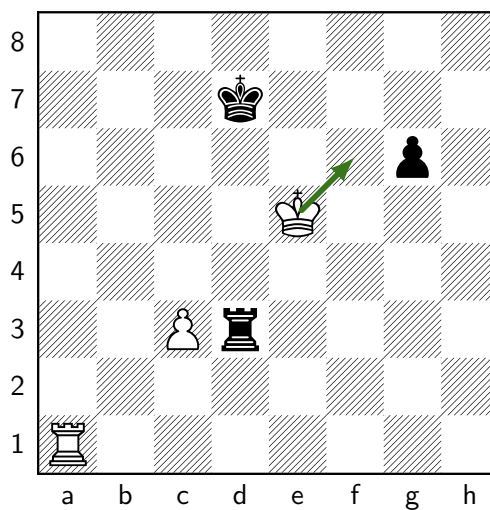


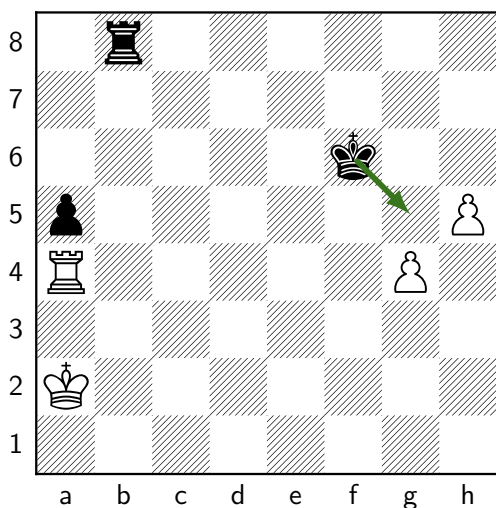
Figure 50: **Move 56, Black: Kg6.** (a) Our position; Melanie plays Kg6 (f5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1732` (ply 127) [structural distance  $d=11$ , QAP obj=69]: the master played Kd7; under  $\mu$  our Kf4 plays the role of that game’s Kd3; cross-type roles: our rf7 as their bf8, our Bh4 as their pe3; transferred move f5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3310` (ply 123) [structural distance  $d=11$ , QAP obj=66]: the master played Kd8; under  $\mu$  our Kf4 plays the role of that game’s Kd2; cross-type roles: our bf8 as their Pb5, our Bh4 as their Rc1; transferred move f5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2824` (ply 108) [structural distance  $d=11$ , QAP obj=66]: the master played Kf1; under  $\mu$  our Kf4 plays the role of that game’s Ke2; cross-type roles: our bf8 as their pf4; transferred move f5g6 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg6 (summed QAP objective 651, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



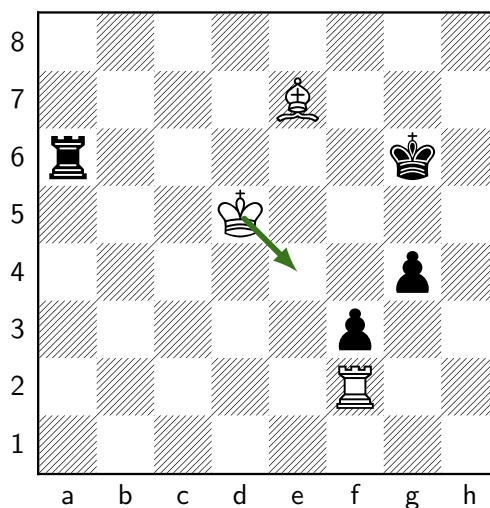
(a)



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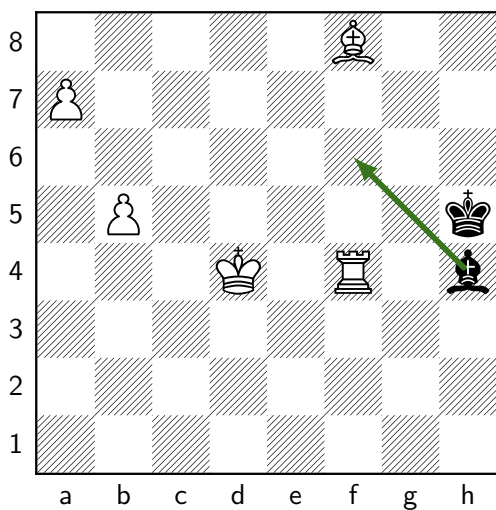


(c)

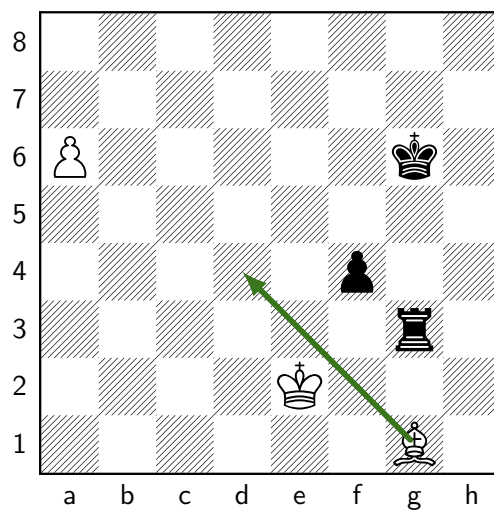


(d)

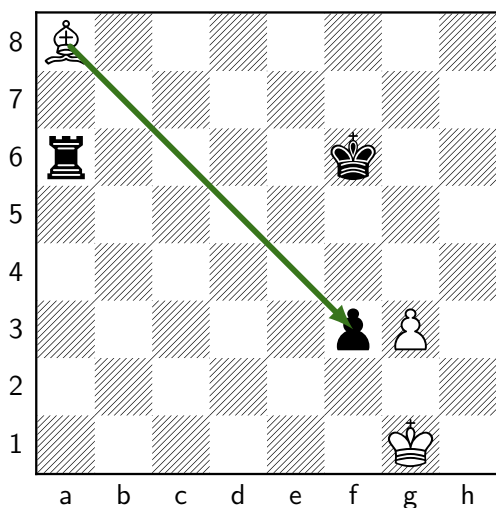
Figure 51: **Move 57, Black: Kh5**. (a) Our position; Melanie plays Kh5 (g6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2148` (ply 104) [structural distance  $d=15$ , QAP obj=16]: the master played Kf6; under  $\mu$  our Kg3 plays the role of that game’s Ke5; cross-type roles: our pa6 as their Ra1, our Bh4 as their Pc3; transferred move g6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3807` (ply 117) [structural distance  $d=15$ , QAP obj=15]: the master played Kg5; under  $\mu$  our Kg3 plays the role of that game’s Kf3; cross-type roles: our pa6 as their Rb8, our Bh4 as their Pa5; transferred move g6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#305` (ply 112) [structural distance  $d=15$ , QAP obj=12]: the master played Ke4; under  $\mu$  our Kg3 plays the role of that game’s Kd5; cross-type roles: our Bh4 as their pf3, our bf8 as their ra6; transferred move g6h5 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kh5 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



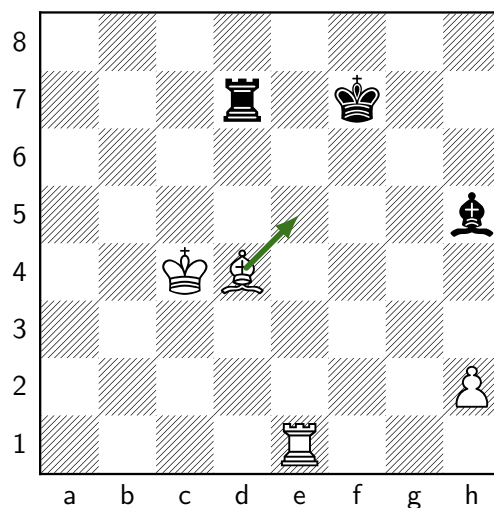
(a)



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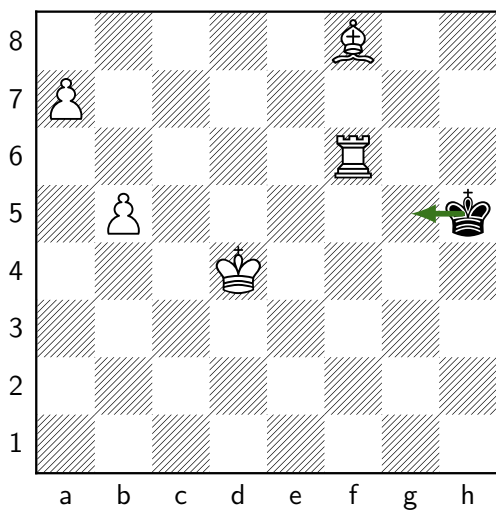


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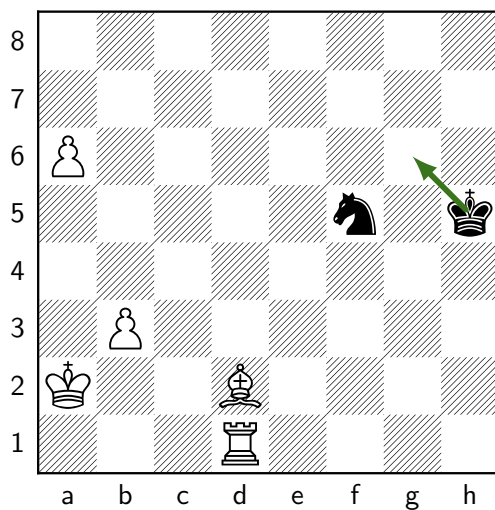


(d)

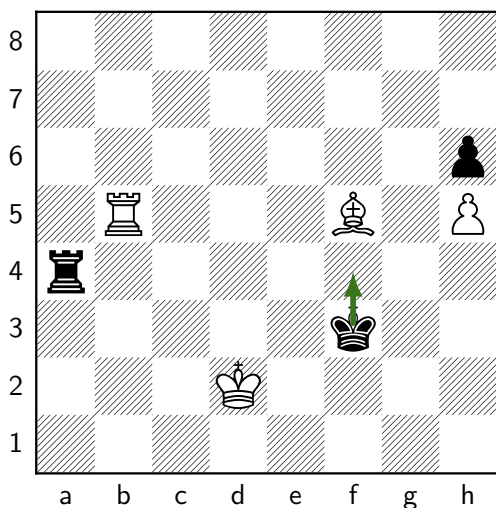
Figure 52: **Move 58, Black: Bf6+.** (a) Our position; Melanie plays Bf6+ (h4f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2824` (ply 106) [structural distance  $d=15$ , QAP obj=22]: the master played Bd4; under  $\mu$  our Bh5 plays the role of that game’s Bg1; cross-type roles: our bf8 as their pf4; transferred move h4f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2960` (ply 106) [structural distance  $d=15$ , QAP obj=21]: the master played Bxf3; under  $\mu$  our Bh5 plays the role of that game’s Ba8; cross-type roles: our bf8 as their pf3; transferred move h4f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#437` (ply 110) [structural distance  $d=16$ , QAP obj=17]: the master played Be5; under  $\mu$  our Bh5 plays the role of that game’s Bd4; cross-type roles: our pb5 as their Re1, our bf8 as their Ph2; transferred move h4f6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bf6+ (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



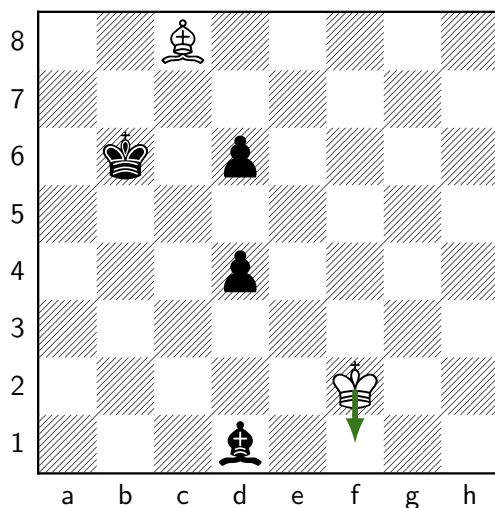
(a)



(b)



(c)



(d)

Figure 53: **Move 59, Black: Kg5.** (a) Our position; Melanie plays Kg5 (h5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#41` (ply 133) [structural distance  $d=7$ , QAP obj=1]: the master played Kg6; under  $\mu$  our  $Kh4$  plays the role of that game’s  $Kh4$ ; transferred move h5g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#810` (ply 129) [structural distance  $d=6$ , QAP obj=1]: the master played Kf4; under  $\mu$  our  $Kh4$  plays the role of that game’s  $Kf6$ ; transferred move h5g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2609` (ply 104) [structural distance  $d=7$ , QAP obj=1]: the master played Kf1; under  $\mu$  our  $Kh4$  plays the role of that game’s  $Kf2$ ; cross-type roles: our  $rf6$  as their  $bd1$ , our  $bf8$  as their  $pd4$ ; transferred move h5g5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kg5 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

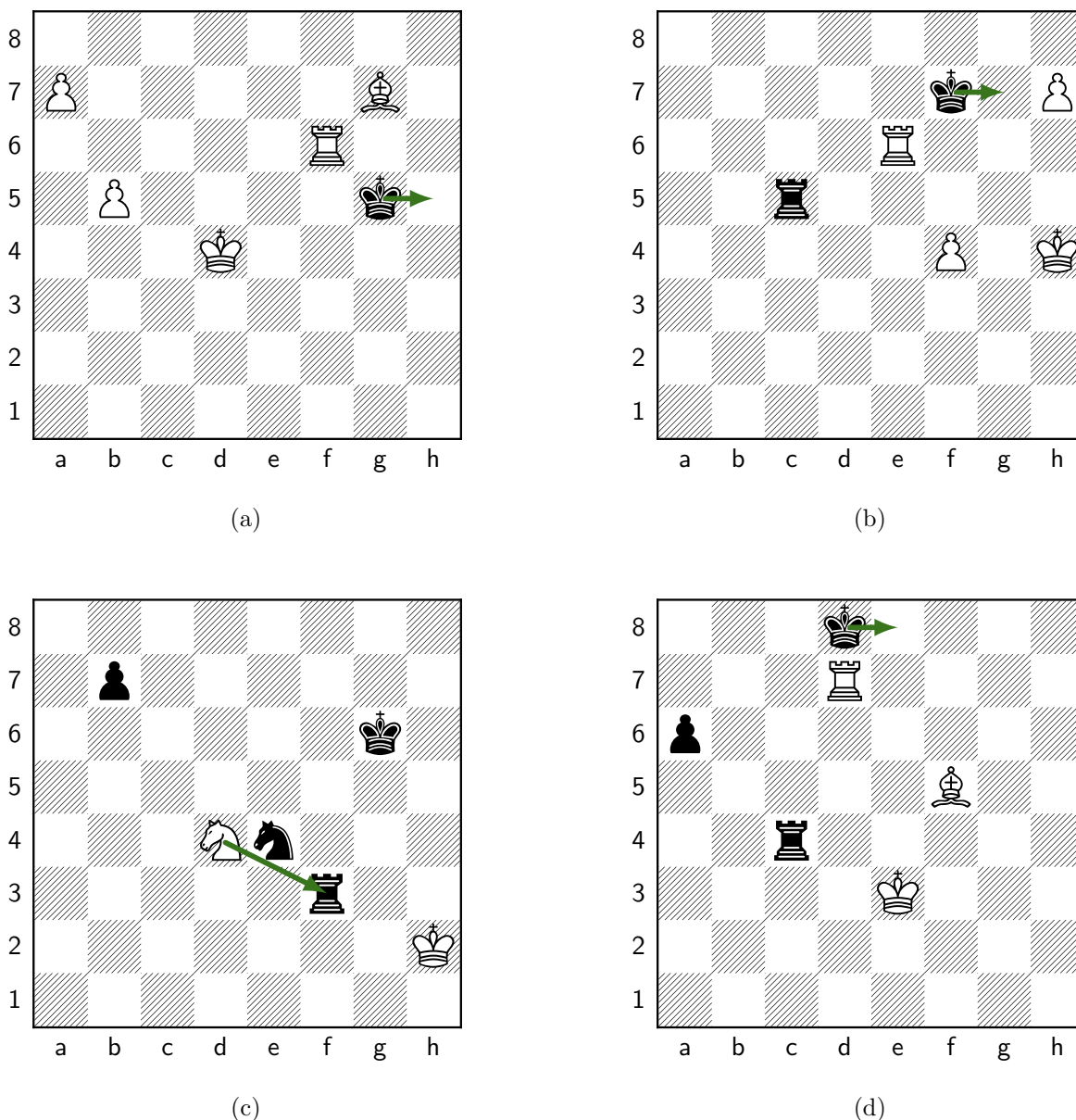
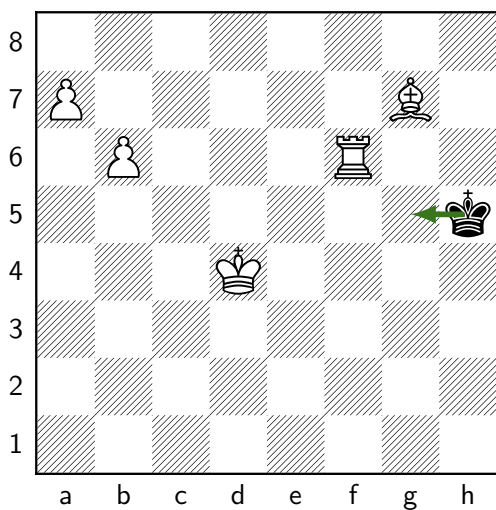
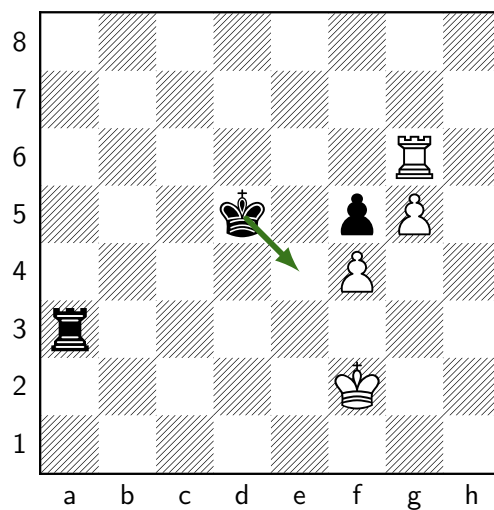


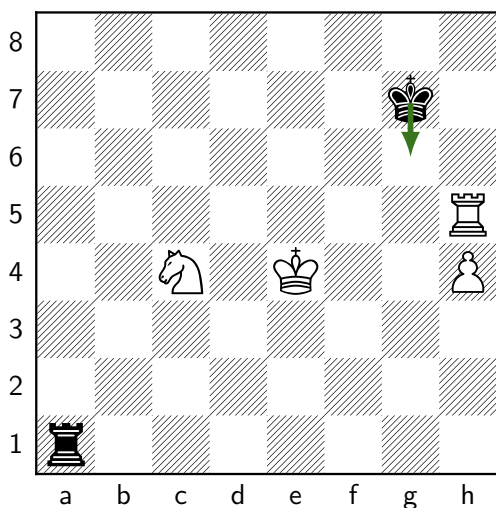
Figure 54: **Move 60, Black: Kh5.** (a) Our position; Melanie plays Kh5 (g5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#1439 (ply 107) [structural distance  $d=11$ , QAP obj=19]: the master played Kg7; under  $\mu$  our Kg4 plays the role of that game’s Kf2; cross-type roles: our bg7 as their ph7, our pa7 as their Rc5; transferred move g5h5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#415 (ply 114) [structural distance  $d=11$ , QAP obj=6]: the master played Nxf3; under  $\mu$  our Kg4 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our pa7 as their ne4, our bg7 as their pb7; transferred move g5h5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#2187 (ply 107) [structural distance  $d=10$ , QAP obj=2]: the master played Ke8; under  $\mu$  our Kg4 plays the role of that game’s Kd1; cross-type roles: our pb5 as their Rc4; transferred move g5h5 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kh5 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



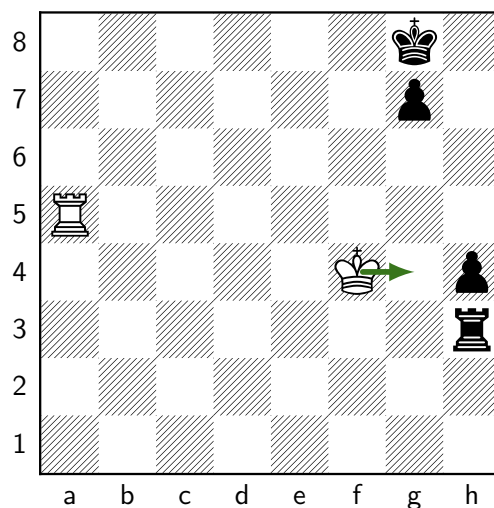
(a)



(b)



(c)



(d)

Figure 55: **Move 61, Black: Kg5.** (a) Our position; Melanie plays Kg5 (**h5g5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2446` (ply 119) [structural distance  $d=10$ , QAP obj=2]: the master played **Ke4**; under  $\mu$  our *Kh4* plays the role of that game’s *Kd4*; cross-type roles: our *bg7* as their *pg5*; transferred move **h5g5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#374` (ply 115) [structural distance  $d=11$ , QAP obj=2]: the master played **Kg6**; under  $\mu$  our *Kh4* plays the role of that game’s *Kg2*; cross-type roles: our *pb6* as their *nc4*, our *bg7* as their *ph4*; transferred move **h5g5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2930` (ply 110) [structural distance  $d=11$ , QAP obj=2]: the master played **Kg4**; under  $\mu$  our *Kh4* plays the role of that game’s *Kf4*; cross-type roles: our *bg7* as their *ph4*, our *pa7* as their *Ra5*; transferred move **h5g5 (endorses this move)**. Synthesis: 14 retrieved analogs transferred to Kg5 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



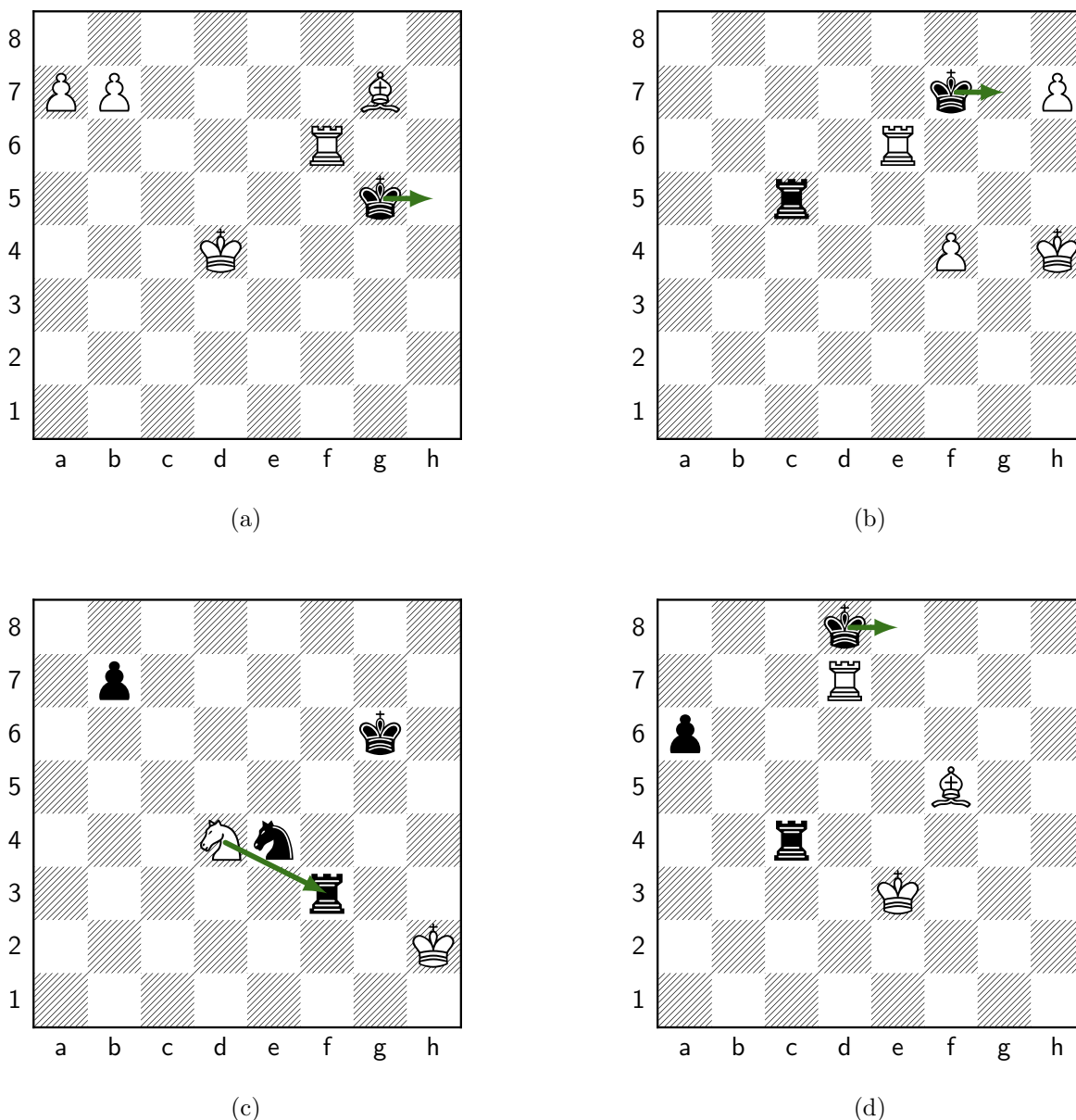
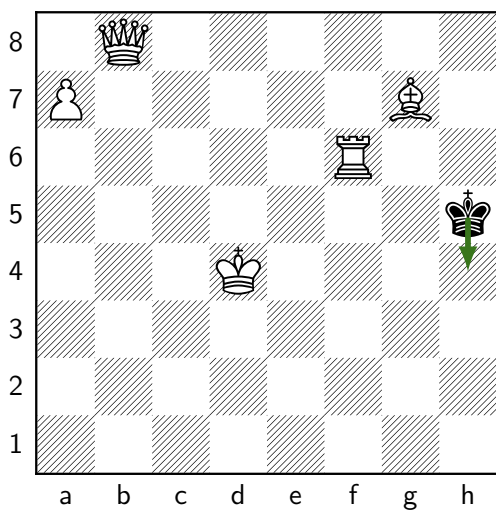
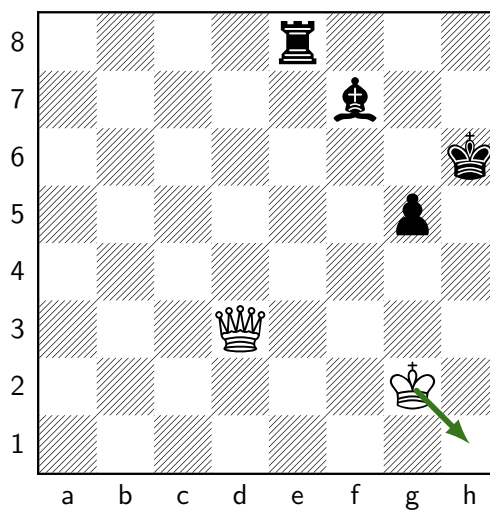


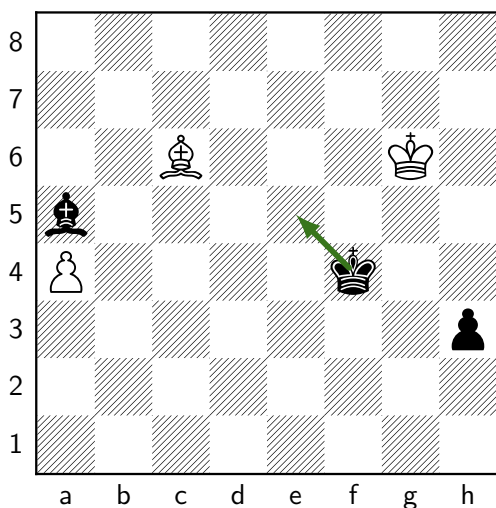
Figure 56: **Move 62, Black: Kh5.** (a) Our position; Melanie plays Kh5 (**g5h5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1439` (ply 107) [structural distance  $d=11$ , QAP obj=19]: the master played Kg7; under  $\mu$  our Kg4 plays the role of that game’s Kf2; cross-type roles: our bg7 as their ph7, our pa7 as their Rc5; transferred move g5h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#415` (ply 114) [structural distance  $d=11$ , QAP obj=6]: the master played Nxf3; under  $\mu$  our Kg4 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our pa7 as their ne4, our bg7 as their pb7; transferred move g5h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2187` (ply 107) [structural distance  $d=10$ , QAP obj=2]: the master played Ke8; under  $\mu$  our Kg4 plays the role of that game’s Kd1; cross-type roles: our pb7 as their Rc4; transferred move g5h5 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kh5 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



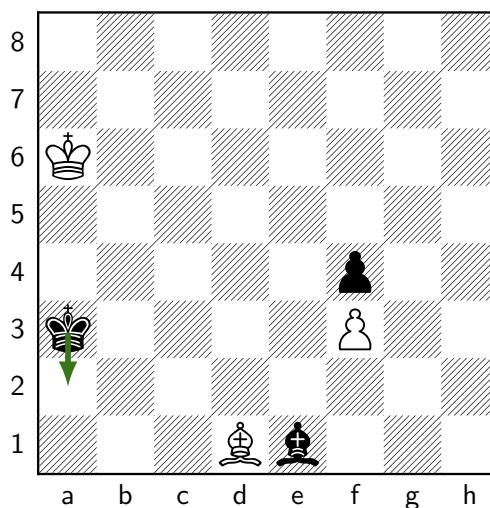
(a)



(b)

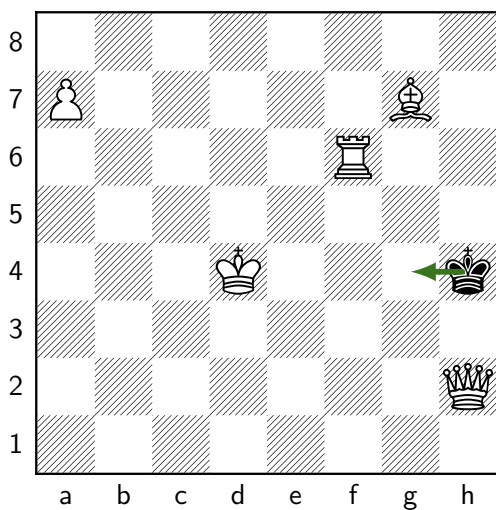


(c)

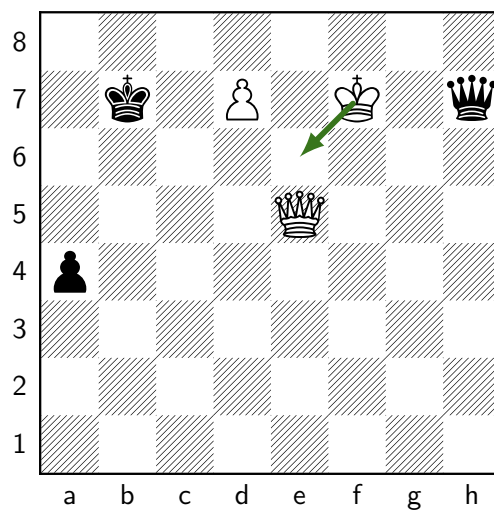


(d)

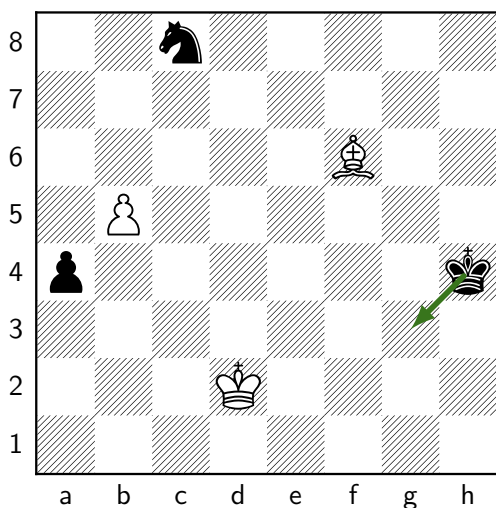
Figure 57: **Move 63, Black: Kh4**. (a) Our position; Melanie plays Kh4 (h5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1694` (ply 122) [structural distance  $d=11$ , QAP obj=2]: the master played Kh1; under  $\mu$  our Kh4 plays the role of that game’s Kg2; transferred move h5h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#8` (ply 119) [structural distance  $d=11$ , QAP obj=1]: the master played Ke5; under  $\mu$  our Kh4 plays the role of that game’s Kf5; cross-type roles: our rf6 as their bc6, our qb8 as their Ba5; transferred move h5h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1866` (ply 141) [structural distance  $d=12$ , QAP obj=1]: the master played Ka2; under  $\mu$  our Kh4 plays the role of that game’s Ka6; cross-type roles: our qb8 as their pf3, our rf6 as their bd1; transferred move h5h4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh4 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



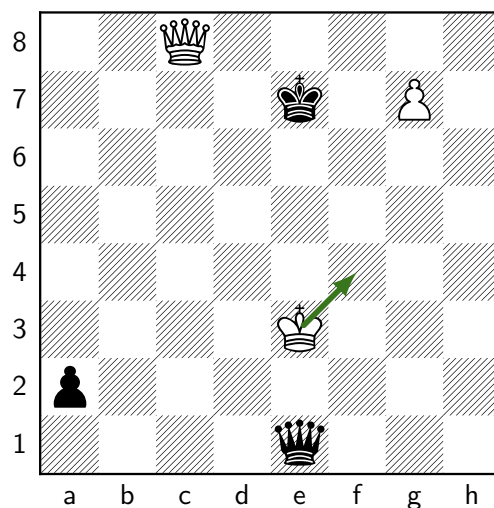
(a)



(b)

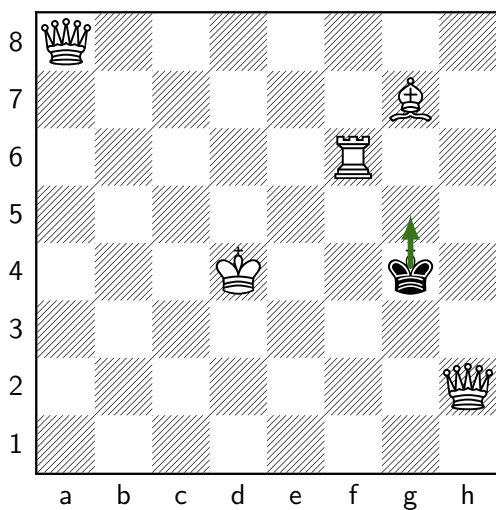


(c)

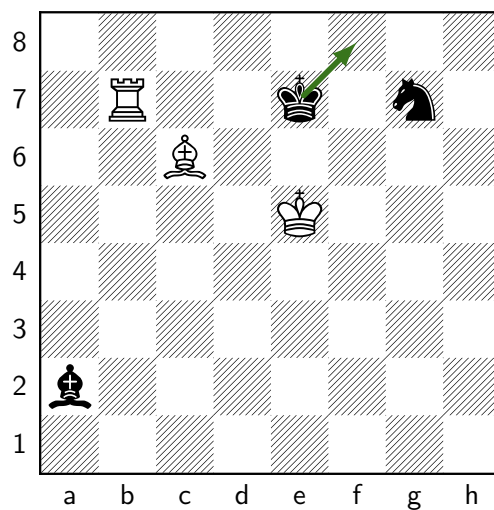


(d)

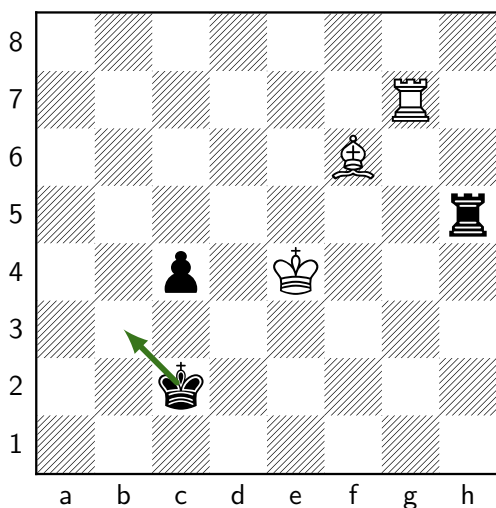
Figure 58: **Move 64, Black: Kg4**. (a) Our position; Melanie plays Kg4 (**h4g4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3920` (ply 134) [structural distance  $d=9$ , QAP obj=67]: the master played Ke6; under  $\mu$  our Kh5 plays the role of that game’s Kf7; cross-type roles: our rf6 as their Qe5, our bg7 as their Pd7; transferred move **h4g4 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#2507` (ply 121) [structural distance  $d=11$ , QAP obj=66]: the master played Kg3; under  $\mu$  our Kh5 plays the role of that game’s Kh5; cross-type roles: our rf6 as their Nc8, our qh2 as their bf6; transferred move **h4g4 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#2429` (ply 128) [structural distance  $d=10$ , QAP obj=66]: the master played Kf4; under  $\mu$  our Kh5 plays the role of that game’s Ke3; cross-type roles: our bg7 as their Pg7, our rf6 as their Qc8; transferred move **h4g4 (endorses this move)**. Synthesis: 31 retrieved analogs transferred to Kg4 (summed QAP objective 1128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



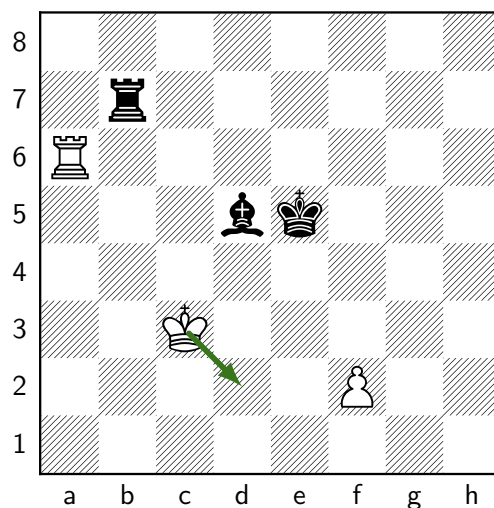
(a)



(b)



(c)



(d)

Figure 59: **Move 65, Black: Kg5**. (a) Our position; Melanie plays Kg5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3612` (ply 117) [structural distance  $d=14$ , QAP obj=2]: the master played Kf8; under  $\mu$  our Kg5 plays the role of that game’s Ke2; cross-type roles: our qa8 as their Ng7, our qh2 as their Ba2; transferred move g4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3326` (ply 145) [structural distance  $d=10$ , QAP obj=2]: the master played Kb3; under  $\mu$  our Kg5 plays the role of that game’s Kc7; cross-type roles: our qa8 as their Rh5, our qh2 as their Pc4; transferred move g4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2447` (ply 136) [structural distance  $d=13$ , QAP obj=1]: the master played Kd2; under  $\mu$  our Kg5 plays the role of that game’s Kc3; cross-type roles: our qh2 as their Pf2, our qa8 as their Ra6; transferred move g4g5 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kg5 (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.